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Teller**

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(54) **MEASUREMENT METHOD AND SYSTEM**

(56) **References Cited**

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(63) Continuation of application No. 18/152,458, filed on Jan. 10, 2023, now Pat. No. 11,892,626, which is a
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(57) **ABSTRACT**

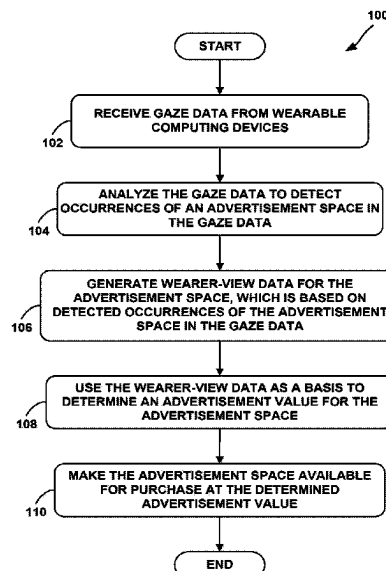
(51) **Int. Cl.**
G02B 27/00 (2006.01)
A61B 3/113 (2006.01)
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Methods and systems for determining an individual gaze value are disclosed herein. An exemplary method involves: (a) receiving gaze data for a first wearable computing device, wherein the gaze data is indicative of a wearer-view associated with the first wearable computing device, and wherein the first wearable computing device is associated with a first user-account; (b) analyzing the gaze data from the first wearable computing device to detect one or more occurrences of one or more advertisement spaces in the gaze data; (c) based at least in part on the one or more detected advertisement-space occurrences, determining an individual gaze value for the first user-account; and (d) sending a gaze-value indication, wherein the gaze-value indication indicates the individual gaze value for the first user-account.

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(58) **Field of Classification Search**
CPC G02B 27/0093; G02B 27/017; G02B 27/0172; G02B 27/14; G02B 2027/0138;
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23 Claims, 22 Drawing Sheets



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continuation of application No. 16/793,175, filed on Feb. 18, 2020, now Pat. No. 11,579,442, which is a continuation of application No. 15/943,346, filed on Apr. 2, 2018, now Pat. No. 10,598,929, which is a continuation-in-part of application No. 15/257,732, filed on Sep. 6, 2016, now Pat. No. 9,952,427, and a continuation-in-part of application No. 15/145,125, filed on May 3, 2016, now Pat. No. 10,354,291, said application No. 15/257,732 is a continuation of application No. 14/524,753, filed on Oct. 27, 2014, now Pat. No. 9,439,563, said application No. 15/145,125 is a continuation of application No. 13/292,898, filed on Nov. 9, 2011, now abandoned, said application No. 14/524,753 is a continuation of application No. 13/292,909, filed on Nov. 9, 2011, now Pat. No. 8,879,155.

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G06F 3/01 (2006.01)
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G06V 40/19 (2022.01)
H04H 60/33 (2008.01)
H04H 60/46 (2008.01)
H04H 60/63 (2008.01)

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See application file for complete search history.

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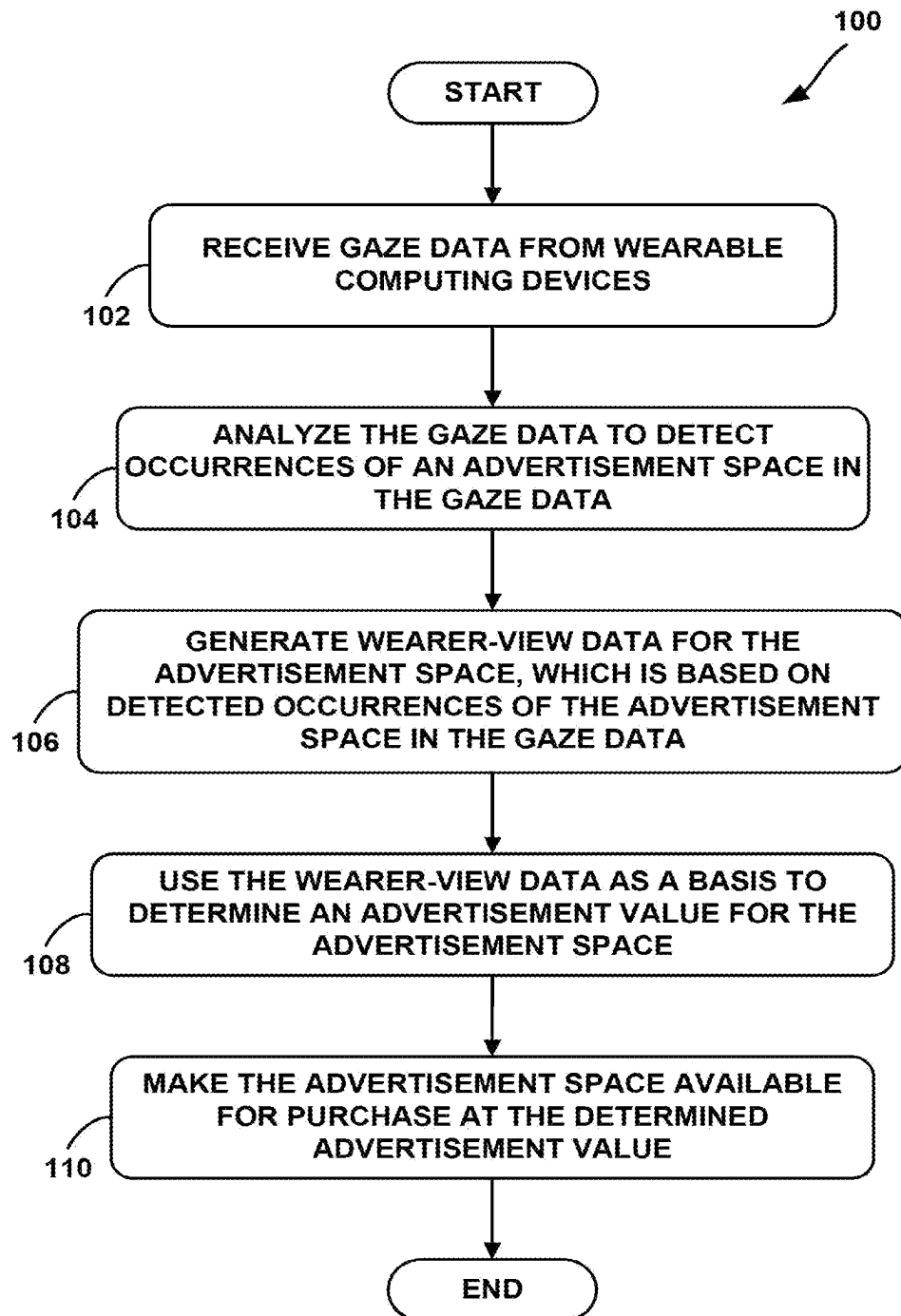
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**FIG. 1**

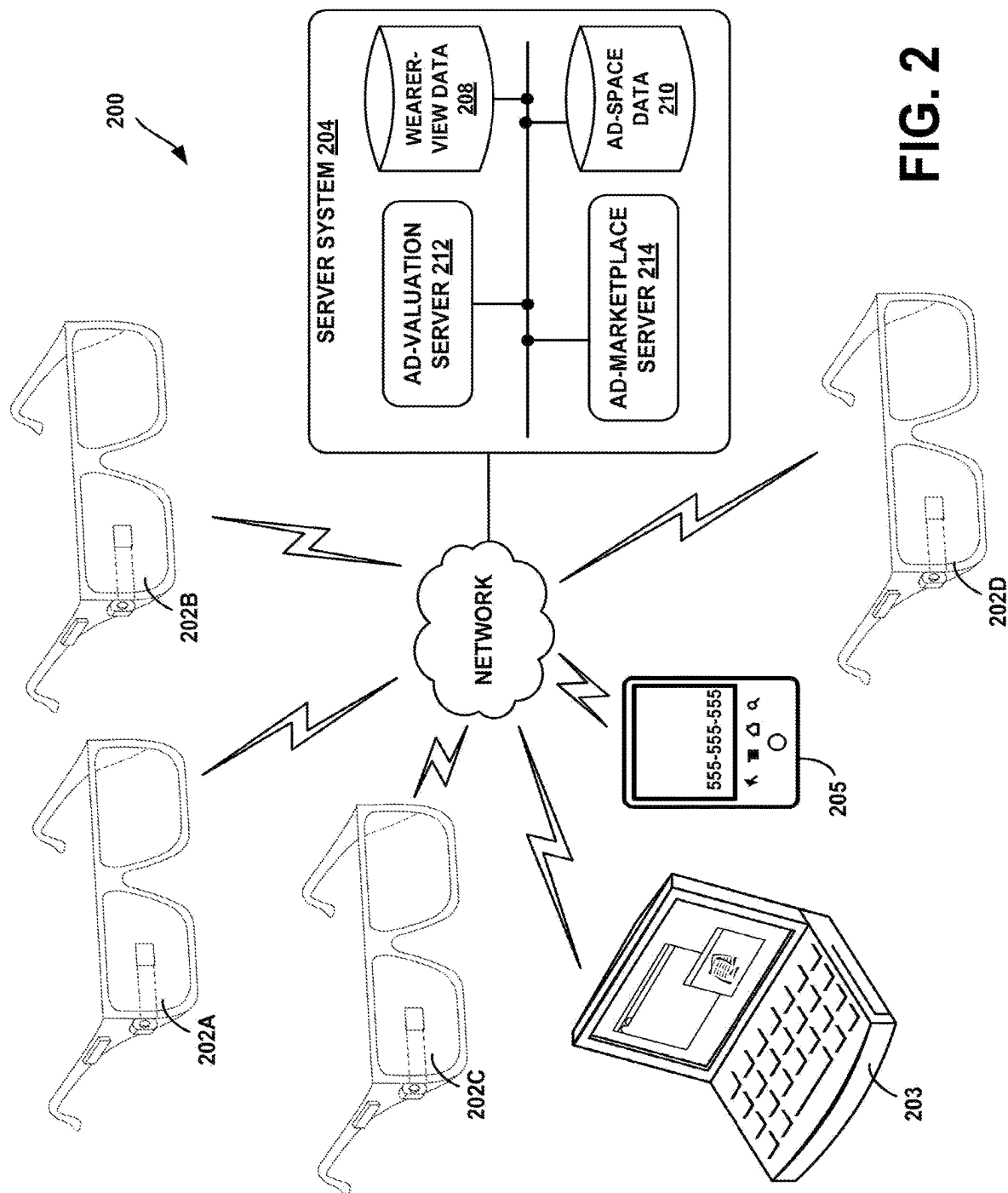
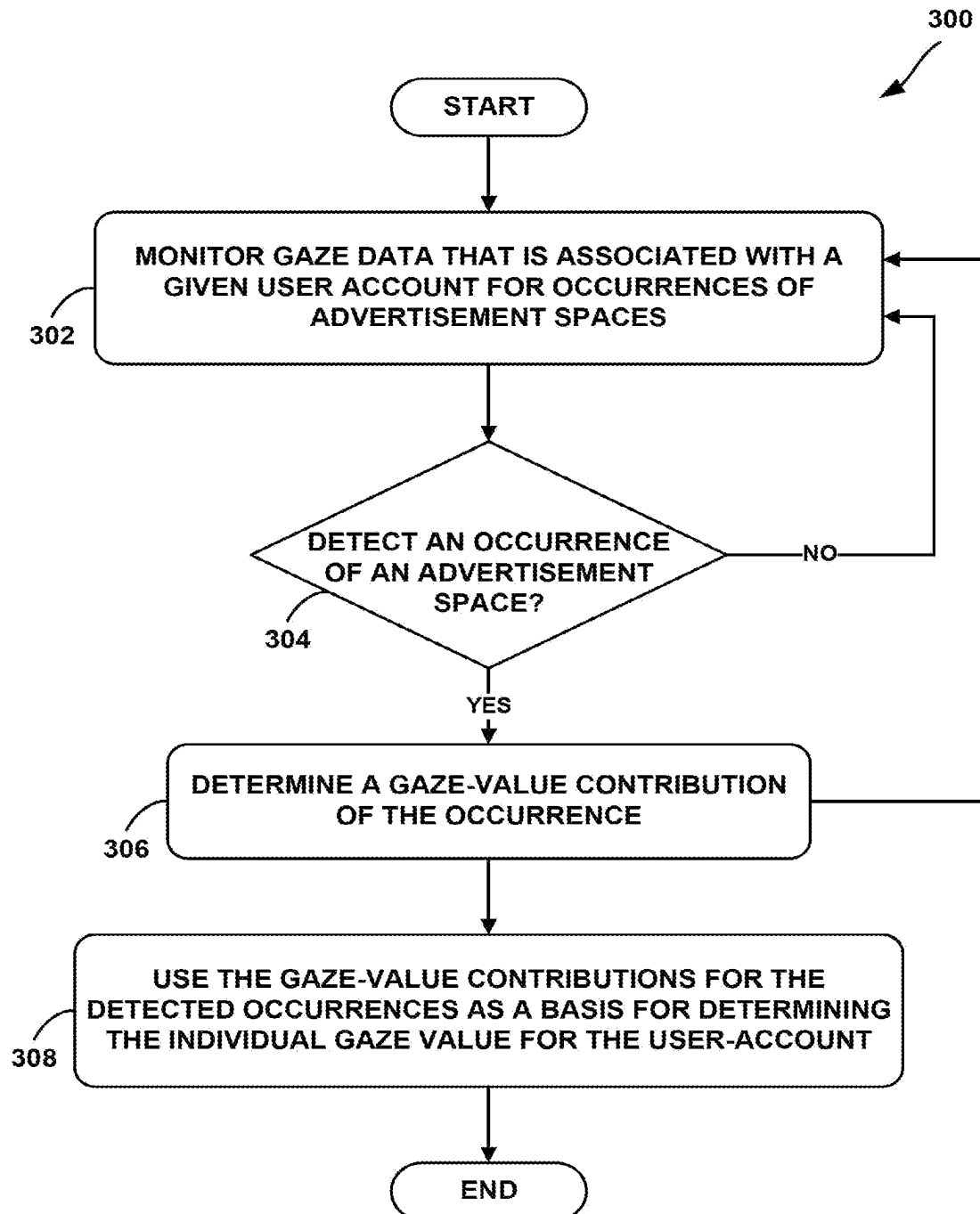


FIG. 2

**FIG. 3A**

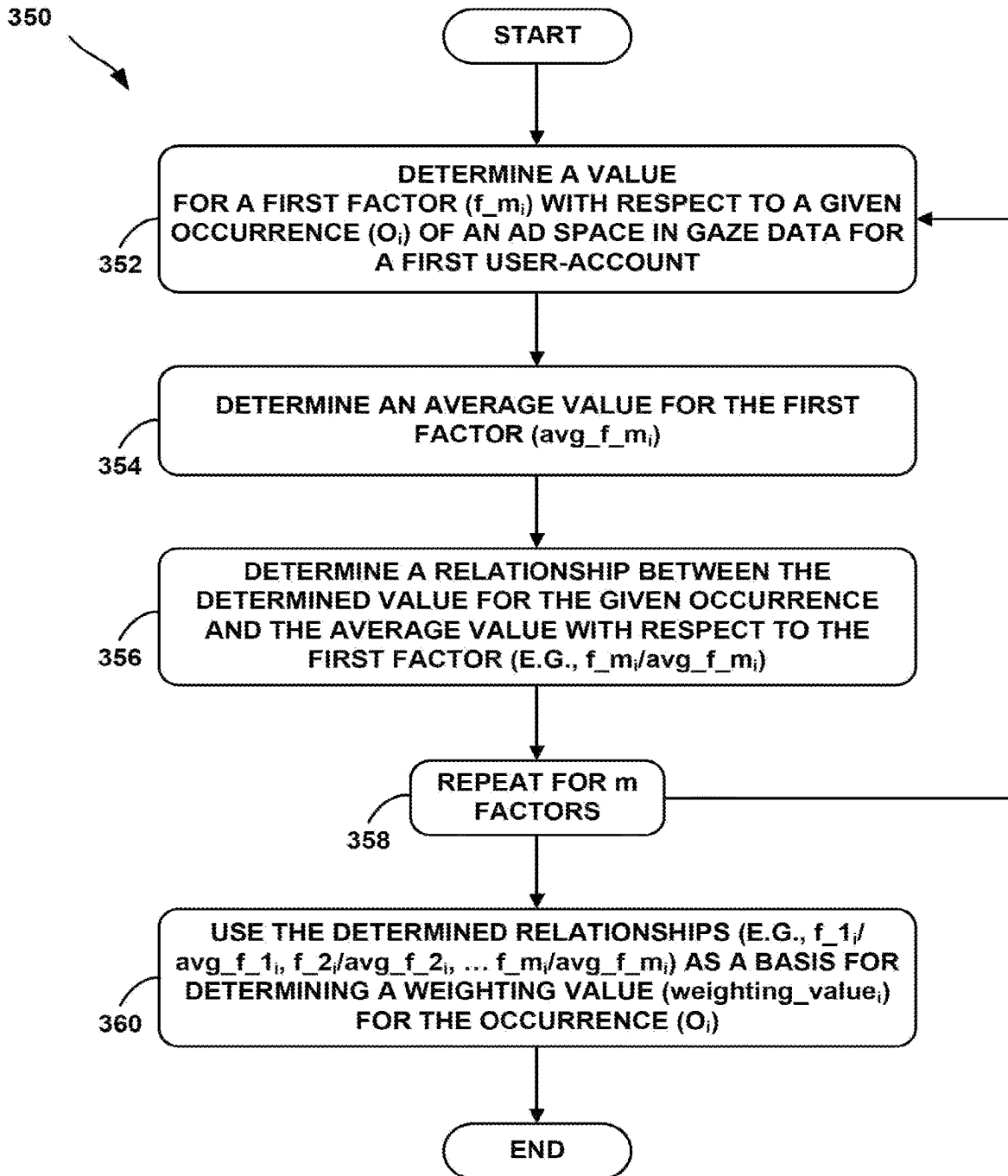
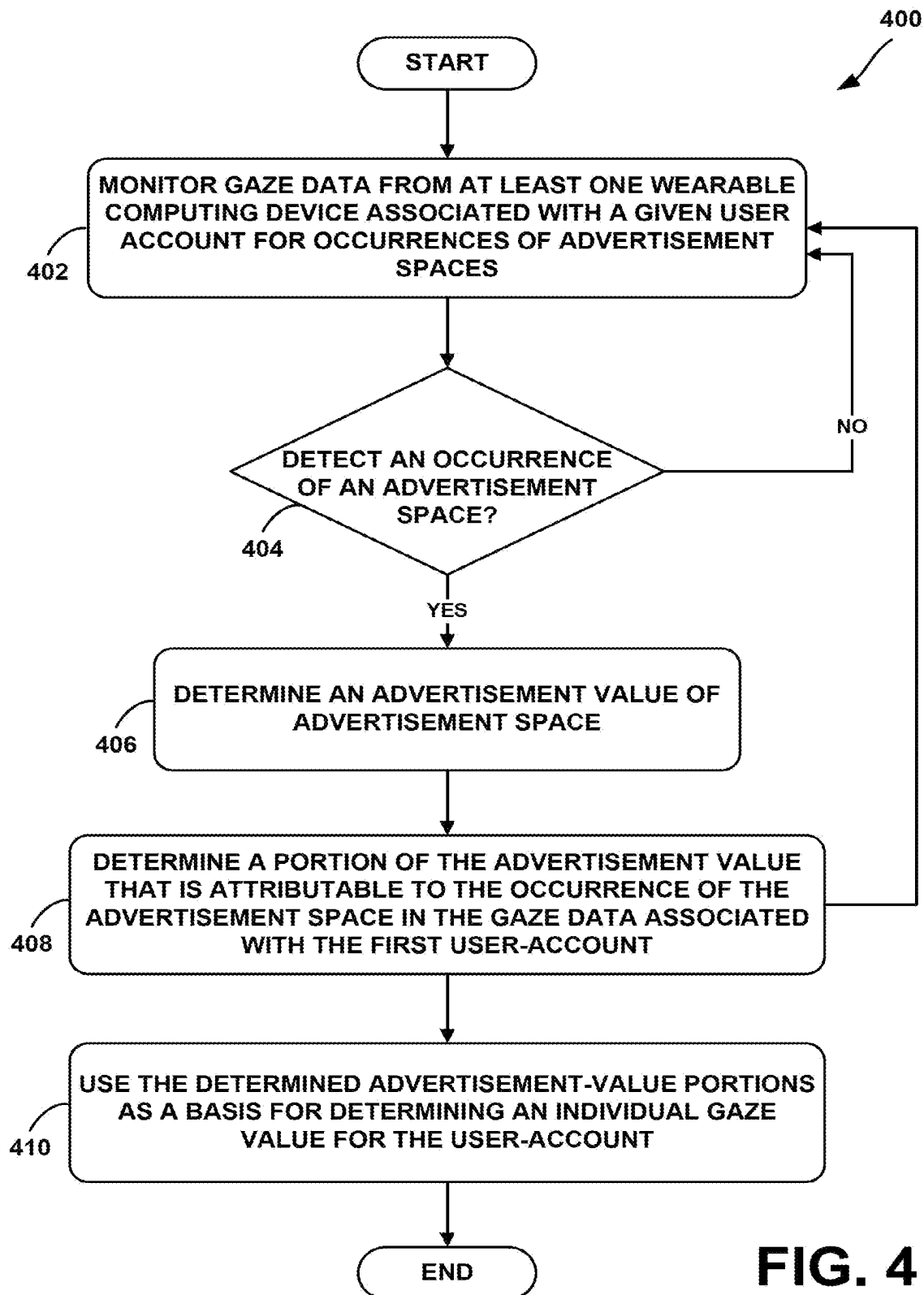
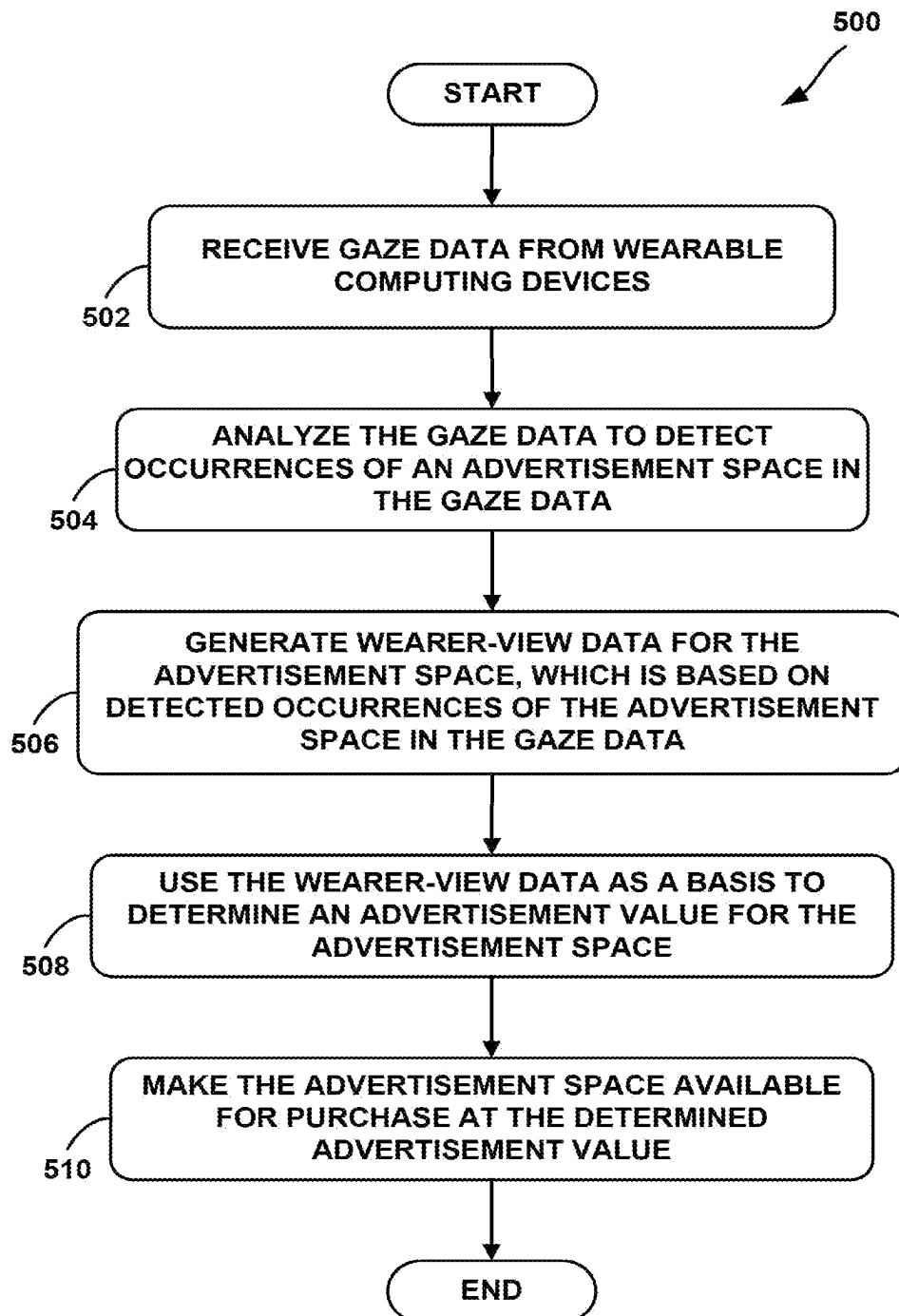
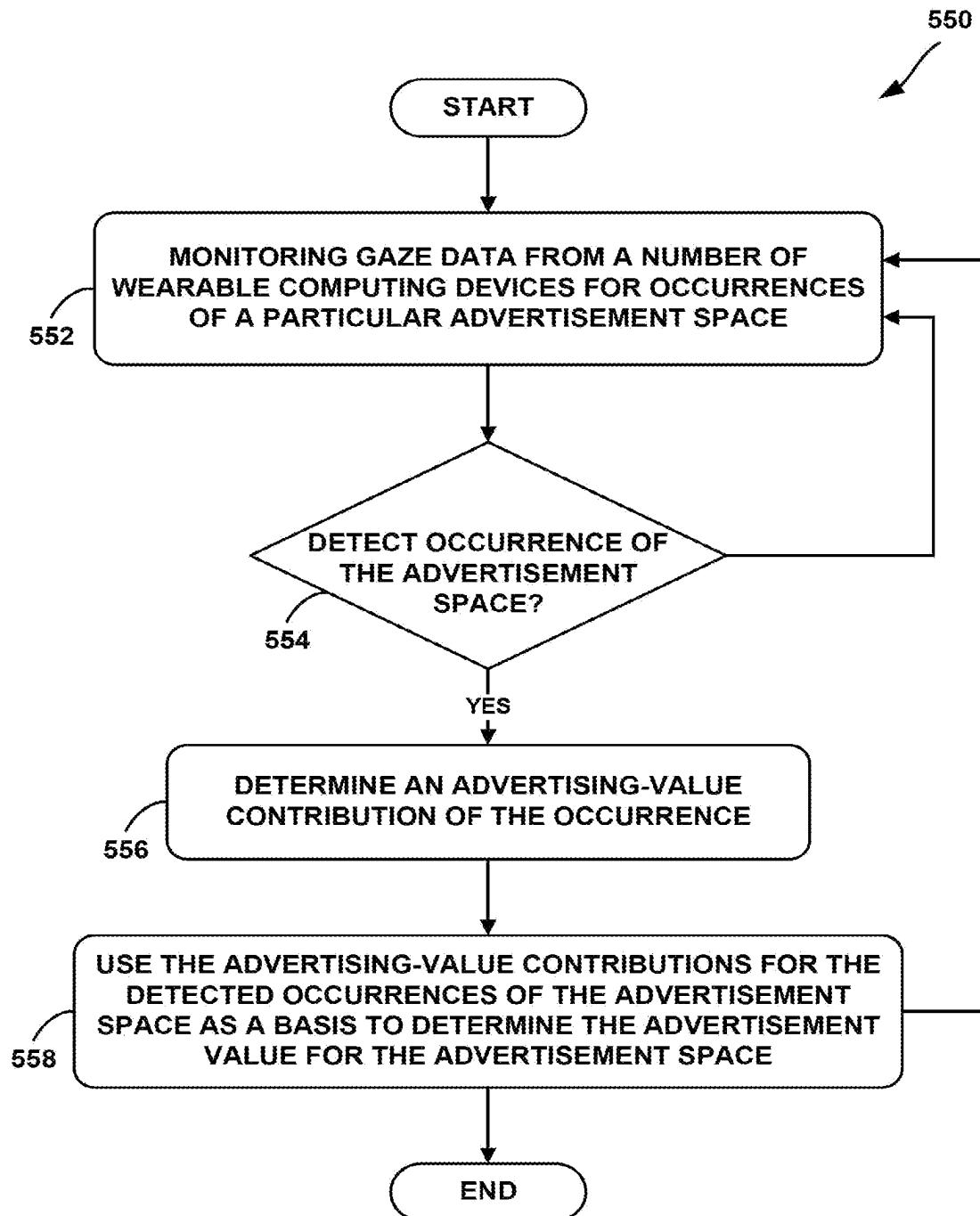
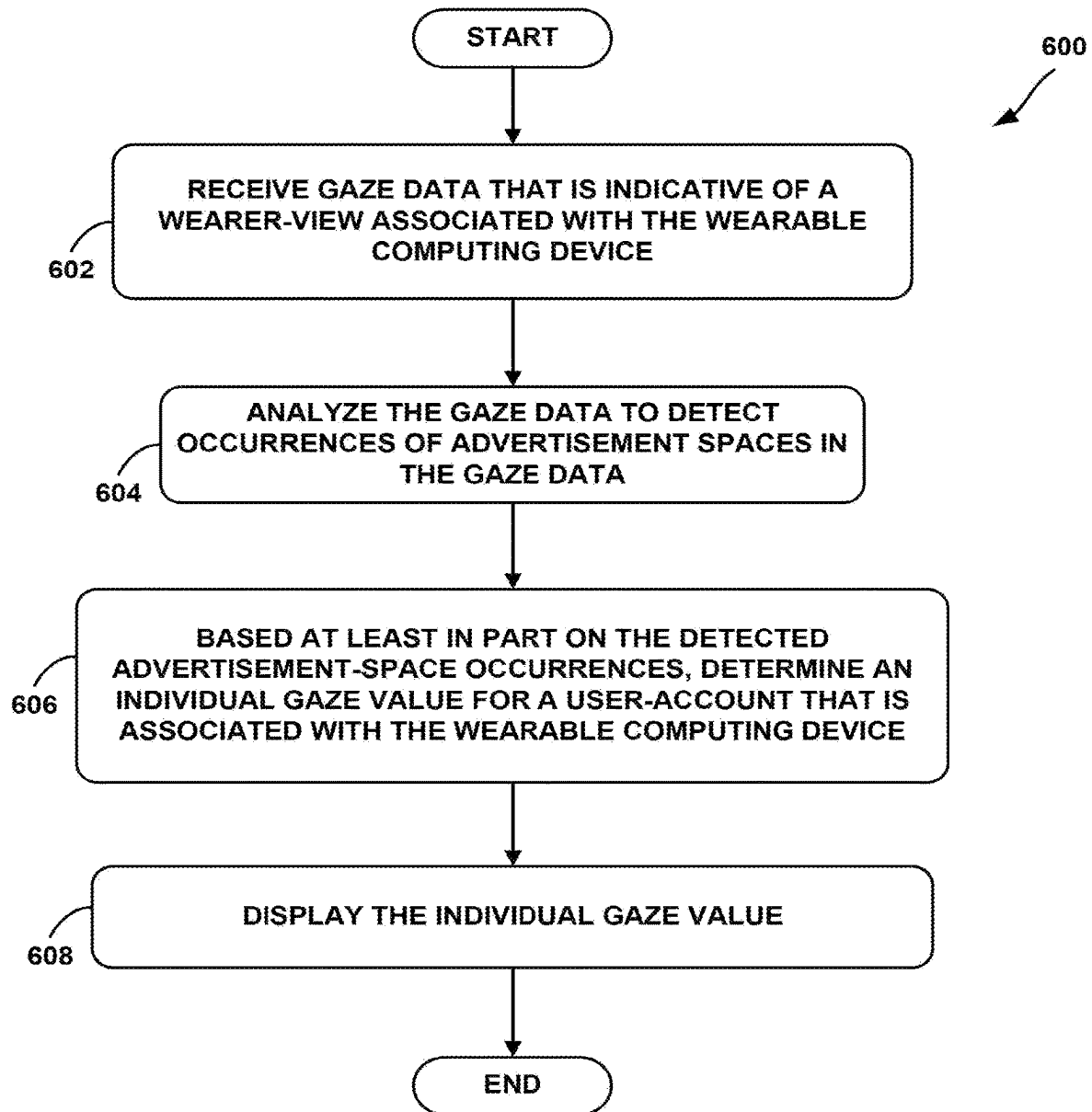


FIG. 3B

**FIG. 4**

**FIG. 5A**

**FIG. 5B**

**FIG. 6**

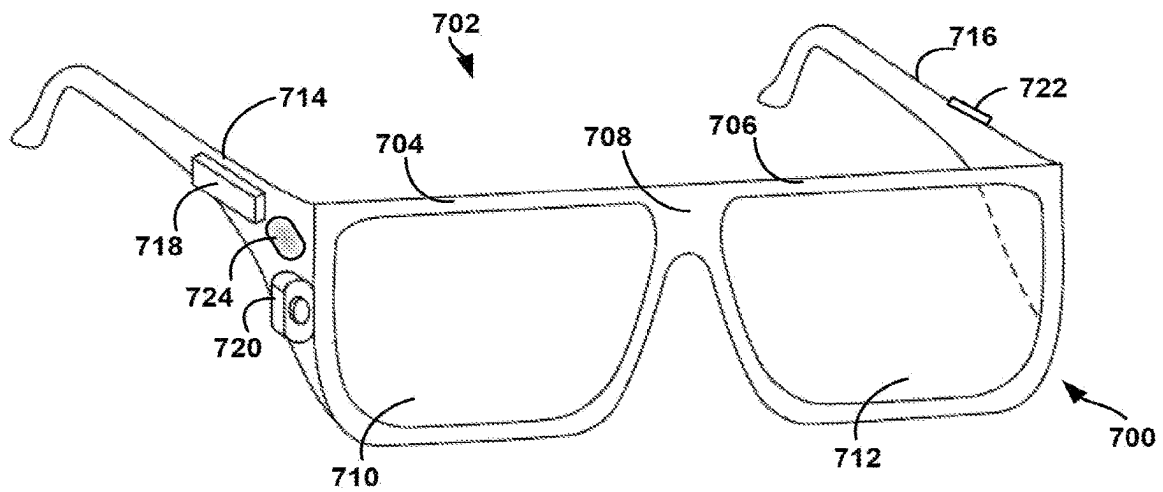


FIG. 7A

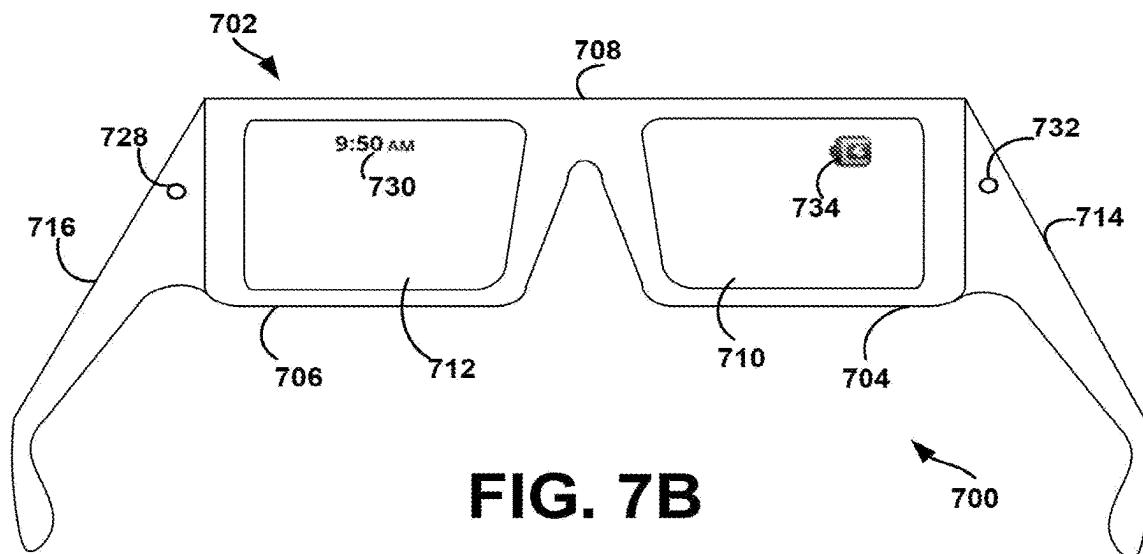


FIG. 7B

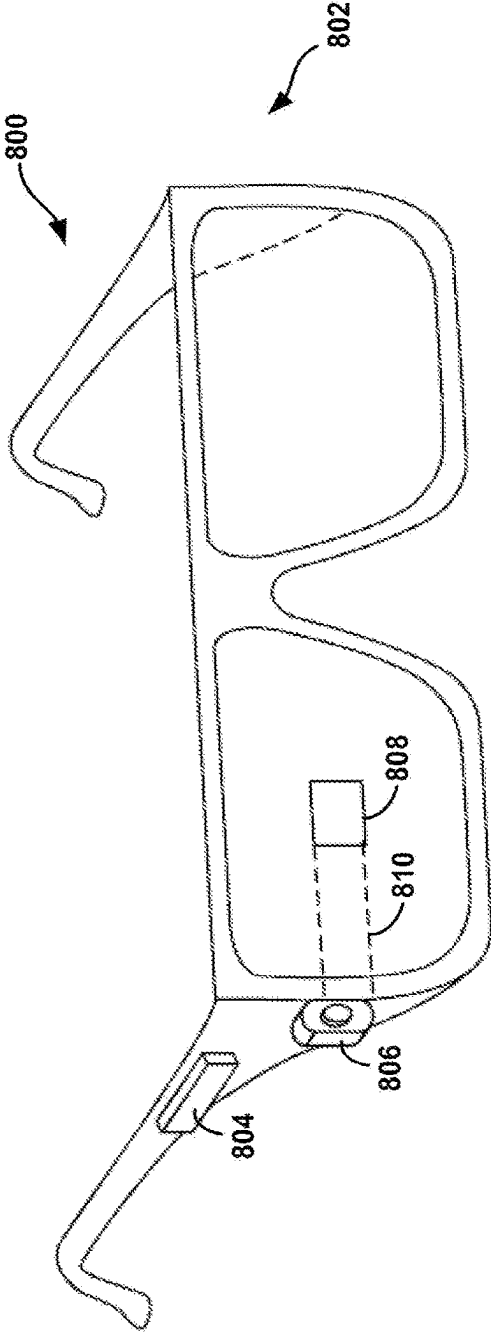


FIG. 8A

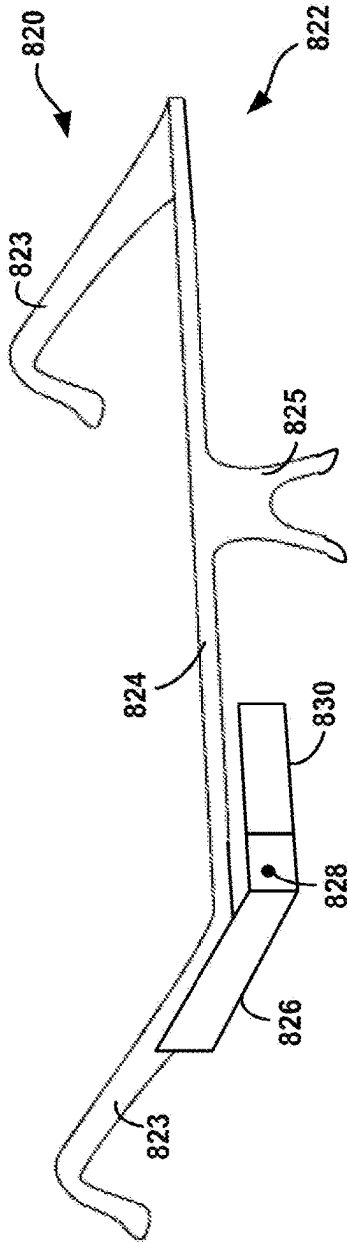


FIG. 8B

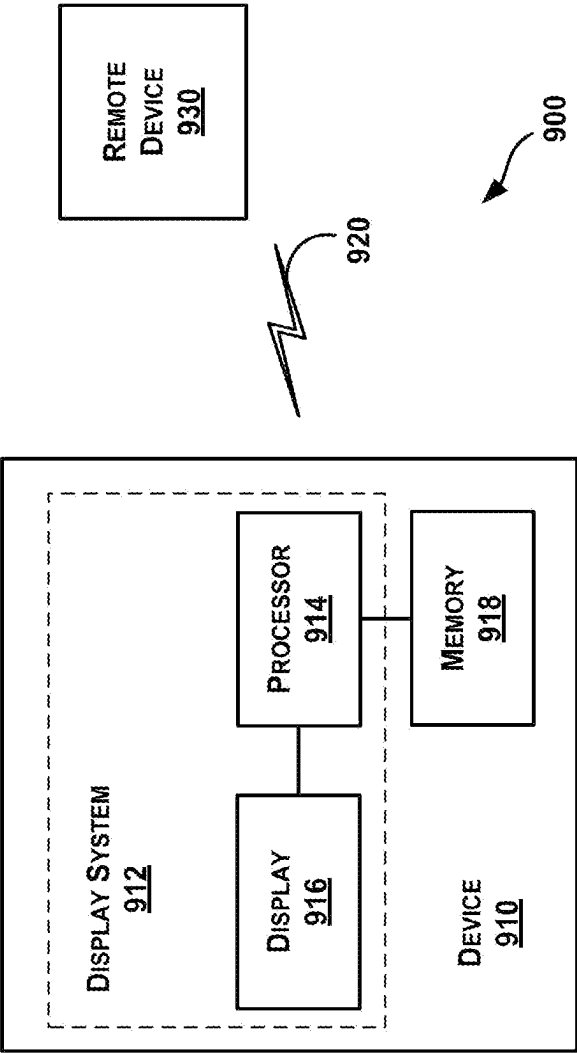
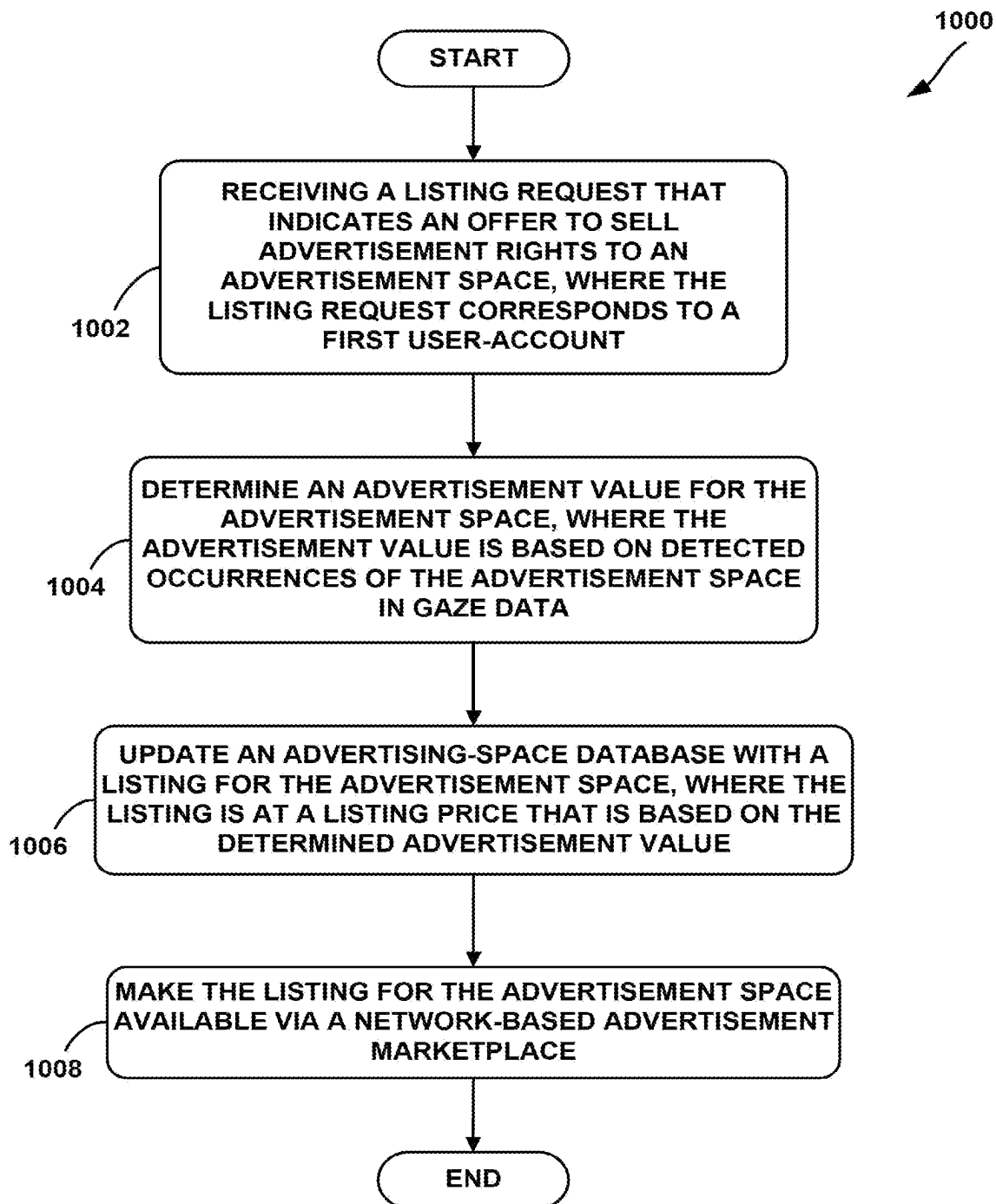
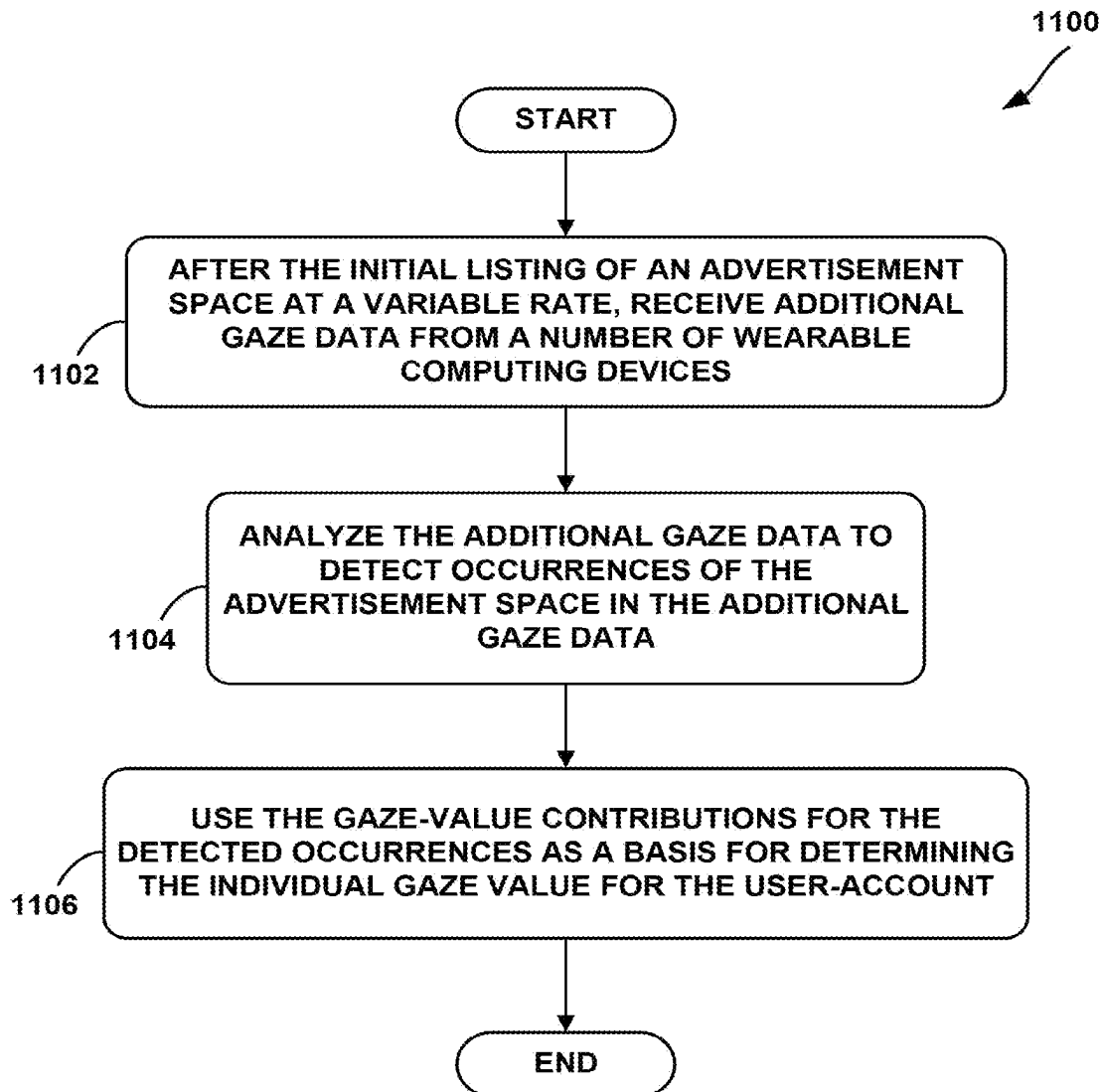
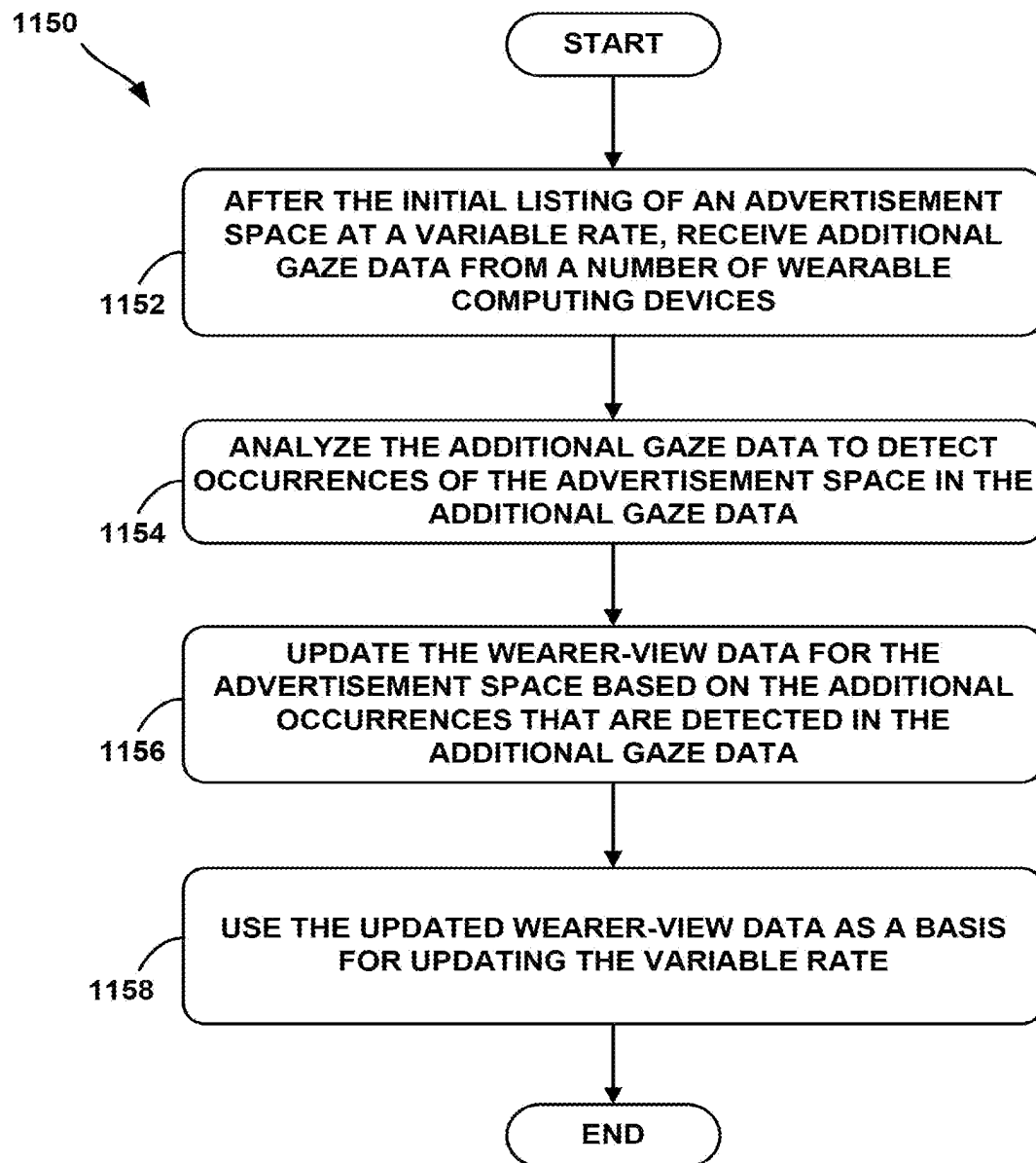


FIGURE 9

**FIG. 10**

**FIG. 11A**

**FIG. 11B**

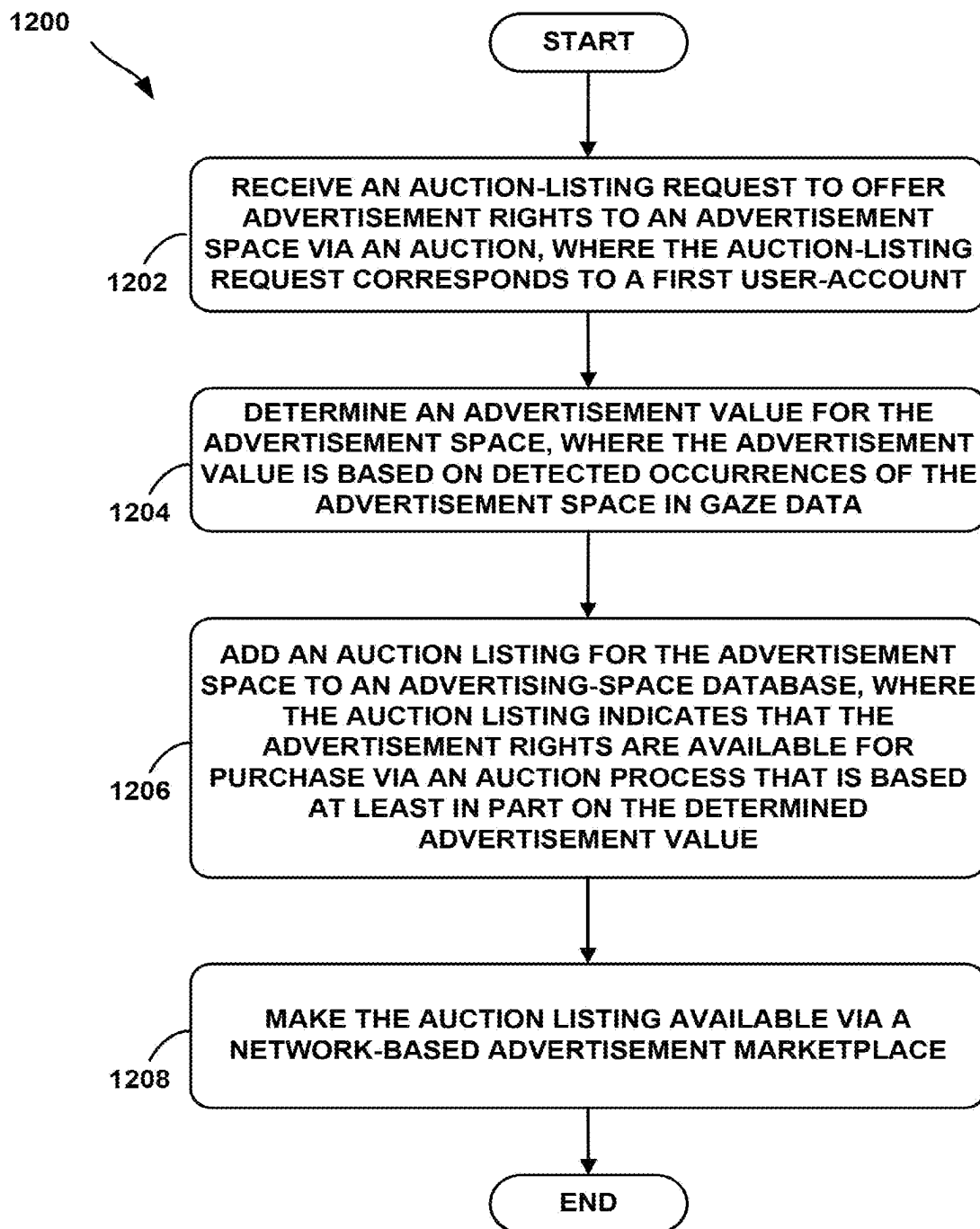
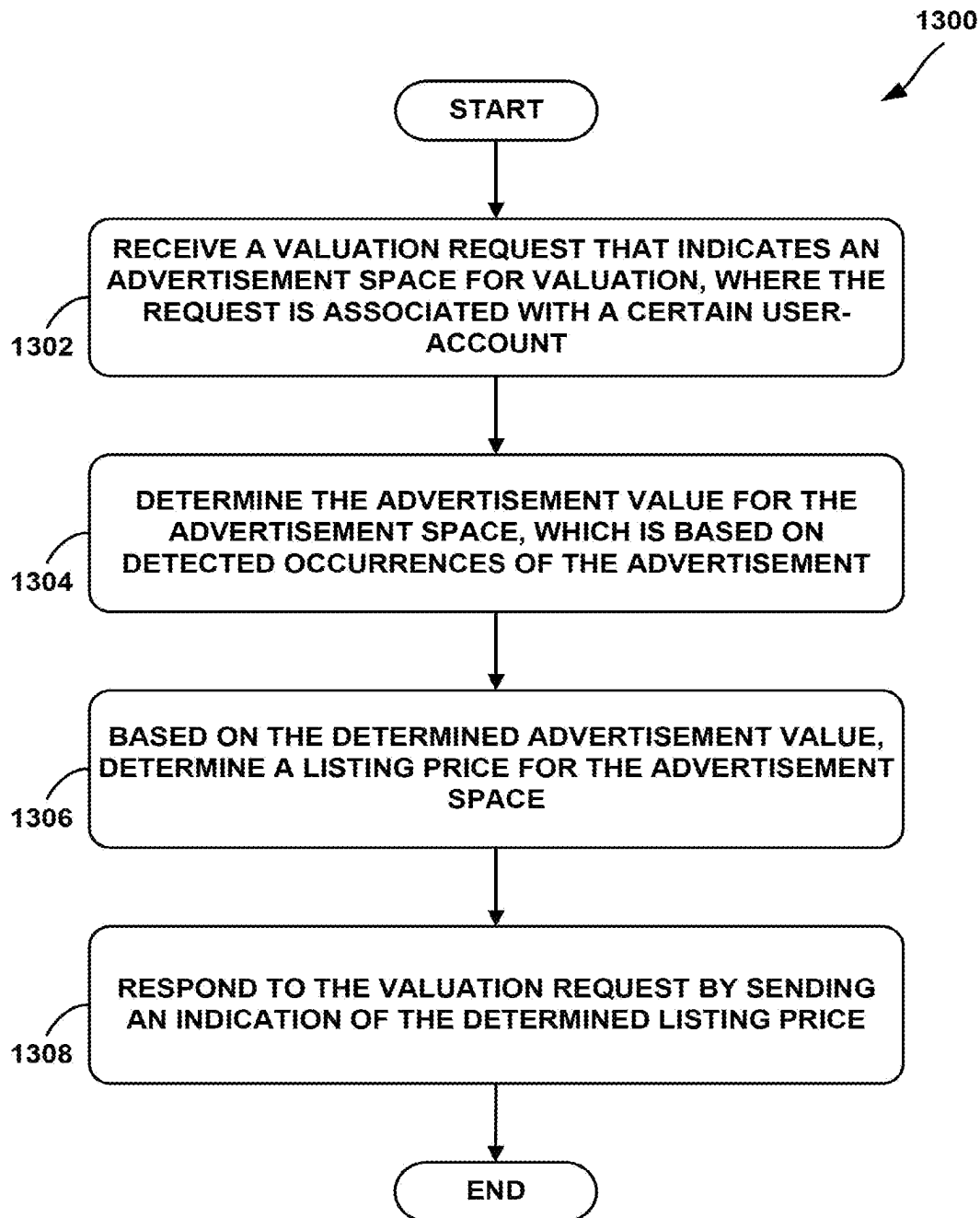
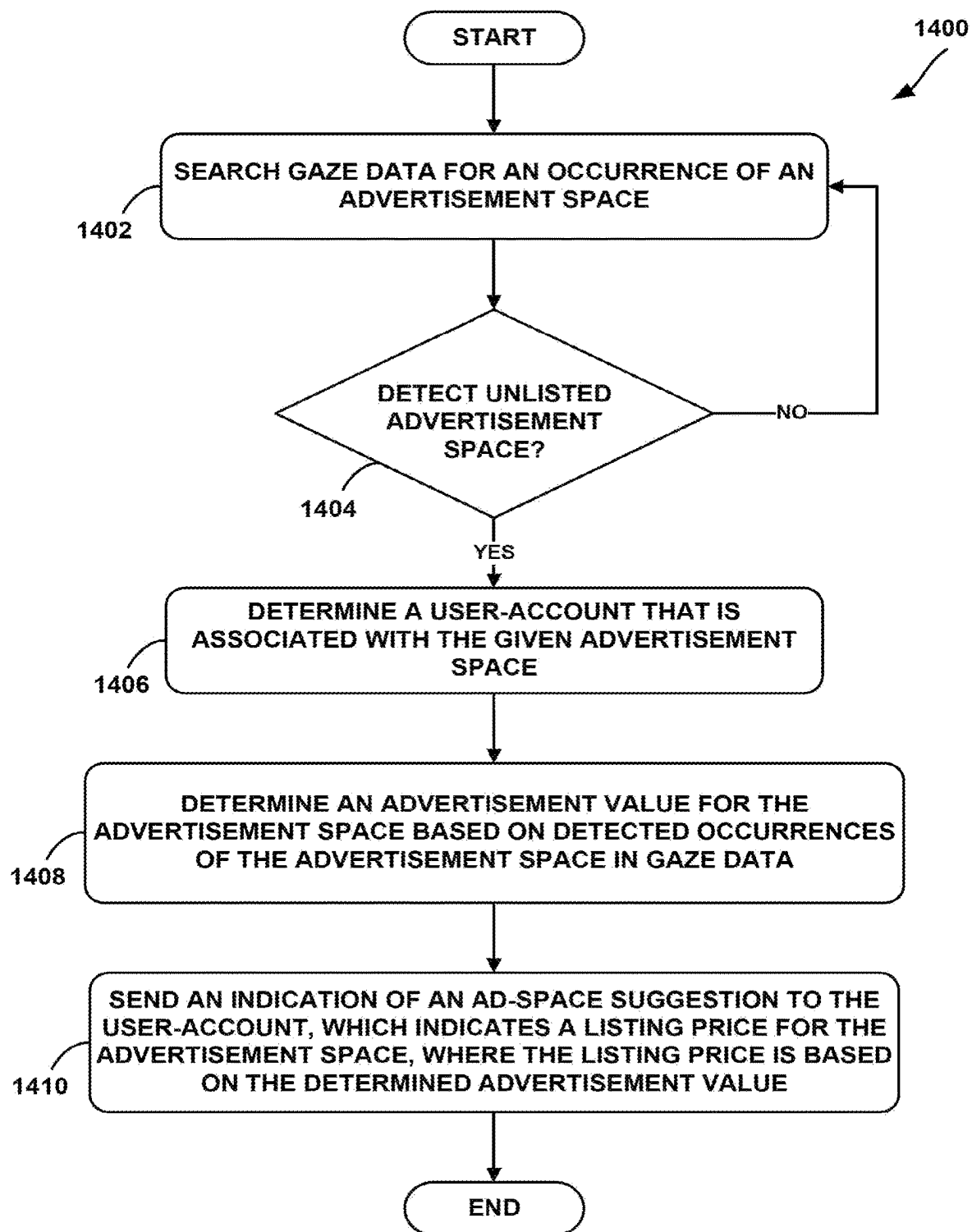
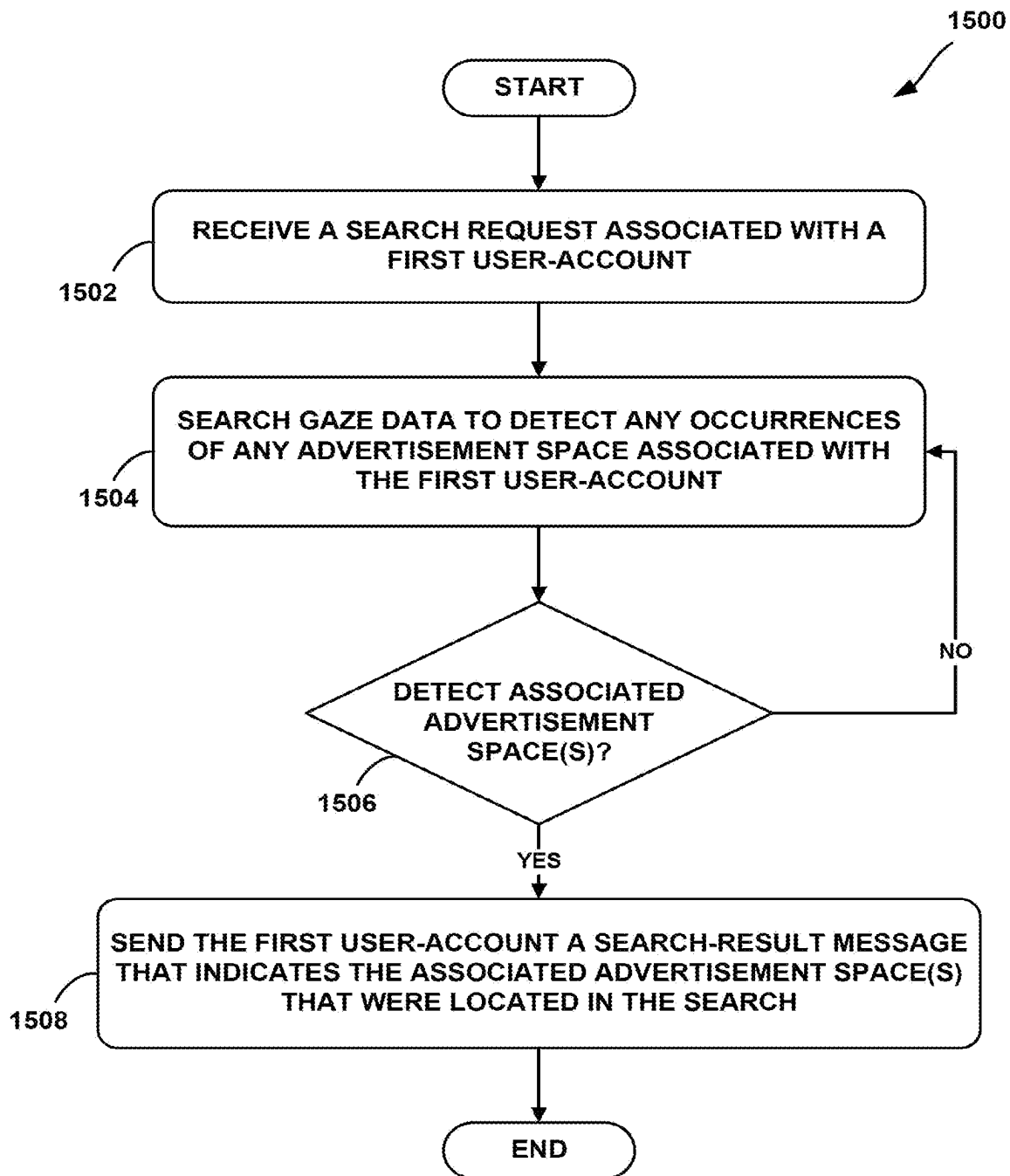
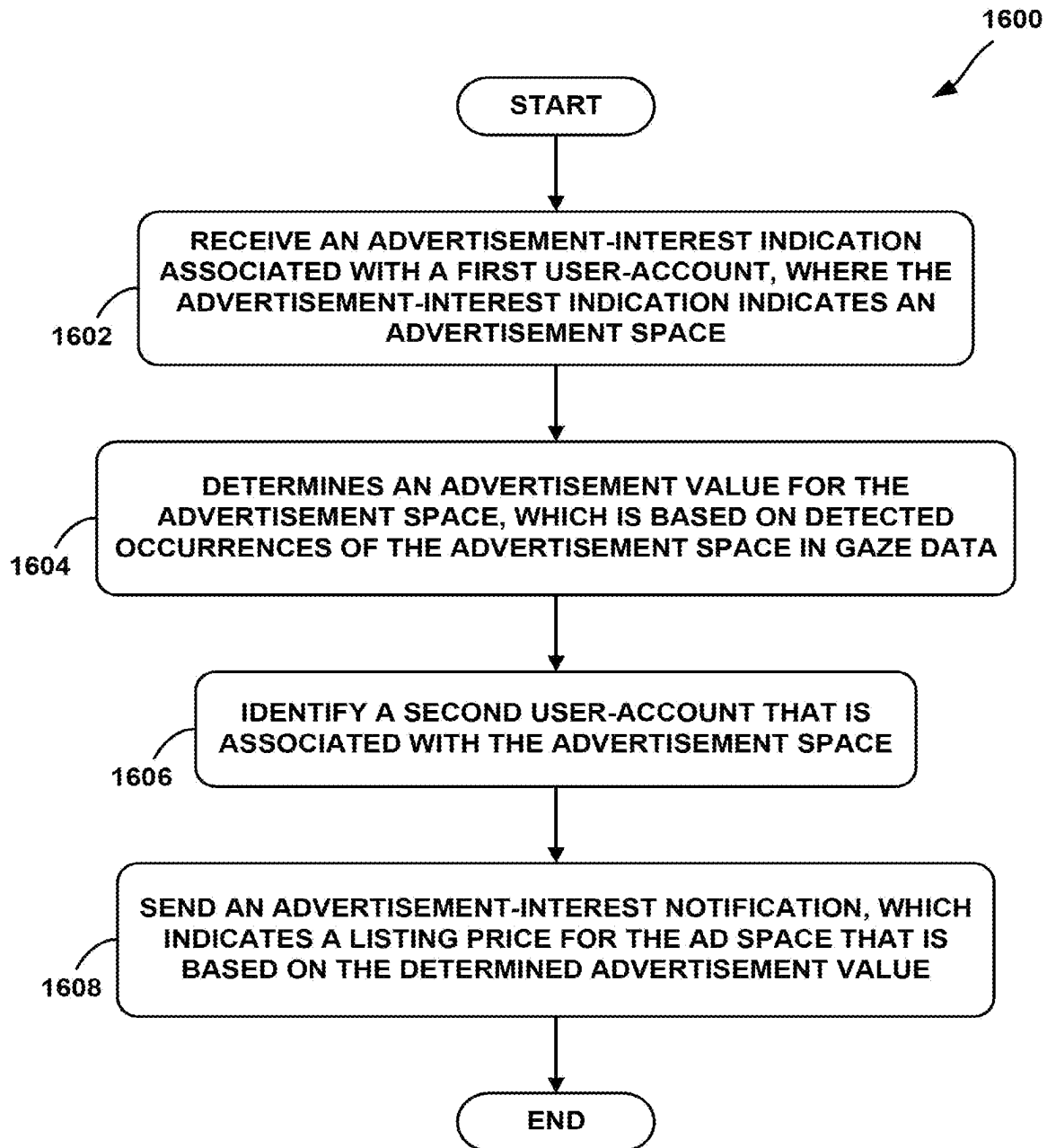


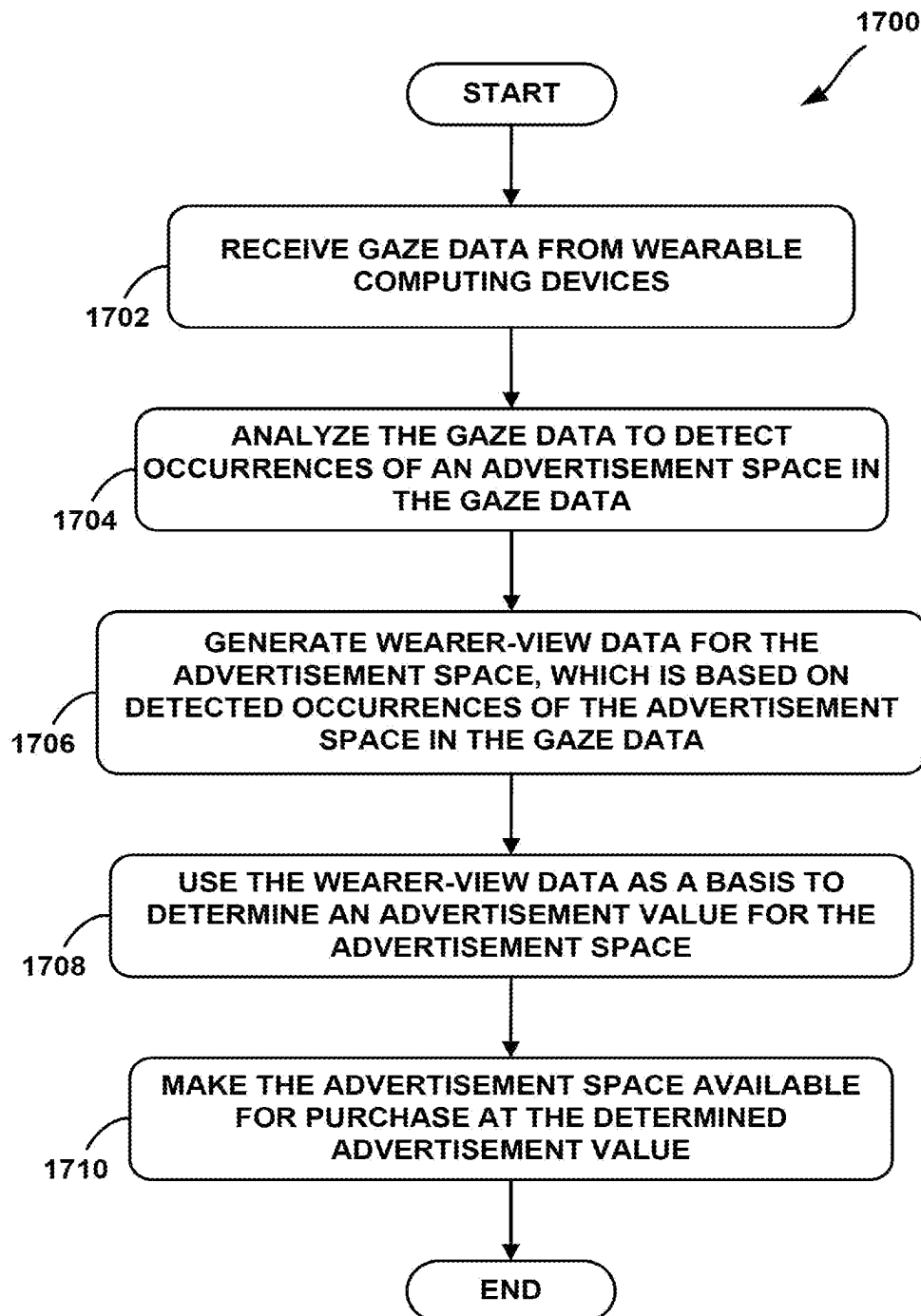
FIG. 12

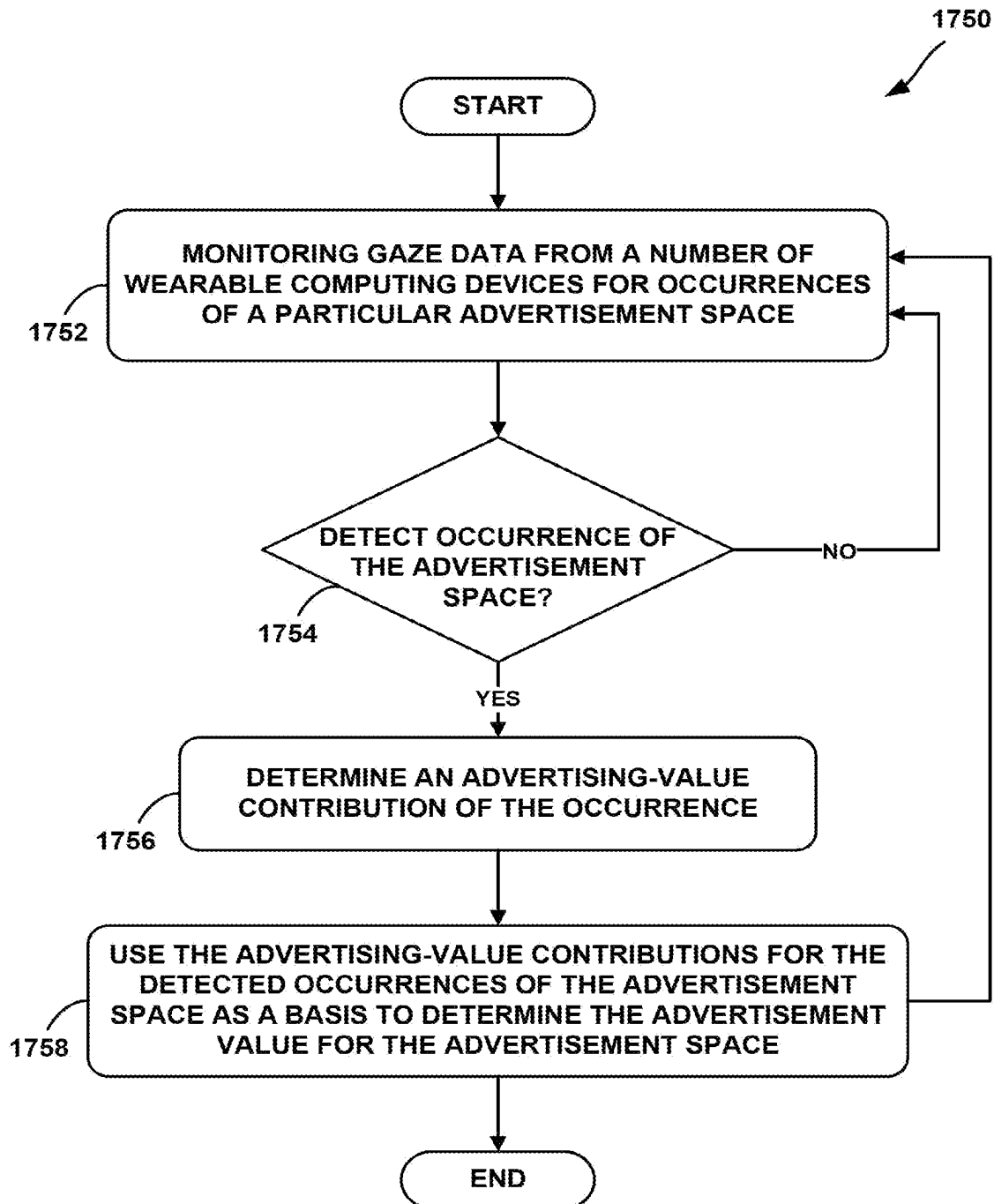
**FIG. 13**

**FIG. 14**

**FIG. 15**

**FIG. 16**

**FIG. 17A**

**FIG. 17B**

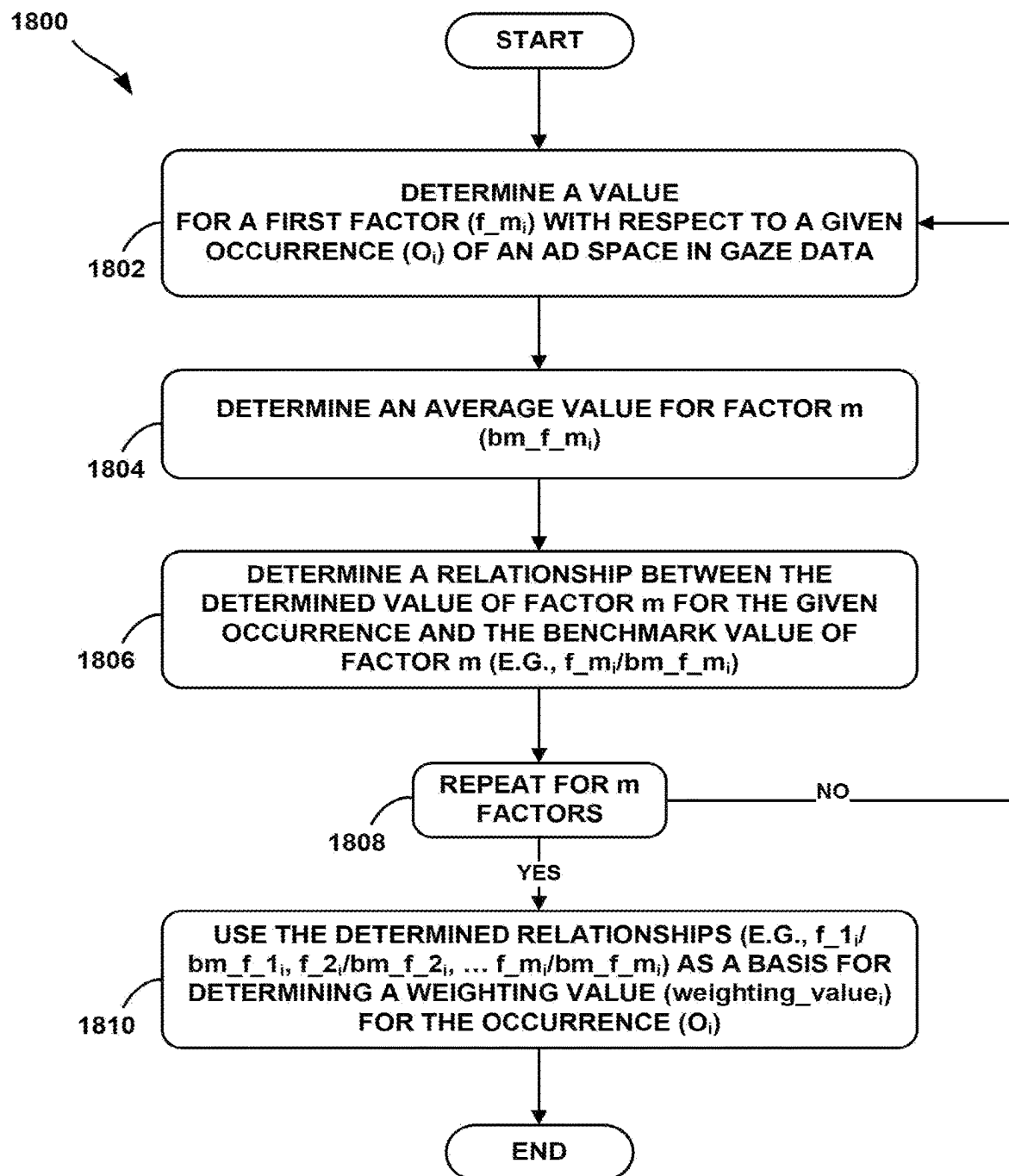


FIG. 18

MEASUREMENT METHOD AND SYSTEM**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 18/152,458, filed Jan. 10, 2023, which is a continuation of U.S. application Ser. No. 16/793,175 filed Feb. 18, 2020, now U.S. Pat. No. 11,579,442, which is a continuation of U.S. application Ser. No. 15/943,346 filed Apr. 2, 2018, now U.S. Pat. No. 10,598,929, which is a continuation-in-part of U.S. application Ser. No. 15/257,732, filed Sep. 6, 2016, now U.S. Pat. No. 9,952,427, which is a continuation of U.S. application Ser. No. 14/524,753, filed Oct. 27, 2014, now U.S. Pat. No. 9,439,563, which is a continuation of U.S. application Ser. No. 13/292,909, filed Nov. 9, 2011, now U.S. Pat. No. 8,879,155. U.S. application Ser. No. 15/943,346 is also a continuation-in-part of U.S. application Ser. No. 15/145,125 filed May 3, 2016, now U.S. Pat. No. 10,354,291, which is a continuation of U.S. application Ser. No. 13/292,898, filed Nov. 9, 2011. The foregoing applications are incorporated herein by reference in their entireties.

BACKGROUND

Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art by inclusion in this section.

Computing devices such as personal computers, laptop computers, tablet computers, cellular phones, and countless types of Internet-capable devices are increasingly prevalent in numerous aspects of modern life. Over time, the manner in which these devices are providing information to users is becoming more intelligent, more efficient, more intuitive, and/or less obtrusive.

The trend toward miniaturization of computing hardware, peripherals, as well as of sensors, detectors, and image and audio processors, among other technologies, has helped open up a field sometimes referred to as “wearable computing.” In the area of image and visual processing and production, in particular, it has become possible to consider wearable displays that place a very small image display element close enough to a wearer’s (or user’s) eye(s) such that the displayed image fills or nearly fills the field of view, and appears as a normal sized image, such as might be displayed on a traditional image display device. The relevant technology may be referred to as “near-eye displays.”

Near-eye displays are fundamental components of wearable displays, also sometimes called “head-mounted displays” (HMDs). A head-mounted display places a graphic display or displays close to one or both eyes of a wearer. To generate the images on a display, a computer processing system may be used. Such displays may occupy a wearer’s entire field of view, or only occupy part of wearer’s field of view. Further, head-mounted displays may be as small as a pair of glasses or as large as a helmet.

Emerging and anticipated uses of wearable displays include applications in which users interact in real time with an augmented or virtual reality. Such applications can be mission-critical or safety-critical, such as in a public safety or aviation setting. The applications can also be recreational, such as interactive gaming.

SUMMARY

In one aspect, an exemplary computer-implemented method may involve: (a) receiving gaze data for a first

wearable computing device, wherein the gaze data is indicative of a wearer-view associated with the first wearable computing device, and wherein the first wearable computing device is associated with a first user-account; (b) analyzing the gaze data from the first wearable computing device to detect one or more occurrences of one or more advertisement spaces in the gaze data; (c) based at least in part on the one or more detected advertisement-space occurrences, determining an individual gaze value for the first user-account; and (d) sending a gaze-value indication, wherein the gaze-value indication indicates the individual gaze value for the first user-account.

In another aspect, an exemplary system may include a non-transitory computer-readable medium and program instructions stored on the non-transitory computer-readable medium. The program instructions may be executable by at least one processor to: (a) receive gaze data for a first wearable computing device, wherein the gaze data is indicative of a wearer-view associated with the first wearable computing device, and wherein the first wearable computing device is associated with a first user-account; (b) analyze the gaze data from the first wearable computing device to detect one or more occurrences of one or more advertisement spaces in the gaze data; (c) based at least in part on the one or more detected advertisement-space occurrences, determine an individual gaze value for the first user-account; and (d) send a gaze-value indication, wherein the gaze-value indication indicates the individual gaze value for the first user-account.

In yet another aspect, an exemplary article of manufacture may include a computer-readable storage medium having instructions stored thereon that, in response to execution by a processor, cause the processor to perform operations. The instructions may include: (a) instructions for receiving gaze data for a first wearable computing device, wherein the gaze data is indicative of a wearer-view associated with the first wearable computing device, and wherein the first wearable computing device is associated with a first user-account; (b) instructions for analyzing the gaze data from the first wearable computing device to detect one or more occurrences of one or more advertisement spaces in the gaze data; (c) instructions for based at least in part on the one or more detected advertisement-space occurrences, determining an individual gaze value for the first user-account; and (d) instructions for sending a gaze-value indication, wherein the gaze-value indication indicates the individual gaze value for the first user-account.

In a further aspect, an exemplary computer-implemented method may involve: (a) receiving gaze data for a first wearable computing device, wherein the gaze data is indicative of a wearer-view associated with the first wearable computing device, and wherein the first wearable computing device is associated with a first user-account; (b) analyzing the gaze data from the first wearable computing device to detect one or more occurrences of one or more advertisements in the gaze data; (c) based at least in part on the one or more detected advertisement occurrences, determining an individual gaze value for the first user-account; and (d) sending a gaze-value indication to the first user-account, wherein the gaze-value indication indicates the individual gaze value for the first user-account.

In yet a further aspect, an exemplary computer-implemented method may involve: (a) receiving, at a wearable computing device, gaze data that is indicative of a wearer-view associated with the wearable computing device; (b) the wearable computing device analyzing the gaze data to detect one or more occurrences of one or more advertisement

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spaces in the gaze data; (c) based at least in part on the one or more detected advertisement-space occurrences, the wearable computing device determining an individual gaze value for a user-account that is associated with the wearable computing device; and (d) the wearable computing device displaying the individual gaze value.

In an additional aspect, an exemplary system may include a non-transitory computer-readable medium and program instructions stored on the non-transitory computer-readable medium. The program instructions may be executable by at least one processor to: (a) receive gaze data that is indicative of a wearer-view associated with the wearable computing device; (b) analyze the gaze data to detect one or more occurrences of one or more advertisement spaces in the gaze data; (c) based at least in part on the one or more detected advertisement-space occurrences, determine an individual gaze value for a user-account that is associated with the wearable computing device; and (d) display the individual gaze value.

These as well as other aspects, advantages, and alternatives, will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flow chart illustrating a method according to an exemplary embodiment.

FIG. 2 is a simplified block diagram illustrating a communication network via which gaze data may be collected, according to an exemplary embodiment.

FIG. 3A is a flow chart illustrating a method for determining gaze value, according to an exemplary embodiment.

FIG. 3B is a flow chart illustrating a method for using multiple factors to determine an individual gaze value for a user-account, according to an exemplary embodiment.

FIG. 4 is a flow chart illustrating a method for determining gaze value, according to an exemplary embodiment.

FIG. 5A is a flow chart illustrating a method for determining advertisement value, according to an exemplary embodiment.

FIG. 5B is a flow chart illustrating a method for determining ad value, according to an exemplary embodiment.

FIG. 6 is a flow chart illustrating a method that may be carried out at a wearable computing device, according to an exemplary embodiment.

FIG. 7A illustrates a wearable computing system according to an exemplary embodiment.

FIG. 7B illustrates an alternate view of the wearable computing device illustrated in FIG. 7A.

FIG. 8A illustrates another wearable computing system according to an exemplary embodiment.

FIG. 8B illustrates another wearable computing system according to an exemplary embodiment.

FIG. 9 illustrates a schematic drawing of a wearable computing device according to an exemplary embodiment.

FIG. 10 is another flow chart illustrating a method according to an exemplary embodiment.

FIG. 11A is a flow chart illustrating a method for updating a variable rate for advertisement rights, according to an exemplary embodiment.

FIG. 11B is a flow chart illustrating another method for updating a variable rate for advertisement rights, according to an exemplary embodiment.

FIG. 12 is a flow chart illustrating an auction process for advertisement rights, according to an exemplary embodiment.

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FIG. 13 is a flow chart illustrating a method for providing an advertisement space valuation in an advertisement marketplace, according to an exemplary embodiment.

FIG. 14 is flow chart illustrating a method for locating potential advertisement spaces, according to an exemplary embodiment.

FIG. 15 is flow chart illustrating a method for providing a search-request feature, according to an exemplary embodiment.

FIG. 16 is flow chart illustrating a method that provides advertiser-request functionality, according to an exemplary embodiment.

FIG. 17A is a flow chart illustrating a method for determining advertisement value according to an exemplary embodiment.

FIG. 17B is a flow chart illustrating another method for determining advertisement value, according to an exemplary embodiment.

FIG. 18 is a flow chart illustrating a method for using multiple factors to determine the ad-value contribution of a given ad-space occurrence in gaze data, according to an exemplary embodiment.

DETAILED DESCRIPTION

Exemplary methods and systems are described herein. It should be understood that the word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or feature described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or features. The exemplary embodiments described herein are not meant to be limiting. It will be readily understood that certain aspects of the disclosed systems and methods can be arranged and combined in a wide variety of different configurations, all of which are contemplated herein.

I. Overview

A. Valuing Ad Space Based on Gaze Data

Many existing methodologies for valuing physical advertising space involve use of different types of data to estimate how many people view an advertisement space (referred to interchangeably as an “ad space”) and/or how effective the advertisement space is at delivering the intended message to the viewer. These methodologies often rely on demographic information and other such indirect measurements of the potential audience for an ad space. Since these methodologies only estimate how many people actually view an ad space and/or who the people are that actually view the ad space, and do not incorporate actual viewership data, the results are often inaccurate.

Some existing valuation techniques do incorporate actual viewership data, which is typically collected for a test group and then extrapolated to the population as a whole (or to a larger group, such as a target market). However, gathering such viewership data with existing methodologies can often be time-consuming and difficult. For example, such techniques often involve polling people in a test group individually or laboriously observing how many people actually view an ad space (e.g., counting vehicles that pass by a billboard). Because of the effort required, advertising is typically limited to certain defined types of spaces (e.g., billboards, television commercials, websites, etc.) for which representative viewership data can be most-readily obtained.

Since most any physical space that is seen by people has some value for purposes of advertising, current valuation

techniques do not allow for capitalization of many would-be advertising spaces. While the individual values of such spaces may be small, the cumulative value of all such spaces may be significant. However, due to the limitations of current advertisement valuation and marketing techniques, much of the potential value of such spaces has not been monetized.

Accordingly, exemplary methods and systems may help to determine the value of advertising spaces in a manner that may be more accurate, and may require less data-collection effort than existing advertisement-valuation techniques. In particular, exemplary methods may utilize “gaze data,” from a number of wearable computers, which is indicative of what the wearers of the wearable computers are actually viewing, in order to value physical spaces.

For example, point-of-view (POV) videos from a number of wearable computers may be analyzed in order to determine how frequently a certain space is captured in the POV videos. Notably, POV video from wearable computers may provide a fairly accurate indication of what a user is actually looking at. Therefore, aggregating such POV videos from a number of users may help to more accurately value advertising rights to physical spaces. Additionally, the wearers of the wearable computing devices may elect to make their respective user-profiles available, such that individual characteristics of each wearer who views an advertisement space may be considered. This information may then be used to determine the value of the physical space.

Furthermore, a cloud-based server system may aggregate gaze data from many wearable computers, and use the gaze data to determine wearer-view data for various advertisement spaces. As such, an exemplary embodiment may provide advertisement valuation that is carried out automatically, without the effort required for manual data collection, and is more accurate, as the valuation is based on data that more-accurately captures what people are actually viewing.

B. Gaze Valuation for Individual Users

In a further aspect, an exemplary method may be implemented to determine what an individual user’s gaze is worth. In particular, a user who is wearing a wearable computer with a head-mounted display (HMD) may send POV video from a camera attached to their HMD to an advertisement server system. Further, the user may opt in to a program where the POV video (and possibly other forms of gaze data) can be used for gaze valuation and/or to value ad spaces. Accordingly, the server system may analyze the gaze data to detect when the user views ad spaces. The individual gaze value for the user can then be determined based on the ad spaces that the user has viewed.

Further, in many instances, additional information, such as consumer data, demographic information, income, job title, hobbies, and/or interests, among others, may also be considered when determining a gaze value. For example, consider two users who view the exact same advertisements. In this scenario, the gaze value for one of these people might still be higher than for the other if, for example, one person has a significantly higher income than the other. Other examples are also possible.

Yet further, the historical efficacy of advertisements may be considered when determining a gaze value. For example, consider again two people who view the exact same advertisements. If, in the past, one user seems to have been influenced more by advertisements (e.g., as indicated by a pattern of purchasing products subsequent to viewing advertisements for the products), then this user’s gaze value may be higher. Other examples are also possible.

To obtain their gaze value, a user may create a user-account and register a wearable computing device (and possibly other devices) to their user-account. As such, when the server receives gaze data from a given device, the server may associate the gaze data with the user-account to which the given device is registered. Then, each time the server detects an advertisement space in gaze data associated with a given user-account, the server may determine what the occurrence of the ad space is worth (e.g., what having the user view the ad space is worth to an advertiser). This value may be referred to as the “gaze-value contribution” for the occurrence of the ad space in the gaze data. As such, the server may determine the user’s individual gaze value based on gaze-value contributions from a number of ad spaces that occur in the user’s gaze data.

As a specific example, the individual gaze value may be calculated as a dollar amount per day. As such, the server may monitor a given user’s gaze data for occurrences of ad spaces, determine a gaze-value contribution for each occurrence of an ad space that is detected during a one-day period, and then determine an individual gaze value by summing gaze-value contributions from the one-day period. Alternatively, the server may sum the gaze-value contributions for ad-space occurrences on a daily basis, and determine the individual gaze value by averaging the daily total over a number of days. Other variations are of course possible.

Providing users with individual gaze values may be useful in various scenarios. In one aspect, the ability to learn an individual gaze value may be used as an incentive for a user to opt in to a program where the user provides and authorizes use of their gaze data. In particular, when gaze data is used to value advertisement spaces, increasing the number of wearable computing devices from which gaze data is available will typically increase the number of ad spaces that can be valued and/or improve how accurately these ad spaces are valued. Accordingly, a user may be provided with access to individual gaze valuation functions only after the user has created a user-account, agreed to provide gaze data from their wearable computing device (and possibly from other devices that the user may optionally associate with their user account), and authorized their gaze data to be used for purposes of advertisement valuation. Since knowing what their individual gaze is worth may be interesting to many users, access to this functionality may be an incentive for such users to provide and authorize use of their gaze data.

Furthermore, in some embodiments, individual gaze value may be more than just a metric that interests users. Rather, individual gaze value functionality may be utilized to determine payments that are actually paid out users who provide gaze data. More specifically, since gaze data is typically indicative of what a user is actually viewing, an ad space marketplace may be established where advertisers pay for rights to ad spaces that are valued using gaze data, and a portion of the money paid by the advertisers is distributed to the users who provide the gaze data. In particular, when an occurrence of an ad space is detected in gaze data for a given user-account, the server system may update wearer-view data used for valuation of the ad space, and may also determine a value to the advertiser of the particular user viewing the ad space (e.g., a gaze-value contribution for the occurrence of the ad space in the user’s gaze data). A portion of this value may then be credited to the user-account and/or paid out to the user.

As an example of one such application, an ad marketplace may be set up to pay users who provide gaze data a 5% commission on ad spaces they view. As such, users who allow their gaze data to be used for ad valuation may be paid

5% of their individual gaze value in exchange for use of their gaze data. Other examples are also possible.

Note that herein, when gaze data is said to be associated with a given user-account, it should generally be understood that this gaze data was sent by a device that is associated with the given user-account (e.g., a device that is registered with the user-account). Further, gaze data and/or other information that is associated with a user-account may also be said to be associated with a user since, functionally, associating gaze data or any other data with a user will generally be accomplished by associating the data with the user's user account.

In a further aspect, when a user creates a user-account for which a gaze value may be determined, a user-profile for the user-account may be created as well. The user-profile may include or provide access to various types of information, from various sources, which is related to the user. For simplicity, examples set forth herein may simply refer to a user-account as including the information included in the associated user-profile. However, this should not be read as requiring that a user-account include a user-profile. It is possible, in some embodiments, that a user-account may not have an associated user-profile.

C. Display on Non-Traditional Surfaces

In a further aspect, exemplary methods and systems may automatically collect wearer-view data for a large number of advertisement spaces. This may allow for valuation and monetization of many physical spaces that have value for purposes of advertising, and might not otherwise be capitalized.

For example, the shortcomings of existing advertisement valuation are especially problematic when it comes to individuals who wish to associate themselves with a brand or product (e.g., individuals who wish to be sponsored). Because of the cost and/or effort involved in using common valuation techniques to gather viewership data for the average person, sponsorship is generally limited to high-visibility individuals (e.g., famous athletes and musicians). Such high-visibility individuals are typically the most valuable to an advertiser, such that the cost of evaluating the sponsorship is justified by the value to the advertiser. For instance, a famous race-car driver can readily sell advertising space on his or her clothing or on their car. However, there is no market for the average person to do the same (in fact, individuals often pay more to have a brand logo on their clothing). Typically, however, there is still value to the average person promoting a product in this manner. Advantageously, by economically and accurately determining the advertisement value of physical spaces associated with the average person, an exemplary embodiment may help open up sponsorship opportunities for almost anyone.

However, it should be understood that exemplary embodiments may generally be used to value almost any type of physical space. To provide just a few examples, advertisement values may be determined for: (a) the back of the screen of a laptop computer, (b) an article of clothing, (c) a billboard, (d) a surface on an automobile, (c) a body surface, (f) a space on a webpage, (g) a print advertisement space, (h) a surface on product packaging, and/or (i) a pet. Many other types of advertisement spaces may also be valued utilizing an exemplary embodiment.

Further, an exemplary methods and systems may be used to value physical spaces types for various different advertisement formats such as: (a) print advertisements, (b) computer-generated images, (c) video, (d) peel and stick paper advertisements, (c) two-dimensional projections, (f) a three-

dimensional projections, (g) iron-on images, (h) temporary tattoos, and/or (i) other advertising formats.

D. Advertisement Marketplace

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B. Advertisement Marketplace

Provided with the ability to determine an advertising value for almost any physical space, there is a need for a marketplace system to facilitate transactions involving use of such physical spaces as advertisement spaces. Accordingly, exemplary methods and systems may help to provide an advertisement marketplace.

In an exemplary marketplace, gaze-data-based valuation may be used to price advertising rights. For instance, in some applications, advertisement rights to an advertisement space may be listed at the advertisement value, or at a fixed rate that is based on the advertisement value. In other instances, advertisement rights may be listed at a variable rate. For example, an initial rate may be based on the advertisement value at or near the time of purchase. The system may then adjust this rate based on views of the advertisement during the advertising period (e.g., based on how often the advertisement space is detected in subsequent gaze data). Other gaze-data-based pricing structures for advertisement spaces are also possible.

Note that in an exemplary embodiment, the advertisement value upon which the listing price is based may be a "relative advertisement value." More specifically, the relative advertisement value may be determined pre-sale, and thus may differ from the price that the advertisement space ultimately sells for in an open market. For example, a relative advertisement value may be determined for an advertisement space before any advertisement is placed, based on gaze data in which the bare advertisement space is detected. As such, the selling price may ultimately differ based on what the market is willing to pay for the advertisement space. Thus, in some embodiments the true or

official advertisement value may be considered to be the market price, and thus may differ from the relative advertisement value.

In some embodiments, an exemplary marketplace may include various features to assist a user who wishes to list an advertisement space for sale in the marketplace. For instance, an exemplary system may support valuation requests, which allows a user to request and be provided with gaze-data-based advertisement values for the user's advertisement spaces. Such valuation requests may help users evaluate whether or not to list an advertisement space in the marketplace. More specifically, once provided with the advertisement value for their advertisement space, a user may then elect to list the advertisement space at a listing price that is based on this advertisement value.

Further, an exemplary system may allow a user to sell advertising rights for a wide variety of physical spaces. For instance, an exemplary system may have pre-defined types of physical spaces that can be valued as an advertisement space and listed for sale (e.g., higher-visibility types of physical spaces such as a billboard, the back of a laptop computer, the front of a shirt, etc.). As such, an exemplary system may automatically search gaze data for the pre-defined types of physical spaces, and in so doing may identify potential advertisement spaces. Once a physical space has been identified as a potential advertisement space, the system may allow a user to sell advertisement rights to the advertisement space via the advertisement marketplace.

An exemplary advertisement marketplace may additionally or alternatively allow a user to dynamically define a physical space as an advertisement space, even if it is not pre-defined as such. For instance, the system may provide features via which a user can submit images, video, and/or other information that allows the system to detect when an advertisement space occurs in gaze data. After receiving such information, the system can search for the user-defined advertisement space in gaze data, so that the advertisement space can be valued and/or listed in the advertisement marketplace.

In some embodiments, an exemplary system may provide a user who wishes to list advertisement spaces with suggestions of advertisement spaces that can be listed by the user. In particular, the marketplace system may search gaze data (and possibly other data sources as well) for physical spaces that are associated with the given user and that are usable as advertisement spaces. The system may do so automatically or on request by a given user. In either case, the user may then be provided with suggestions of unlisted advertisement spaces (possibly including valuations of each advertisement space), which the user can list in the advertisement marketplace. This may be particularly useful in the scenario where a user is unaware that a certain physical space is usable as an advertisement space. In this scenario, the suggestions may inform a user of the existence of advertisement spaces that the user was previously unaware of. Of course, ad-space suggestions may be useful in other scenarios as well.

An exemplary system may also include various features to assist an advertiser who wishes to purchase an advertisement space. For instance, a marketplace system may allow an advertiser to search, browse, and/or purchase advertisement spaces that are listed in the marketplace.

Furthermore, an exemplary system may allow an advertiser to identify or define an advertisement space for which they are interested in purchasing advertisement rights. In this regard, the advertiser may be provided with similar features as those provided to a user wishing to list an advertisement space, which allow the advertiser to dynami-

cally define a physical space as an advertisement space, and/or to identify a pre-defined type of advertisement space. Once an advertisement space has been identified, the marketplace system may identify a user that is authorized to list the advertisement space, and notify the user that there is interest in their advertisement space.

In a further aspect, an exemplary marketplace may include features to facilitate a transaction to purchase advertisement rights between the user who listed the advertisement space and the advertiser who is purchasing the advertisement space. The advertisement marketplace may also facilitate completion of the contract created between the seller and the purchaser after such a transaction. In particular, when an advertiser purchases an advertisement space, an exemplary marketplace system may create and maintain a record of a contract between the advertiser and user who sold the advertisement. The system may also provide features to facilitate performance of the contract. For instance, a server may search gaze data received post-contract to determine that the advertisement specified by the advertiser is being displayed in the advertisement space in accordance with the contract. Further, the system may facilitate billing the advertiser, and possibly even transferring funds between user-accounts for the advertiser and the seller.

Note that herein, when gaze data is said to be associated with a given user-account, it should generally be understood that this gaze data was sent by a device that is associated with the given user-account (e.g., a device that is registered with the user-account). Further, gaze data and/or other information that is associated with a user-account may also be said to be associated with a user since, functionally, associating gaze data or any other data with a user will generally be accomplished by associating the data with the user's user account.

In a further aspect, when a user creates a user-account for which a gaze value may be determined, a user-profile for the user-account may be created as well. The user-profile may include or provide access to various types of information, from various sources, which is related to the user. For simplicity, examples set forth herein may simply refer to a user-account as including the information included in the associated user-profile. However, this should not be read as requiring that a user-account include a user-profile. It is possible, in some embodiments, that a user-account may not have an associated user-profile. Furthermore, herein, the term user-profile may more generally be understood to refer to any information or collection of information related to a given user. As such, a user-profile may be specifically created for a user-account or may simply take the form of data that is associated with a given user.

II. Exemplary Methods

FIG. 1 is a flow chart illustrating a method according to an exemplary embodiment. The method 100 shown in FIG. 1 may be implemented by a computing device, and in particular, by a server system, in order to determine a gaze value for a wearable computing device and/or for a user-profile associated with the wearable computing device. According to an exemplary embodiment, the gaze value is based on point-of-view gaze data received from a wearable computing device (which may be referred to interchangeably as a wearable computing device or a wearable computer). Further, a server system that implements an exemplary method may be referred to as a gaze-valuation system, as a gaze-valuation server, as an ad-valuation server, or simply as a server system or server.

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As shown by block **102**, method **100** involves a gaze-valuation server receiving gaze data for a first wearable computing device, which is associated with a first user-account. The server may analyze the gaze data from the first wearable computing device to detect occurrences of advertisement spaces in the gaze data, as shown by block **104**. Then, based at least in part on the detected ad-space occurrences, the server may determine an individual gaze value for the first user-account, as shown by block **106**. The server may then send a gaze-value indication, which indicates the individual gaze value, to the first user-account, as shown by block **108**.

In an exemplary method **100**, the gaze data received for a given wearable computing device is generally indicative of the wearer-view associated with the wearable computing device. For example, server may receive gaze data from a wearable computing device in the form of point-of-view (POV) video that is captured at the wearable computing device. As such, the POV video may be monitored in order to detect when advertisement spaces occur in the video.

In an exemplary method, such as method **100**, gaze data may additionally or alternatively take forms other than point-of-view video. For example, the gaze data may take the form of point-of-view images captured by a forward- or outward-facing camera on a wearable computing device or another device. As a specific example, a given wearable computing device may periodically take a picture using a camera on an HMD that is generally aligned with the wearer's field of view. The wearable computing device may send these periodically-captured pictures to the server system for use in an exemplary method. Other examples are also possible.

Since the gaze data from a given wearable computing device is generally indicative of the wearer-view of the wearable computing device's wearer, the gaze data may be interpreted to be generally indicative of what the wearer of the device is actually looking at. For instance, the gaze data may be analyzed to determine information such as when a wearer is looking at a particular ad space and/or how long the wearer was looking at the particular ad space, among other information. Accordingly, the gaze data may be used to determine a gaze value for the wearer.

In an exemplary embodiment, the individual gaze value for a user (which may also be referred to as the gaze value) is indicative of the value of the user's gaze to advertisers. More specifically, the value of the user's gaze may represent the cumulative value to advertisers of the user viewing advertisements over time. As such, determining the individual gaze value may involve determining what each view is worth to an advertiser (e.g., what each occurrence of an ad space in gaze data associated with the given user is worth) and calculating a total value for all of the user's views.

Exemplary systems will now be described before further details of exemplary methods are set forth.

III. Exemplary Server Systems

FIG. **2** is a simplified block diagram illustrating a communication network via which gaze data may be received, according to an exemplary embodiment. As shown, communication network **200** includes a number of wearable computing devices **202A** to **202D**, which are configured to communicate with a server system **204** via one or more networks **206**. An exemplary network, such as communication network **200**, may also include computing devices other than wearable computing devices, such as laptop computer **203** and mobile phone **205**, for instance. As such, an

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exemplary advertisement marketplace may be implemented in a network such as communication network **200**.

In order to facilitate an exemplary method, the users of wearable computing devices **202A** to **202D** may register their respective devices and opt in to programs via which the users submit gaze data from their respective devices. As such, wearable computing devices **202A** to **202D** may send gaze data to the server system **204**, which the server system **204** may then analyze to help determine advertisement values for advertisement spaces, possibly to value advertisement spaces as well. Further, in some embodiments, laptop **203**, mobile phone **205**, and/or other computing devices may provide supplemental gaze data, which may be used by server system **204** to supplement the gaze data from wearable computing devices **202A** to **202D**.

In an exemplary embodiment, the server system **204** may be a computing system including one or more computing devices. In particular, server system **204** may be a cloud-based server system that is configured to receive gaze data, and to determine a gaze value for at least one of wearable computing devices **202A** to **202D**. In a further aspect, the server system **204** may also be configured to utilize the gaze data to value advertising spaces and/or support an advertisement-space marketplace for advertising spaces.

As noted, the gaze data in an exemplary embodiment may include point-of-view videos captured by a number of wearable computing devices. For example, some or all of the wearable computing devices **202A** to **202D** may include or take the form of glasses-style HMDs that each include a forward-facing video camera for taking point-of-view video (e.g., video that generally captures the perspective of a person wearing the HMD). As such, when the HMD is worn, the forward-facing camera will capture video and/or images that are generally indicative of what the wearer of the HMD sees. Note that exemplary glasses-style HMDs will be described in greater detail with reference to FIGS. **7A**, **7B**, **8A**, **8B**, and **9**.

Further, server system **204** may include or be in communication with an ad-valuation server **212** and an ad-marketplace server **214**. In some embodiments, ad-valuation server **212** and ad-marketplace server **214** may be separate server systems, which each include one or more computing devices. In other embodiments, some or all of the functionality attributed to ad-valuation server **212** and ad-marketplace server **214** may be provided by a single server system, which may include one or more computing devices.

In an exemplary embodiment, ad-valuation server **212** may be configured to receive gaze data from wearable computing devices **202A** to **202D**. Further, ad-valuation server **212** may analyze the received gaze data for occurrences of one or more ad spaces, and generate wearer-view data for the ad spaces based on occurrences of the ad spaces in the gaze data.

In a further aspect, the server system **204** may include or have access to a wearer-view database **208** that includes wearer-view data for a number of advertisement spaces (e.g., ad spaces indicated by ad-space database **210**). When ad-valuation server **212** generates wearer-view data, ad-valuation server **212** may store the generated data in wearer-view database **208**. Accordingly, server system **204** may access the wearer-view database **208** to retrieve wearer-view data for a given ad space, which in turn may be used to determine the ad value for the given ad space.

To assist the server in detecting occurrences of various ad spaces in gaze data, advertisement server **204** may include or have access to an ad-space database **210** that includes information that can be used to identify various ad spaces.

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Accordingly, ad server system **204** may use the identifying information from ad space database **210** to determine when ad spaces occur in gaze data from wearable computing devices **202A** to **202D**. Further, in embodiments that utilize location data for ad spaces, ad-space database **210** may also store location information for individual ad spaces.

In another aspect, ad-valuation server **212** and/or other components of system **204** may additionally or alternatively be configured to use gaze data to determine an individual gaze value for a given user-account. As such, ad-valuation server **212** and/or other components of system **204** may include program instructions stored in a tangible computer-readable medium that are executable to provide the functionality described herein, and possibly to provide other functionality as well.

In another aspect, server system **204** may include ad-marketplace server **214**, which is configured to provide an advertisement marketplace via which advertisement spaces that are valued by ad-valuation server **212** can be bought and sold. Further, ad-marketplace server **214** may facilitate transactions between parties in such an advertisement marketplace. As such, computing devices such as wearable computing devices **202A** to **202D**, laptop computer **203**, and/or mobile phone **205**, may be provided with advertisement marketplace functions via the server system **204**, and in particular, via ad-marketplace server **214**.

Data related to advertisement space in the advertisement marketplace may be stored in ad-space database **210**. For instance, ad-marketplace server **214** and/or other components of server system **204** may support an advertisement marketplace including features for listing advertisement rights to an advertisement space for sale, a bidding system for physical spaces (e.g., auction functionality), organization and indexing of available physical spaces, searching and/or browsing listings for advertisement spaces, tracking usage of advertisement spaces, calculation of fees and other billing functions, and/or portfolio management for sellers and purchasers of advertisement space, among other features.

Further, server system **204** may provide access to an advertisement marketplace via a website, via a standalone application, and/or via other points of access. Data related to listings and/or transactions in the advertisement marketplace may then be stored in ad-space database **210**.

IV. Detecting Advertisement Spaces in Gaze Data

As noted above, an exemplary method **100** may involve analysis of gaze data to detect when advertisement spaces occurs in the gaze data. To do so, an exemplary server system **200** may employ various types of video and/or image-processing techniques. For instance, advertisement server system **204** may implement various well known and yet-to-be-developed techniques for object recognition in video and/or still images in the process of recognizing advertising spaces.

In some cases, an ad space may be identified in gaze data by way of the advertisement that is displayed in the ad space. For example, ad-space database **210** may include data related to which specific advertisements are being displayed in which ad spaces (and may further indicate when there is no advertisement being displayed in a given ad space). As such, ad-valuation server **212** may search for advertisements that are currently being displayed in gaze data. To do so, the ad-valuation server may use various visual search techniques that are now known or yet to be developed in order to identify an advertisement in gaze data.

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In other cases, an ad space may itself be identified, without necessarily relying on the particular ad that is being displayed in the ad space. (Note that this functionality may be particularly useful in cases where an ad space is empty.)

In such an embodiment, detecting that an advertisement space occurs in gaze data may involve recognizing when the gaze data includes an object or a certain combination of objects that are associated with a particular advertisement space. For example, to recognize advertisement space on the bumper of a particular car, gaze data may be analyzed for an object shaped and/or having coloration that is characteristic of a car bumper. Further, the gaze data may be analyzed for an object having such a shape and/or coloration in conjunction with a license plate having a certain license plate number. In such an embodiment, the server system may consider an occurrence of a bumper in combination with the license plate number for the specific car to be an occurrence of the ad space on the car's bumper. Many other examples are also possible.

In some cases, searching gaze data from a large number of wearable computing devices for a large number of ad spaces may be data intensive. Accordingly, an exemplary server and/or wearable computing devices may implement pre-processing techniques to tag and ID certain types of objects or certain types of information in gaze data, which may help to speed up the process of detecting ad spaces. In some instances, wearable computing devices and/or the server may also store gaze data for processing when, e.g., a wearable computing device is offline, or when the amount of real-time data being collected is generally less. For example, a server may use certain processing resources to receive incoming gaze data during the day, when more gaze data may be received, and then re-assign these processing resources to analyze stored gaze data for ad spaces at night, when less new gaze data may be received.

In some embodiments, a server system may utilize location data to detect an occurrence of an ad space in gaze data. For example, a server system may determine or be provided with the geographic location of a particular ad space (e.g., the GPS coordinates of the ad space). Then, when the ad space is detected in gaze data from a particular wearable computing device, the server may determine the location of the wearable computing device. If this wearable computing device is located such that the ad space could be visible to the wearer of the wearable computing device (e.g., within a predetermined distance from the location of the ad space), then the server system may consider this an occurrence of the ad space. However, if the wearable computing device that provided the gaze data is located such that the ad space could not be viewed by the wearer (e.g., not within a predetermined distance from the location of the ad space), then the server system may not consider this an occurrence of the ad space.

As another example, a server system may use the geographic location of a particular ad space to limit the gaze data that is monitored for the ad space. For instance, the server may determine the locations of wearable computing devices from which gaze data is received. As such, the server may only monitor gaze data that is received from wearable computing devices that are located within a predetermined distance from the ad space. Other methods that utilize the location of an ad space when detecting occurrences of the ad space in gaze data are also possible.

In some embodiments, radio frequency identification (RFID) may be used to help detect occurrences of an ad space in gaze data. In particular, an ad space may be associated with a certain RFID tag, and wearable computing

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devices may be configured with RFID readers. As such, when a wearable computing device detects an RFID tag from an ad space, the wearable computing device may relay this to the server system. For instance, when the wearable computing device detects an RFID that is associated with an ad space, it may insert metadata into the gaze data which indicates the RFID tag and the time at which the RFID tag was detected. Alternatively, the wearable computing device may send a separate message indicating that the RFID tag was detected at a particular time. In either case, the server system can then search for the associated ad space in gaze data that is received from the wearable computing device at or near the time when the RFID tag is detected. This may help the server system to more efficiently detect occurrences of ad spaces, as the timing with which the RFID tags are detected may indicate, for example, times in corresponding point-of-view video where the ad space is likely to occur. Further, various types of RFID may be utilized, such as near-field communications (NFC) and/or other types of RFID, depending upon the implementation.

In some embodiments, barcodes may be used to help detect occurrences of an ad space in gaze data. For instance, a barcode that identifies an ad space may be displayed within or near to an ad space. The server system may then search for barcodes within gaze data. When a barcode associated with a particular ad space is detected, the server may consider this to be an occurrence of the ad space, or may treat this as a factor that, along with other factors, can indicate that there is an occurrence of the ad space in the gaze data. Various types of barcodes, such as high capacity color barcodes (HCCBs) and/or quick response (QR) codes may be utilized in such an embodiment. Other types of barcodes are possible as well.

It should be understood that the above techniques for detecting occurrences of ad spaces are not intended to be limiting. Other techniques are also possible.

V. Determining a Gaze Value

As noted in reference to block 106 of FIG. 1, an exemplary method 100 may involve a wearable computing device and/or a server system determining an individual gaze value for a given user-account, based on advertisements that are detected from a device or devices associated with the user-account. The gaze value may be determined using a number of different techniques, which may vary from implementation to implementation.

In a basic embodiment, the server system may value each view of an advertisement space equally, without regard to who is viewing the ad space, how long they view it, what (if any) is being displayed in the ad space, or the characteristics of the ad space itself. In this scenario, each occurrence of an ad space may be valued equally. Therefore, the server system may determine the gaze value for a given user-account by determining the total number of ad-space occurrences during the period over which a user's gaze value is being calculated, and multiplying the total number by a universally-defined value for a single view.

In other embodiments, the server system may determine an individual gaze-value contribution for each occurrence of an ad space. As such, the individual gaze-value contribution may vary from occurrence to occurrence, depending upon which user is viewing the ad space, how long the user views the ad space, characteristics of an ad that is being displayed in the ad space, characteristics of the ad space itself, and/or other factors.

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A. Gaze Value Based on Per-Occurrence Gaze-Value Contributions

FIG. 3A is a flow chart illustrating a method for determining gaze value, according to an exemplary embodiment. In particular, FIG. 3A illustrates a method 300 in which the gaze value for a given user-account is based on individual gaze-value contributions of occurrences of the ad space in gaze data associated with the given user-account.

More specifically, method 300 involves monitoring gaze data that is associated with a given user account for occurrences of advertisement spaces, as shown by block 302. Each time an advertisement space is detected, as shown by block 304, the server system may determine a gaze-value contribution of the occurrence, as shown by block 306. The server system may further determine gaze-value contributions for a number of occurrences by repeating blocks 304 and 306 as additional ad-space occurrences are detected. The server system may then use the gaze-value contributions for the detected occurrences as a basis for determining the individual gaze value for the user-account, as shown by block 308.

In some embodiments, such as method 300 of FIG. 3A, the gaze-value contribution for each occurrence may be determined in real-time as each occurrence is detected. However, it should be understood that the gaze-value contribution for some or all occurrences of an ad space may be calculated at a later time, based on wearer-view data that is stored as the occurrences are detected in the gaze data.

A server may use various techniques to determine the individual gaze-value contribution for a given occurrence of an ad space in gaze data. Such techniques may utilize various different factors to determine the gaze-value contribution for a given occurrence. In some embodiments, the server may determine a weighting value for an occurrence, which may then be applied to a "standard" gaze-value contribution to determine the gaze-value contribution for the particular occurrence. In such an embodiment, the weighting value may be based on various factors or combinations of factors, such as the particular wearable computing device from which the gaze data including the particular occurrence was received, the duration of the occurrence, characteristics of the person who viewed (e.g., as indicated by the user-profile associated with the occurrence), the focus value of the occurrence, and/or other factors.

Once a server system has determined the individual gaze-value contributions for a number of occurrences, the server may use various techniques to determine the gaze value for the user. For example, in some embodiments, the server may determine the gaze value by summing the gaze-value contributions of some or all of the detected occurrences. As another example, in some embodiments, the server may determine the gaze value by averaging the gaze-value contributions of some or all of the detected occurrences. Other examples are also possible.

In some cases, the gaze-value contribution for each occurrence of an ad space may be a dollar amount that is attributed to the occurrence. As such, the server may determine a dollar amount for the gaze value by summing the gaze-value contributions. In other cases, the gaze-value contribution for each occurrence of the ad space may be a price rate (e.g., dollars per month, dollars per week, etc.) that is attributed to the occurrence of the ad space. As such, the server may determine the ad value by summing or averaging the gaze-value contributions to get a total or an average price rate, respectively. Other examples are also possible.

It should be understood that techniques described herein for determining an individual gaze value and/or determining

gaze-value contributions of individual occurrences of an ad space are not intended to be limiting. Other techniques for determining an individual gaze value and/or determining gaze-value contributions of individual occurrences are also possible, without departing from the scope of the invention.

B. Gaze-Value Contribution Based on Duration

As noted, the server system may consider the duration of an occurrence when determining the portion of the ad value that is attributable to a given ad-space occurrence associated with a given user-account. In some embodiments, a predetermined rate may be defined for gaze valuation (e.g., dollars per minute), and this rate may be used in conjunction with the duration of an occurrence to determine the gaze-value contribution of the occurrence (e.g., by multiplying the rate by the duration).

In other embodiments, the server system may determine a total view time for an ad space by summing the respective durations of all occurrences associated with a number of different user-accounts (e.g., all user-accounts that have authorized collection and use of gaze data for such a purpose). In some cases, the server system may extrapolate from the gaze data to estimate a total view time for all views of an ad space (whether captured in gaze data or not). In either case, the ad-value portion that is attributable to the given occurrence may be based on the ratio of the duration of the occurrence to the total view time. For instance, if an ad space is valued at a rate of \$12 per day, and the server estimates that an ad space averages two hours of viewing per day, a \$0.10 portion may be attributed to an occurrence lasting for one minute. Other examples are also possible.

In a further aspect, when a gaze-value contribution of an occurrence accounts for the duration of the occurrence, it may also take into account a diminishing return of viewing duration. For example, if a person views an advertisement for one minute, the first twenty seconds that a person views the advertisement may be considered more valuable than the next twenty seconds during which the person continues to view the advertisement, which in turn may be considered more valuable than the final twenty seconds in the minute-long view. Similarly, having a user view one advertisement may be considered less valuable overall than either the same user view five advertisements for one minute each or five different users viewing the same or different advertisements for one minute each. As such, when duration is considered the diminishing returns of extended viewing periods may be taken into account.

C. Gaze-Value Contribution Based on Focus Value for an Occurrence

While detecting an ad space in gaze data from a wearable computing device may generally indicate that the ad space was within the field of view of the wearer, it is possible that the ad space was in the periphery of the wearer's field of view, was in the center of the wearer's field of view, or somewhere in between. Furthermore, the wearer can move their eyes such that they are focusing on the ad space or such that they are focusing elsewhere in their field of view. Accordingly, in some applications, an exemplary method may vary the gaze-value contribution for a given occurrence based on the amount of attention paid to the ad space during the occurrence. Generally, the higher the focus value for a given occurrence, the higher the gaze-value contribution of a given occurrence. However, there may be exceptions to this general principal, without departing from the scope of the invention.

An exemplary server system may use various techniques to calculate a focus value for a given occurrence of an ad space in gaze data. In some implementations, the focus value

may be a numeric value that increases as more attention is paid to an ad space. However, in alternative implementations, it is possible that the focus value may be inversely proportional to the amount of attention paid to the ad space.

In some embodiments the focus value may be based at least in part on a location of the ad space in the gaze data. For example, consider an embodiment where the gaze data from a given wearable computing device includes POV video from the device. The server system may determine the location of the advertisement space in the point-of-view video, and then use the location of the advertisement space in the point-of-view video as a basis to determine the focus value. For instance, the server system may determine coordinates of the ad space (e.g., the coordinates of the center of the ad space) within one or more video frames. The server may then use the determined coordinates as input when determining a focus value for the detected occurrence. In particular, the closer the location of the advertisement space is to the center of the video frame, the greater the determined focus value, and vice versa. Thus, in practice, the server may increase the focus value as the distance between the location of the ad space and the center of the video frame decreases.

Note that if multiple frames with the object are used to calculate the location of the ad space, the server system may determine the coordinates of the object in each frame and then average the coordinates from the frames to determine the location of the object in the video frame. In some embodiments, the server may implement Visual Simultaneous Localization and Mapping (V-SLAM) to track the location of an object from frame to frame. V-SLAM can provide a registered 3D point cloud of the world, which identifies the general shape of objects and the respective distances to the objects, and allows for pixels in one frame to be related to pixels in another frame. V-SLAM is well known in the art and therefore is not discussed in further detail herein. Furthermore, it should be understood that other techniques may also be utilized instead of or in addition to V-SLAM.

In some embodiments, the server may utilize eye-tracking data corresponding to the occurrence of the ad space when determining a focus value for the occurrence. The eye tracking data may generally be indicative of a direction that the wearer is looking. Accordingly, the server may determine the location of the advertisement space in the point-of-view video, and then use eye-tracking data from the wearable computing device that provided the gaze data to determine a wearer-gaze location in the point-of-view video at or near the time when the ad space occurs in the video. The server may then determine the proximity of the wearer-gaze location to the location of the advertisement space in the point-of-view video, and use the proximity as a basis for determining the focus value for the particular occurrence of the advertisement space. In an exemplary embodiment, the server may generally increase the focus value as the distance between the ad-space location and the wearer-gaze location decreases.

In some embodiments, the server may use movement of the ad space during a given occurrence in POV video as a basis for determining the focus value for the occurrence. For instance, if an ad space is detected in a number of consecutive video frames, the server may determine the location of the ad space in each of the frames (or possibly in a representative subset of the frames in which the ad space is detected). The server may then compare the determined locations of the ad space to determine how much the ad space moved during the occurrence of the ad space. If the ad space is relatively still and does not move significantly

within the frame, this may be indication that the wearer focusing on the ad space. On the other hand, more movement of the ad space may indicate that the wearer is focusing on something other than the ad space. Accordingly, the server may generally increase the focus value of the occurrence as the amount of movement during the occurrence decreases, and vice versa.

In some embodiments, the server may use the amount of the point-of-view video frame that is occupied in POV video as a basis for determining the focus value for the occurrence. In particular, if the ad space occupies a large amount of the video frame, this may be an indication that the ad space is more prominent in the wearer's field of view and/or that the user is closer to the ad space. As such, if the ad space occupies a large amount of the video frame, the server may interpret this as an indication that the user is paying more attention to the ad space. Accordingly, the server may generally increase the focus value as the percentage of the video frame that is occupied by the ad space increases.

It should be understood that the focus value may be based on just one of the above factors or another factor altogether. Further, the focus value may also be based on a combination of some or all of the above factors and/or other factors.

D. Gaze-Value Contribution Based on User-Characteristics.

In some embodiments, an exemplary method may help account for the fact that views of an ad space by certain people may be considered more valuable than views of the same ad space by other people. As such, various types of information provided by and/or related to a given user-account may be used to determine a gaze-value contribution for an ad-space occurrence in gaze data for the given user-account.

For instance, a user-account may include or provide access to: (a) consumer information such as spending habits, locations of purchases, amounts of purchases, types or categories of purchases, timing of purchases, etc., (b) demographic information such as age or age group, ethnicity, nationality, sex, location of residence, and/or location of workplace, (c) contact and/or social networking information such as a user's contacts, and possibly data indicating a purchasing influence of the user with regard to their contacts (e.g., data indicating any correlation of the user's purchasing history to the wearers' friends' purchasing histories), and/or (d) other information such as income, job or job type, other job details, hobbies, interests, and so on. When a user has given permission for information from their user-account to be used for purposes of determining their individual gaze value, the server system may use such information to determine the gaze-value contribution that is attributable to a given occurrence of an ad space in gaze data associated with the user's account.

To provide one specific example, the server may determine an income level for a given user-account and then increase or decrease the gaze-value contribution for occurrences in the user's gaze data according to their income level. For example, number of income ranges may be mapped to certain adjustments that should be applied when determining gaze-value contribution. The adjustments may generally increase the gaze-value contribution for higher income ranges, and decrease the gaze-value contribution for higher income ranges.

In another application, the server may determine an average income level for a relevant group (e.g., users who have viewed the advertisement, a target user group for the ad displayed in the ad space, the population as a whole, etc.). The server may then adjust gaze-value contribution for an occurrence in gaze data associated with the user based on the

relationship between the user's income level and the average income level (e.g., whether and/or by how much the user's income level is above or below the average income level). Other examples are also possible.

Furthermore, in some applications, the gaze-value contribution for an occurrence may not increase or decrease in proportion to associated user's income level. For example, a given advertisement may be targeted at a specified income range. As such, views by users that do not fall within the specified income range may be considered less valuable. Therefore, gaze-value contributions for user-accounts having an income level outside the income range may be reduced, regardless of whether the income level is above or below specified income range. Other examples are also possible.

As another specific example, the server may base the gaze-value contribution on how well demographic information from a user-profile matches a targeted demographic profile for the advertisement displayed in a detected ad space. The targeted demographic profile may indicate a single type of demographic information to be evaluated (e.g., male or female) or a combination of various types of demographic information (e.g., males between 30 and 45 years of age who live in an urban location). Other examples are also possible.

E. Gaze-Value Contribution Based on Context

In a further aspect, an exemplary method may help account for the fact that in some cases, a certain user viewing a certain advertisement in one context, may be considered more valuable than the same user viewing the same advertisement in another context. Accordingly, the gaze-value contribution for a given advertisement space that is detected in a user's gaze data may vary based on the advertisement that is displayed and context associated with detecting the advertisement space.

For instance, consider a user-profile that indicates the particular user is a scientist who is interested in the latest research in their field. Therefore, it may be determined that this user will likely be interested in an advertisement for an upcoming conference in their field. As such, gaze-value contributions of certain advertisements may be adjusted depending on whether or not the context in which the advertisement is viewed makes it more or less likely that this user will be interested in the advertisement and/or be prompted to act when they view the advertisement.

As a specific example, consider a scenario where the above-described user is driving to work at their laboratory, and sees a billboard while driving. In this scenario, context signals such as the day of the week, time, and/or the user's location may be used to determine that the user is "driving to work at their lab" (e.g., based on context signals indicating that the user is located on a highway on the laboratory, on a weekday, at 8:00 am). If the advertisement for upcoming conference in the user's field is displayed on the billboard in this context, it may be inferred that the user is likely to have work on their mind, and thus be more likely to be interested in and/or to act on this advertisement. However, if this same advertisement is displayed to the same user when the user is on vacation, context may be used to reduce the gaze-value contribution of this user viewing the advertisement.

In a further aspect, gaze-value contribution in a given context may vary between different users (e.g., between different user-accounts). For example, in some cases, the gaze-value contribution corresponding to a first user viewing a certain advertisement in a certain context may be greater than the gaze-value contribution for another user viewing

the same advertisement in same context. In particular, the relationship between characteristics of a user indicated by their user-account and a given context may be evaluated to determine whether an adjustment to a gaze-value contribution is appropriate in the given context.

For example, consider a first user and a second user who are both at a department store. An exemplary system and/or the users' respective devices may evaluate various context signals, such as each user's location (e.g., at the location of the department store), the time of day (e.g., a time during the business hours of the department store), and so on, to determine that the first user's context is "at the department store." However, employment information in the first user's user-account may indicate that the first user is an employee of the department store, while employment information (and other types of information) in the second user's user-account may reveal no such connection to the department store.

Note also that in some instances, context signals may be used to infer characteristics of a given user, such that use of information from a user-account may not be necessary. For instance, in the above example, it might be inferred that the first user is an employee of the department store from context signals that indicate, e.g., that the first user is typically located in the department store for eight hours per day, five days per week, during store business hours. Other examples are also possible.

Based on these relationships between the context and the information from the respective user characteristics of these users, the gaze-value contribution for the first user viewing a particular advertisement while located in the department store may be lower than the gaze-value contribution for the second user viewing the same advertisement while located in the department store. More specifically, the gaze-value contribution for the first user may be lower because the first user is an employee and thus may be considered unlikely to act on an advertisement while on the job, whereas the second user may be inferred to be shopper at the department store, since available information does not indicate to the contrary. This may be the case even if most or all other information regarding the two users is the same (e.g., the same or similar demographic information, same income range, etc.).

To determine a context associated with a given advertisement being detected in gaze data for a given user-account, a cloud-based server may be configured to use context data from a single device that is associated with the user-profile. For instance, referring back to FIG. 2, server system 204 may use context data from a single device, such as wearable computing device 202A, to determine context for the user-account that is associated with that device. Alternatively, the server may be configured to aggregate context data from two or more devices that are associated with the user-account, and use the aggregate context data to determine context for the user-account. For example, if a number of devices are all associated with the same user-account, such as wearable computing device 202A, a mobile phone, and a laptop computer, for example, then context data provided by some or all of the associated devices may be aggregated when determining context for the associated user-profile.

In an exemplary embodiment, the context associated with a given user-profile may be determined using various techniques. In general, a "context" may be determined based on various "context signals" or combinations of context signals. A context signal may be any signal that provides a measurement or otherwise provides information pertaining to the state or the environment associated with a certain subject (e.g., with a certain user, device, event, etc.). In this case, the context signals associated are generally pertain to a user-

profile for a wearer of a wearable computing device. As such, the context signals may generally provide some type of information pertaining to the state or the environment of the wearer.

In some instances, a context may be a state associated with a particular context signals or set of context signals. However, a context may also be abstracted from the context signals upon which it is based. As such, a "context" may also be a data-based description or characterization of an environment or state that is determined or derived from one or more context-signals. For example, contexts may take the form of data indicating environment or state information such as "at home," "at work," "in a car," "indoors," "outside," "in a meeting," etc. Furthermore, a context may be a qualitative or quantitative indication that is determined based on one or more context signals. For example, context signals indicating that it is 6:30 AM on a weekday and that a user is located at their home may be used to determine the context that the user is "getting ready for work."

Many types of information, from many different sources, may be used as context signals or provide information from which context signals may be derived. For example, context signals may include: (a) the current time, (b) the current date, (c) the current day of the week, (d) the current month, (e) the current season, (f) a time of a future event or future user-context, (g) a date of a future event or future user-context, (h) a day of the week of a future event or future context, (i) a month of a future event or future user-context, (j) a season of a future event or future user-context, (k) a time of a past event or past user-context, (l) a date of a past event or past user-context, (m) a day of the week of a past event or past user-context, (n) a month of a past event or past user-context, (o) a season of a past event or past user-context, ambient temperature near the user (or near a monitoring device associated with a user), (p) a current, future, and/or past weather forecast at or near a user's current location, (q) a current, future, and/or past weather forecast at or near a location of a planned event in which a user and/or a user's friends plan to participate, (r) a current, future, and/or past weather forecast at or near a location of a previous event in which a user and/or a user's friends participated, (s) information on user's calendar, such as information regarding events or statuses of a user or a user's friends, (t) information accessible via a user's social networking account, such as information relating a user's status, statuses of a user's friends in a social network group, and/or communications between the user and the users friends, (u) noise level or any recognizable sounds detected by a monitoring device, (v) items that are currently detected by a monitoring device, (w) items that have been detected in the past by the monitoring device, (x) items that other devices associated with a monitoring device (e.g., a "trusted" monitoring device) are currently monitoring or have monitored in the past, (y) information derived from cross-referencing any two or more of: information on a user's calendar, information available via a user's social networking account, and/or other context signals or sources of context information, (z) health statistics or characterizations of a user's current health (e.g., whether a user has a fever or whether a user just woke up from being asleep), and (aa) a user's recent context as determined from sensors on or near the user and/or other sources of context information, (bb) a current location, (cc) a past location, and (dd) a future location, among others. Those skilled in the art will understand that the above list of possible context signals and sources of context information is not intended to be limiting,

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and that other context signals and/or sources of context information are possible in addition, or in the alternative, to those listed above.

In some embodiments, the detection or observation of a certain event in data from a certain data source may itself be interpreted as a context signal. For example, the fact that a certain word is detected in an audio signal from a microphone may be interpreted as a context signal providing context to the event of that word being spoken. Other examples are also possible.

In some embodiments, context signals may be obtained or derived from sources such as a user's computer-based calendar, blog, webpage, social network account, and/or e-mail account, among others. For instance, context signals may be provided by user's calendar entries, e-mail messages, and social-network profile, messages, posts, and/or tweets. Further, in some embodiments, similar context signals may be obtained or derived from other users' computer-based calendars, blogs, webpages, social network accounts, and/or e-mail accounts, who are listed in a user's electronic contact list, listed as a "friend" in a user's social network, or otherwise associated with the user (provided such users have opted in to share such context information).

It should be understood that the above examples of contexts, context signals, techniques for determining a context, and/or techniques for using context when selecting an advertisement are provided for illustrative purposes, and are not intended to be limiting. Other examples and/or techniques are also possible.

F. Combining Various Factors to Determine a Gaze Value

FIG. 3B is a flow chart illustrating a method for using multiple factors to determine an individual gaze value for a user-account, according to an exemplary embodiment. In method 350 of FIG. 3B, a pre-determined base contribution for an ad space is weighted according to multiple factors in order to determine a gaze-value contribution for each occurrence of an ad space in gaze data for a given user-account. The individual gaze value for the user-account may then be determined from the collective knowledge provided by the gaze-value contributions of the ad-space occurrences detected in gaze data associated with the user-account. Method 350 is described by way of example as being implemented by a server system, but could also be implemented by a wearable computing device, by another device or system, or by a combination of devices and systems.

Method 350 involves the server system determining a weighting value (weighting_value_i) to be applied for an occurrence of an ad space (O_i) in gaze data from a given user account, based on m different factors (f_{1i} to f_{m_i}). In particular, the server system may determine a value for a first factor (f_{1i}) with respect to a given occurrence of an ad space in gaze data for a first user-account, as shown by block 352. The server system may then determine an average value for the first factor (avg_f_{1i}), as shown by block 354. As such, the server system may determine a relationship between the determined value for the given occurrence and the average value with respect to the first factor (e.g., f_{1i}/avg_f_{1i}), as shown by block 356. If there are additional factors to consider (e.g., if m is greater than one), as indicated by block 358, then the server may repeat blocks 352 to 356 for the additional factors, until the relationship between the given user-account and relationship between the value for the given occurrence and the average value has been determined for each factor. Once all the factors have been evaluated, the server system may use the determined relationships as a basis for determining a weighting value for the occurrence, as shown by block 360.

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Thus, for a given occurrence O_i, and values for a set of n factors f₁ to f_n, method 350 may be implemented to determine a weighting_value; as a function of the respective relationships between the values of factors for the given occurrence f_{1i} to f_{m_i} and the average value for factors avg_f_{1i} to avg_f_{m_i}. For example, weighting_value; may be calculated as:

$$\text{weighting_value}_i = F\left[\left(f_{1i}/\text{avg_f}_{1i}\right), \left(f_{2i}/\text{avg_f}_{2i}\right), \dots \left(f_{mi}/\text{avg_f}_{mi}\right)\right]$$

Note that the particular function used may vary from implementation to implementation, depending upon the design goals.

Once the server has determined the weighting_value; for a given occurrence O_i of an ad space, the server may use a base contribution for the ad space in the ad space detected in occurrence O_i to calculate the gaze value contribution for the occurrence as:

$$\text{gaze_value_contribution}_i = \text{base_contribution}_i * \text{weighting_value}_i$$

Further, the server system may repeat the above process to determine a gaze-value contribution for n occurrences O_i in the gaze data for the user-account. As such, the individual gaze value for the user-account may be determined as a function of gaze_value_contribution_i for i equal 1 to n. For example, the individual gaze value may be calculated as the sum of gaze_value_contribution₁ to gaze_value_contribution_n. Other examples are also possible.

It should be understood that many other types of information provided by and/or related to a given user-account may be considered, alone or in combination, when determining the gaze-value contribution of an ad space in gaze data for the given user-account. Further, information provided by and/or related to a given user-account may be considered in combination with other factors, such as duration of an occurrence and/or a focus value associated with the occurrence, when determining the gaze-value contribution for an occurrence.

G. Fitting Gaze-Value Contributions to Predetermined Ad Values

In some of the described embodiments, the gaze-value contribution for a given occurrence of an ad space may be calculated as the portion of a known ad value for the ad space that is attributable to the given occurrence.

FIG. 4 is a flow chart illustrating a method for determining gaze value, according to an exemplary embodiment. In particular, FIG. 4 illustrates a method 400 in which the gaze-value contribution for each occurrence of an ad space is a portion of the ad value that is attributable to the particular occurrence.

More specifically, method 400 involves monitoring gaze data from at least one wearable computing device associated with a given user account (and possibly other devices associated with the same user-account) for occurrences of advertisement spaces, as shown by block 402. Each time an occurrence of an advertisement space is detected, as shown by block 404, the server system may determine an advertisement value of the advertising space, as shown by block 406. Further, the server system may determine a portion of the advertisement value that is attributable to the occurrence

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of the advertisement space in the gaze data associated with the first user-account, as shown by block 408. The first server system may repeat block 404 to 408 for a number of occurrences of advertisement spaces to determine portions of the respective advertisement values that are attributable to the respective occurrences number of ad-space occurrences. Accordingly, the server system may use the determined advertisement-value portions as a basis for determining an individual gaze value for the given user-account, as shown by block 410.

In an exemplary method 400, the function of determining the advertisement value of a given advertising space may be accomplished using various techniques. In some instances, there may be a built-in assumption that the ad value for a given ad space is equal to whatever is being paid for the ad space by the advertiser. Accordingly, the server system may simply determine a price that was paid for the ad space by an advertiser. For example, the server system may determine a fixed price that was paid for the ad space, or may determine a price rate (e.g., dollars per month, dollars per view, etc.).

However, in some cases, it may not be assumed that the ad value for an ad space is equal what the advertiser paid; or in other words, some embodiments may allow for the possibility of an advertiser “getting more than what they paid for.” For example, if each view by a certain type of person (e.g., the target market) is considered to be worth a certain amount to an advertiser, the advertiser may pay an amount based on the expected number of views by the target market. However, if an ad space receives more than the expected number of views from the target market, then from the advertiser’s perspective, the ad space is worth more than they paid for it. Furthermore, there may be cases where an ad space has been valued or could be valued, but has not been purchased by an advertiser.

Accordingly, the server system may additionally or alternatively use a measure of advertisement value other than what was actually paid for the advertisement space. For example, the server system may query an advertisement-value database, which indicates advertisement values that are based on detected occurrences of the advertisement space in gaze data from a plurality of wearable computing devices (and possibly gaze data from other types of devices as well). Methods for determining an ad value for and ad space based on gaze data from a number of wearable computing devices are described in greater detail with reference to FIGS. 5A to 7.

Further, in some embodiments, individual gaze valuation may be implemented in conjunction with an ad marketplace where advertisement spaces are generally valued based on gaze data. In such an embodiment, the server system may also use as the ad value, the price at which an ad space is being offered for sale in the ad marketplace.

It should be understood that techniques described herein for determining an ad value for a given ad space are not intended to be limiting. Other techniques for determining an ad value based on the ad-value contributions of individual occurrences are also possible, without departing from the scope of the invention. Further, it should be understood that the technique and/or format of the ad value may vary from one occurrence to another within the gaze data associated with a given account. For example, the server system may might use the actual purchase price for an ad space, when the purchase price is available, but use the ad value based on gaze data when a purchase price is not available or is otherwise deemed to be less accurate (e.g., when the purchase price is considered out of date).

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In an exemplary method 400, the function of determining the portion of the advertisement value that is attributable to a given occurrence of a given advertisement space may be accomplished using various techniques.

In some embodiments, the server system may attribute an equal portion of the advertisement value to each occurrence of a particular ad space. For example, the portion of an ad space’s value that is attributed to a given occurrence may be calculated by dividing the advertisement value by a total number of occurrences detected in gaze data from a number of devices associated with a number of different user-accounts. For instance, if an advertisement space is valued at \$100 per month and the advertisement space averages ten occurrences per month in all available gaze data, a \$10 portion may be attributed to a given occurrence of the ad space.

In some cases, the portion of an ad space’s value that is attributed to a given occurrence may be based on an estimated number of total views of the ad space. In particular, the server system may extrapolate from the occurrences that are detected to determine an estimated number of total views. This estimation may account for all views of an ad space, regardless of whether the view was captured in gaze data. This may be useful as there may be many cases where those that view an ad space are not wearing a wearable computer that is configured to capture and/or provide gaze data. As one specific example, consider the case where it is assumed that one out of every thousand views will be captured in gaze data. In this case, if an advertisement space is valued at \$100 per month and the advertisement space averages ten occurrences per month in all available gaze data, the server system may calculate that there are 10,000 views per month. Accordingly, a \$0.01 portion may be attributed to a given occurrence of the ad space. Other examples are also possible.

In some embodiments, an exemplary method may help account for the fact that views of an ad space by certain people may be considered more valuable than views of the same ad space by other people. As such, various types of information provided by and/or related to a given user-account may be used to determine an ad-value portion that is attributable an occurrence of an ad space in gaze data associated with the given user-account. For instance, a user-account may include or provide access to: (a) consumer information such as spending habits, locations of purchases, amounts of purchases, types or categories of purchases, timing of purchases, etc., (b) demographic information such as age or age group, ethnicity, nationality, sex, location of residence, and/or location of workplace, (c) contact and/or social networking information such as a user’s contacts, and possibly data indicating a purchasing influence of the user with regard to their contacts (e.g., data indicating any correlation of the user’s purchasing history to the wearers’ friends’ purchasing histories), and/or (d) other information such as income, job or job type, other job details, hobbies, interests, and so on. When a user has given permission for information from their user-account to be used for purposes of determining their individual gaze value, the server system may use such information to determine an ad-value portion that is attributable to a given occurrence of an ad space in gaze data associated with the user’s account.

It should be understood that many other types of information provided by and/or related to a given user-account may be considered, alone or in combination, when determining the portion of the ad value to attribute to an occurrence of an ad space in gaze data for the given user-account. Further, information provided by and/or related to a given

user-account may be considered in combination with other factors, such as duration of an occurrence and/or a focus value associated with the occurrence, when determining the portion of the ad value to attribute to the occurrence.

H. Gaze-Value Contributions on Per-Ad Basis

In the above examples, the gaze value is based upon ad-value portions that are determined on a per-occurrence basis. However, in some embodiments, the gaze value may only consider each ad space once when determining a gaze value for a given user-account. As such, the server system may effectively ignore subsequent occurrences of the same ad space in gaze data associated with a given user-account. For example, the server system may determine that a given ad space has occurred in gaze data for a certain number of user-accounts. The server system may then divide the ad value for the ad space by this number to determine the portion that is attributable to one user-account. As another example, the server may extrapolate from the total number of user-accounts have provided gaze data including an occurrence of a given ad space to determine an estimated number of users that have viewed (or will view) the ad space, and divide the ad value by the estimated number of users who have viewed the ad space, in order to determine the portion of the ad value that is attributable to one user-account. Other examples are also possible.

VI. Sending a Gaze-Value Indication

Referring back to method **100** of FIG. **1**, once a gaze value for a given user-account has been determined, the server may send a gaze-value indication to the given user-account, which indicates the individual gaze value that was determined. This may be accomplished in various ways. For example, the server system may (a) send an e-mail message to at least one e-mail account associated with the first user-profile, (b) send a text or multimedia message to at least one phone number associated with the first user-profile, (c) initiate an automated phone call to at least one phone number associated with the first user-profile, and/or (d) display the determined gaze value in a web browser or another application via which the user has accessed their user-account. Other techniques for providing a user with their individual gaze value are also possible.

VII. Determining the Value of an Advertisement Space

As noted, in some embodiments, an exemplary system may also be configured to use gaze data to determine advertisement values for ad spaces, in addition to using the gaze data to determine individual gaze values for users. In such an embodiment, an exemplary system may be configured to determine an advertisement value for almost any time of physical space.

FIG. **5A** is a flow chart illustrating a method according to an exemplary embodiment. This method may be implemented by a computing device, and in particular, by a server system, in order to value an advertisement space based on point-of-view gaze data received from a number of wearable computing devices (which may be referred to interchangeably as wearable computing devices). Note that wearable computing devices may also be referred to as wearable computers herein. Further, a server system that implements an exemplary method may be referred to as an ad-valuation system, as an ad-valuation server, or simply as a server.

As shown by block **502**, method **500** involves a server system receiving gaze data from a number of wearable

computing devices. The server system analyzes the gaze data from the wearable computing devices to detect occurrences of an advertisement space in the gaze data, as shown by block **504**. The server system then generates wearer-view data for the advertisement space, which is based on detected occurrences of the advertisement space in the gaze data, as shown by block **506**. The wearer-view data can then be used as a basis for determining an advertisement value for the advertisement space, as shown by block **508**. Once the advertisement value is determined, the server system may cause a computing system to make the advertisement space available for purchase at the determined advertisement value, as shown by block **510**.

In an exemplary method **500**, the gaze data is received from a number of wearable computing devices. Further, the gaze data from each wearable computing device is generally indicative of a respective wearer-view associated with the given wearable computing device. For example, the gaze data from each wearable computing device may take the form of point-of-view video that is captured at the wearable computing device. As such, the gaze data that is analyzed by the server system may include a number of point-of-view videos (e.g., a respective point-of-view video from each of the wearable computing devices).

The gaze data from some or all of the wearable computing devices that provide gaze data may additionally or alternatively take forms other than point-of-view video. For example, the gaze data from some or all of the wearable computing devices may take the form of respective point-of-view images captured by a forward- or outward-facing camera on the respective wearable computing device. As a specific example, a given wearable computing device may periodically take a picture, and then send the picture to the server system for use in generating wearer view data. To do so, the wearable computing device may analyze point-of-view video for one or more ad spaces, and generate a screen capture of the video when an ad space detected. The wearable computing device may then send the screen capture to the server system. Other examples are also possible.

Since the gaze data from a given wearable computing device is generally indicative of the wearer-view of the wearable computing device's wearer, the gaze data is generally indicative of what the wearer of the device is actually looking at. Further, since the wearer-view data is based on the gaze data, the wearer-view data is indicative of actual views of the ad space by wearers. For instance, the wearer-view data may provide an indication of how many people are looking at a particular advertisement space, which people are actually looking at a particular ad space, when people are looking at a particular ad space, and/or how long people are actually looking at a particular ad space, among other information. As such, the wearer-view data may help to more accurately determine what an advertising space is worth.

As noted above, when occurrences of an ad space are detected in gaze data, an exemplary method **500** may involve generating wearer-view data that is based on the detected occurrences. As such, an exemplary server system **204** may be configured to carry out an exemplary method **500** or portions thereof for many different advertisement spaces. Generally, the accuracy of the ad-space valuation will typically increase as the number of wearable computing devices providing gaze data increases. However, the gaze data may be collected from any number of wearable computing devices without departing from the scope of the invention.

To facilitate determining an advertisement value for a given ad space, the wearer-view data may provide various types of information. For example, the wearer-view data for a given ad space may include, for each detected occurrence of the given ad space: (a) data indicating the particular wearable computing device that provided the gaze data in which the ad space occurred, (b) data indicating a user-profile associated with the particular wearable computing device, (c) data indicating a time of the detected occurrence, (d) a duration of the detected occurrence, and/or (c) other information.

Generally, the function of generating wearer-view data for the advertisement space, as shown in block 506 of method 500, may vary depending upon the information to be included in the wearer-view data. In an exemplary embodiment, detecting an occurrence of an advertising space in the gaze data may serve as a trigger for the server system to generate wearer-view data recording the fact that the occurrence was detected. Further, to generate the wearer-view data for a given occurrence, the server system may extract information from the gaze data in which the occurrence was detected. The extracted information (or information derived from the extracted information) may be included in the wearer-view data generated for the detected occurrence.

A. Per-Occurrence Data for an Ad Space

In some embodiments, the server system 204 may update the wearer-view database 308 upon each detected occurrence of an ad space. For example, the server system may generate a record in the wearer-view database for each detected occurrence of an ad space. In such an embodiment, the record for a given occurrence of an ad space may include: (a) an indication of the particular wearable computing device that provided the gaze data in which the ad space occurred, (b) an indication of a user-profile associated with the particular wearable computing device, (c) a time of the occurrence, and/or (d) a duration of the occurrence.

The wearer-view data for a given occurrence of an ad space may indicate the corresponding wearable computing device that provided the gaze data in which the ad space occurred. In such an embodiment, the server system may determine the corresponding wearable computing device in various ways. For instance, consider an embodiment where the server system receives point-of-view (POV) video stream from a number of wearable computing devices. In such an embodiment, the server system may establish a communication session to receive the video stream from a given one of the wearable computing devices, and as part of establishing and/or participating in the session, may receive an identifier of the wearable computing device. (Note that various protocols, which are well known in the art, may be used to receive a POV video stream and/or to receive other forms of gaze data.) Additionally or alternatively, metadata in the gaze data itself may include an identifier of the wearable computing device that is providing the gaze data. Other techniques for determining which wearable computing device corresponds to a particular occurrence of an ad space are also possible.

As further noted above, the wearer-view data for a given occurrence of an ad space may indicate an associated user-profile, which is associated with the wearable computing device that provided the gaze data having the particular occurrence. The server system may determine the associated user-profile in various ways. For example, the server may determine the identifier for the corresponding wearable computing device in a manner such as described above or otherwise. The server may then look up a user-profile of a user that is registered to use or is otherwise associated with

the corresponding wearable computing device (e.g., by querying a user database that indicates which users are associated with which wearable computing devices). Alternatively, a user-identifier may be provided in the course of receiving the gaze data (e.g., in a communication session or in metadata). In such an embodiment, the server system may use the user-identifier to access a user-profile for the user. As another alternative, the user-profile itself may be received directly from the device (e.g., during the communication session in which the gaze data is received, as metadata included in the gaze data, or in a separate message that is associated with the gaze data). Other techniques for determining a corresponding user-profile for a particular occurrence of an ad space are also possible.

In a further aspect, when the wearer-view data for a given occurrence indicates the associated user-profile, the wearer-view data may simply include an identifier of the associated user-profile. In such an embodiment, the data from such user-profiles may be stored in one or more separate user-profile databases. In this case, the server may use the identifiers of the associated user-profiles to retrieve the data from the actual user-profiles. Alternatively, some or all of the data from the associated user-profile may be included in the wearer-view data for the ad space (e.g., in wearer-view database 308).

In a further aspect, the server system may include a time stamp in the wearer-view data that is generated for a given occurrence. The timestamp may indicate the time at which the occurrence of the ad space was detected. Additionally or alternatively, the timestamp may indicate a time that is derived from time data included in the gaze data. For example, point-of-view video from a given wearable computing device may include time data indicating when the video was recorded by the wearable computing device. As such, the server system may use this time data to generate a timestamp for an occurrence that is detected in such point-of-view video. For instance, the server system may determine a frame or frames of the video that include the ad space, and use a time stamp or time stamps of the frame or frames to generate the timestamp for the detected occurrence. Other techniques for generating a timestamp for a particular occurrence of an ad space are also possible.

In another aspect, the wearer-view data for a given occurrence of an ad space may indicate the duration of the given occurrence. Accordingly, the server system may be configured to determine the duration of a given occurrence of an ad space. For instance, in the above example where POV video includes time data, the server system may use timestamps on frames of the video to determine the duration of time the first frame of the video that includes the ad space and the last subsequent and consecutive frame that includes the ad space. Alternatively, the server system may implement its own timer to determine the duration of a given occurrence of an ad space. Other techniques for determining the duration of a particular occurrence of an ad space are also possible.

In a further aspect, when generating wearer-view data for a given occurrence, the server may consider whether the wearable computing device that corresponds to a given occurrence was being worn during the occurrence. In particular, if the corresponding wearable computing device is not being worn at the time of the detected occurrence, the server may adjust or change the wearer-view data that is generated in response to detecting the occurrence. For example, when the wearable computing device is not being worn, the server may interpret this to mean that the gaze data from the wearable computing device is unlikely to represent

what the wearer is actually viewing. Accordingly, the server may include an indication that the wearable computing device was not being worn in the wearer-view data that is created for such an occurrence. Further, server may adjust the wearer-view data so as to decrease the weight of such an occurrence when determining the ad value for the ad space, or may ignore the occurrence entirely (e.g., by refraining from generating any wearer-view data for the occurrence).

B. Summary Data for an Ad Space

In some embodiments, the wearer-view data for a given ad space may include summary data for the ad space such as: (a) a list of which wearable computing devices viewed the ad space (e.g., which wearable computing devices provided gaze data in which one or more occurrences were detected), (b) a list of the user-accounts or the user-profiles that are associated with the wearable computing devices that have viewed the ad space, (c) a total view count indicating the total number of detected occurrences of the ad space, (d) a total view duration of the ad space, (e) an average view duration for occurrences of the ad space, and/or (f) a view rate that indicates how frequently the advertisement space occurs in the gaze data (e.g., occurrences/hour, occurrences/month, etc.). The wearer-view data for a given ad space may additionally or alternatively include other types of summary data for the ad space.

In order to keep the above and other such summary data substantially current, the server system may update the wearer-view data for an ad space each time the ad space is detected in gaze data. For example, when the server system detects an ad space in gaze data from a given wearable computing device, the server system may update the wearer-view data by: (a) adding the given wearable computing device to a list of wearable computing devices that have viewed the ad space (if the wearable computing device is not on the list already), (b) adding the user-account or the user-profile that is associated with the given wearable computing device to a list of user-accounts or user-profiles that have viewed the ad space, (c) incrementing the total view count for the ad space, (d) determining the duration of the occurrence and adding the determined duration to a total view duration for the ad space, and/or (e) determining the duration of the occurrence and adding the determined duration and recalculating the average view duration to account for the determined duration. Other examples are possible as well.

In some embodiments, the wearer-view data for each ad space may include only summary data such as that described above, and thus may not include per-occurrence data for each detected occurrence of an ad space. However, it is also possible that the wearer-view data for a given ad space may include only per-occurrence data, or may include both per-occurrence data and summary data for the ad space.

C. Focus Data for an Occurrence

In some embodiments, the wearer-view data for a given ad space may include focus data, which is generally indicative of the amount of attention paid to an ad space by viewers of the ad space. The focus data may help to provide a more accurate valuation for the ad space by helping take into account the fact that not all views are necessarily equal, since the amount of attention paid to the ad space may vary between views. In such an embodiment, the server system may determine a focus value for a detected occurrence (as described above) when it generates wearer-view data for the occurrence, or may determine a focus value at a later time.

D. Use of Summary Data for Advertisement Valuation

As noted above, an exemplary method 500 may involve using the wearer-view data for an ad space to determine an

advertisement value for the advertisement space. Various types of wearer-view data may be utilized when determining an advertisement value. For instance, various types of the summary data described above and/or various types of the per-occurrence data described above may be used to determine the advertisement value for a given ad space. An exemplary valuation method may also incorporate other types of data in addition to wearer-view data. Further, the manner in which a given type of wearer-view data is used to determine an advertisement value may vary depending upon the implementation.

In some embodiments, the ad value for a given ad space may be based on summary data for the ad space. For example, the ad value may be based at least in part on the total view count for an ad space (e.g., the total number of occurrences that are detected in the gaze data). In such an embodiment, the total number of occurrences may be tracked over all time. Alternatively, the total number of occurrences may be tracked over a predetermined period of time (e.g., a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates total view count, the determined advertisement value will typically increase as the total number of occurrences increases. Further, the manner in which the total view count is used to determine ad value may vary, depending upon the implementation.

As another example, the ad value for a given ad space may be based at least in part on a view rate for the advertisement space (e.g., the rate at which occurrences of the ad space are detected in the gaze data). For instance, the wearer-view data may indicate a number of views per month, per week, per day, per hour, etc. In such an embodiment, the rate may be based on detected occurrences over all time. Alternatively, the rate may be based on occurrences during a predetermined period of time (e.g., during a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates view rate, the determined advertisement value will typically increase as the view rate increases. Further, the manner in which the view rate is used to determine ad value may vary, depending upon the implementation.

In the above examples, the ad value is determined based on summary data that generally does not differentiate one detected occurrence from another. However, some embodiments may apply further intelligence to account for the fact that some views of an ad space may be more valuable to an advertiser than others.

For example, the ad value for a given ad space may be based at least in part on a total view duration and/or an average view duration for the ad space. In such an embodiment, the total view duration and/or the average view duration may be calculated from all detected occurrences of the ad space or from a representative sample of occurrences. In either case, the total view duration and/or the average view duration may be calculated over all time, or may be calculated over a predetermined period of time (e.g., a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates total view duration and/or the average view duration, the determined advertisement value will typically increase as the total view duration and/or the average view duration increases. Accordingly, views that last longer will generally contribute more to the ad value and/or be weighted more heavily when determining the ad value. It should be understood that the manner in which the total view duration and/or the average view duration is used to determine ad value may vary, depending upon the implementation.

As another example, the server may determine focus values for all or a representative sample of the detected occurrences of an ad space. The server may then average the focus values for the detected occurrences to determine an average focus value for the ad space. The server can then use the average focus value to determine the ad value for the ad space.

In a further aspect, an exemplary embodiment may help account for the fact that views of an ad space by certain people may be considered more valuable than views of the same ad space by other people. More specifically, in an exemplary embodiment, wearers may opt-in to a program or otherwise give permission for information from their user-profile to be used to value ad spaces. Various types of information from an associated user-profile may then be used to determine how valuable a given occurrence of an ad space is. For instance, a user-profile for a wearer may include: (a) consumer information such as spending habits, locations of purchases, amounts of purchases, types or categories of purchases, timing of purchases, etc., (b) demographic information such as age or age group, ethnicity, nationality, sex, location of residence, and/or location of workplace, (c) contact and/or social networking information such as a wearer's contacts, and possibly data indicating a purchasing influence of the wearer with regard to their contacts (e.g., data indicating any correlation of the wearer's purchasing history to the wearers' friends' purchasing histories), and/or (d) other information such as income, job or job type, other job details, hobbies, interests, and so on.

Therefore, since the occurrence of an ad space in gaze data from a given wearable computing device may be interpreted to mean that the wearer of the given wearable computing device has viewed or is viewing the ad space, information from user-profiles that wearer-view data associates with a given ad space may provide information about the type or types of people that an ad space reaches and/or of the characteristics of people that the ad space reaches. As a result, this information may be used to more accurately determine what types of people are viewing the ad space, and value the ad space accordingly. In particular, an exemplary server may place greater weight on occurrence of an ad space associated with certain people and/or certain types of people when determining the ad value for a given ad space.

For example, the server may determine a respective income level for the user-profile associated with each occurrence. The server may then average the determined income levels to calculate an average income level for viewers of the ad space, and use the average income level as input data to determine the ad value for the ad space. Alternatively, the server may determine an income range of the determined income levels, and use the income range as an input to the ad-value calculation for the ad space. Other examples are also possible.

It should be understood that the ad value for a given ad space may be based upon one type of summary data or a combination of various types of summary data. For example, in one implementation, the total number of views, the view rate, the average view duration, and one or more characteristics of the associated user-profiles, could all be used as inputs when calculating ad value. Many other examples are also possible.

E. Use of Per-Occurrence Ad-Value Contributions for Advertisement Valuation

In some embodiments, a server system may determine an advertisement value for an ad space by first determining an individual advertisement-value contribution for each detected occurrence of the advertisement space. The adver-

tisement-value contribution for a given occurrence may be based on information from the user-profile associated with the occurrence and/or on other information related to the occurrence. The collective knowledge provided by all the individual advertisement-value contributions may then be used to determine the advertisement value for the advertisement space and/or be used to determine summary data for the ad space, which may in turn be used to determine the ad value.

FIG. 5B is a flow chart illustrating a method for determining advertisement value, according to an exemplary embodiment. In particular, FIG. 5B illustrates a method 550 in which the advertisement value for an ad space is based on individual ad-value contributions of occurrences of the ad space in gaze data.

More specifically, method 550 involves monitoring gaze data from a number of wearable computing devices for occurrences of a particular advertisement space, as shown by block 552. Each time an occurrence of the advertisement space is detected, as shown by block 554, the server system may further determine an advertising-value contribution of the occurrence, as shown by block 556. The server system may then determine advertising-value contributions for a number of occurrences by repeating blocks 554 and 556 for a number of detected occurrences of the advertisement space. The server system may then use the advertising-value contributions for the detected occurrences of the advertisement space as a basis for determining the advertisement value for the advertisement space, as shown by block 558.

In some embodiments, such as method 550 of FIG. 5B, the ad-value contribution for each occurrence of an ad space may be determined upon detecting the occurrence in the gaze data. However, it should be understood that the ad-value contribution for some or all occurrences of an ad space may be calculated at a later time, based on wearer-view data that is stored as the occurrences are detected in the gaze data.

A server may use various techniques to determine the individual ad-value contribution for a given occurrence of an ad space in gaze data. Such techniques may utilize various different factors to determine the ad-value contribution for a given occurrence. In some embodiments, the server may use a weighting value for an occurrence to determine the ad-value contribution of the occurrence. In particular, the server may determine a weighting value that generally indicates the particular occurrence's value relative to a "standard" occurrence of the ad space. The weighting value for the particular occurrence may then be applied to a base advertising-value contribution to determine the advertising-value contribution for the particular occurrence. In such an embodiment, the weighting value may be based on various factors or combinations of factors, such as the particular wearable computing device from which the gaze data including the particular occurrence was received, the duration of the occurrence, characteristics of the person who viewed (e.g., as indicated by the user-profile associated with the occurrence), the focus value of the occurrence, and/or other factors.

As a specific example, the ad-value contribution for each occurrence of the ad space may be a dollar amount that is attributed to the occurrence. As such, the server may determine a dollar amount for the ad value by summing the ad-value contributions. As another example, the ad-value contribution for each occurrence of the ad space may be a price rate (e.g., dollars per month, dollars per view, etc.) that is attributed to a respective occurrence of the ad space. As

such, the server may determine the ad value by summing the ad-value contributions to get an overall price rate. Other examples are also possible.

Once a server system has determined the individual ad-value contributions for a number of occurrences of the particular ad space, the server may use various techniques to determine the ad value for the ad space. For example, in some embodiments, the server may determine an average advertising-value contribution by averaging the advertising-value contributions of some or all of the detected occurrences. The server may then use the average advertising-value contribution as a basis for determining the advertisement value for the ad space. As a specific example, the server may determine an ad-value contribution for each occurrence in the same manner as the overall price or rate for the ad space, but using the assumption that all occurrences of the ad space are identical to the occurrence. The server may then determine a dollar amount or price rate for the ad space by averaging the ad-value contributions determined in this manner. Other examples are also possible.

It should be understood that techniques described herein for determining an ad value based on the ad-value contributions are not intended to be limiting. Other techniques for determining an ad value based on the ad-value contributions of individual occurrences are also possible, without departing from the scope of the invention.

F. Valuation of an Ad Space on a Per-Advertisement Basis

Some embodiments may involve determining a value an ad space that is specific to a particular type of advertisement. For example, an ad server may determine a value for an ad space when the ad space is used to display an ad for a particular type of product (e.g., for clothing or for a movie).

Further, in some embodiments, an exemplary method may be implemented for ad-specific valuation of an ad space based on the extent to which the ad space reaches the target market of the advertisement. For instance, wearer-view data may be used to determine who is viewing an ad space. The ad space valuation may then be based on the correlation between those who view the ad space and the target market of the specific advertisement.

In such an embodiment, an exemplary method may utilize wearer-view data indicating user-profiles associated with occurrences of an ad space in the gaze data. As such, the server may analyze the associated user-profiles to determine one or more characteristics of those who have viewed the ad space. More specifically, an exemplary method may involve determining a group of user-profiles associated with the advertisement space (e.g., user-profiles that are associated with wearable computing devices that captured the ad space in their respective gaze data). Then, based on characteristics of the associated user-profiles, the server may determine one or more viewer characteristics of the group. The viewer characteristics of the group may be interpreted as indicative of a "standard" viewer of the ad space. As such, the viewer characteristics of the group may incorporate when determining the advertisement value for a specific type advertisement.

For example, some embodiments may involve determining both: (a) the viewer characteristics of the group of associated user-profiles and (b) one or more target-viewer characteristics for the particular advertisement. The server may then compare the viewer characteristics of the group to the target-viewer characteristics and, based on the comparison, determine the advertisement value for the particular advertisement in the advertisement space.

In some embodiments, the ad value for the particular advertisement may be further based on the location of the ad

space. In particular, there may be a relationship between the characteristics of a particular advertisement and the location of ad space, and an exemplary method may help to account for such a relationship. In such cases, a weighting factor may be applied to increase or decrease the ad value depending upon the relationship between the location of the ad space and the characteristics of the advertisement.

For example, consider an advertisement for a clothing product and an ad space that is located near to a shopping area and/or near to a store where the clothing product can be purchased. This ad space may generally be considered more valuable when used to display the advertisement for the clothing product than when used to display an ad for a type of product that cannot be purchased nearby. Accordingly, a weighting factor may be applied to increase the ad value for the clothing product in the ad space. Similarly, the weighting factor may function to decrease the ad value for a product that cannot be purchased nearby.

As another example, consider an advertisement for a movie and an ad space that is located near to a movie theater that is showing the movie. This ad space may generally be considered more valuable when used to display the advertisement for the movie than when used to display an ad for a movie that is not in any nearby theaters. Accordingly, a weighting factor may be applied to increase the ad value of the ad space for a movie that is showing in the nearby theater. Similarly, the weighting factor may decrease the ad value for the movie that is not in any nearby theaters. Other examples are also possible.

In a further aspect, some implementations of method 500 may utilize the type of advertisement as the characteristic of the advertisement upon which the ad value is based. In such an embodiment, all advertisements of the same type may be evaluated in the same way. As such, the ad value in such an embodiment may in effect be determined for the type of advertisement in the ad space (rather than specifically for an individual advertisement). Alternatively, the type of advertisement may be one of a number of factors, such that an ad space may be valued differently for different advertisements that are classified as being the same type of advertisement.

G. Adjusting the Ad Value Based on Other Factors

In some embodiments, ad valuation may be based on other types of data, in addition to wearer-view data. In such an embodiment, the ad server may determine a base value for the advertisement, or a weighting to be applied to the ad value based on an intrinsic value of the ad space, which accounts for the characteristics of the ad space itself, and then adjust the intrinsic value according to the wearer-view data.

For example, in some embodiments, an exemplary method may use the geographic location of the advertisement space as a further basis for determining the advertisement value for the advertisement space. For example, an advertisement that is located in a shopping area may have a greater intrinsic value than one that is located in an alley. Accordingly, an ad value that is determined based on wearer-view data may be adjusted based on the location of the ad space. Other examples are also possible. Generally, the type and/or the amount of adjustment that is applied due to the location of an ad space may vary depending upon the particular implementation.

Further, in some embodiments, the server may consider the type of advertisement space and/or the format in which advertisements can be displayed in the ad space when determining the advertisement value for the advertisement space. For example, an LCD billboard might generally be considered more valuable than an equivalent print billboard.

As such, when an ad value is determined for a billboard based on wearer-view data, the determined ad value may be adjusted based on whether the billboard is an LCD or a print billboard. Other examples are also possible. Generally, the type and/or the amount of adjustment that is applied based on the type of advertisement space may vary depending upon the particular implementation. Further, other adjustments, based on other characteristics of an ad space, are also possible.

In another aspect, an ad space may be blank (e.g., not displaying an advertisement) during some or all of the period in which gaze data is being collected for purposes of determining the ad value. The fact that an ad space is blank, as opposed to displaying an advertisement, may affect the gaze data for the ad space because a blank space might attract less attention from nearby people. Further, different advertisements may attract different amounts of attention. Therefore, when an advertisement is displayed while gaze data is being collected, the particular advertisement may itself affect the gaze data. As such, it is possible that wearer-view data for an ad space may be effected by whether or not an ad space is blank, and if something is displayed in the ad space, what specifically is displayed.

Accordingly, an exemplary method may further involve determining a pre-sale weighting factor for an advertisement space, which is based on: (a) whether the ad space is blank while gaze data is being collected and/or (b) the characteristics of what is displayed in the ad space while gaze data is being collected. A server may then use the pre-sale weighting factor for the ad space as a further basis for determining the advertisement value for the advertisement space.

As a specific example, an exemplary method may further the server determining whether or not the advertisement space had an advertisement in place while receiving and analyzing the gaze data. Then, if the advertisement space had an advertisement in place, the server may apply a first adjustment to the wearer-view data before using the wearer-view data to determine the advertisement value (e.g., an adjustment or weighting factor that corresponds to the particular advertisement that is displayed). On the other hand, if an advertisement was not displayed in the advertisement space, then the server may apply a second adjustment to the wearer-view data (e.g., an adjustment that accounts for the fact that no advertisement was displayed).

In such an embodiment, the server may determine whether or not the advertisement space had an advertisement in place in various ways. For example, the server may query an ad space database to determine whether the ad space is in use and if so, what advertisement is being displayed. Additionally or alternatively, the server may analyze the gaze data itself (e.g., by analyzing point-of-view video in which the ad space is detected). Other examples are also possible.

In yet another aspect, some embodiments may implement gaze-data requirements that require a certain amount of gaze data be analyzed before an ad space can be offered for sale in an ad-marketplace system. For instance, an ad-marketplace system may require that gaze data from a threshold number of devices have been analyzed before the determined ad value is deemed accurate enough to offer the ad space for sale via the marketplace. Additionally or alternatively, ad-marketplace system may require that gaze data be monitored for a certain period of time (e.g., at least a week) before the determined ad value is deemed accurate enough to offer the ad space for sale.

Further, in some embodiments, wearer-view data requirements may require that a certain amount of wearer-view data be generated before an ad space can be offered for sale in an

ad-marketplace system. For example, an ad-marketplace system may require that a certain number of occurrences of an ad space be detected before the determined ad value is deemed accurate enough to offer the ad space for sale via the marketplace (or in other words, require that the wearer-view data takes into account at least a threshold number of occurrences). Other examples are also possible.

VIII. Use of Supplemental Gaze Data from Non-Wearable Computing Devices

In some embodiments, an exemplary method may incorporate supplemental gaze data from one or more non-wearable computing devices. In such an embodiment, the supplemental gaze data may include media captured by the non-wearable computing devices. For example, supplemental gaze data may be received from mobile phones, tablet computers, network-enabled video and/or still cameras, and/or other non-wearable computing devices.

Similar to the gaze data from wearable computing devices, the supplemental gaze data is generally indicative of a respective user-view associated with a device that provides supplemental gaze data. Accordingly, an exemplary method may further involve receiving supplemental gaze data from one or more non-wearable computing devices that are registered to a given user-account. A server may then detect additional occurrences of advertisement spaces in the supplemental gaze data, and factor the additional occurrences in when determining the gaze value for the user-account.

However, because supplemental gaze data is captured at non-wearable devices, supplemental gaze data may less reliably represent what the user actually sees, as compared to gaze data captured by a wearable device that is physically worn on the user's person. Accordingly, in an exemplary method, the server may weight supplemental occurrences that are detected in the supplemental gaze data in order to account for the increased probability that the supplemental gaze data does not represent what the user actually sees. For example, the server may weight a supplemental occurrence by a significance factor corresponding to the likelihood that the corresponding supplemental gaze data is indicative of a respective user-view associated with the non-wearable computing device that provided the supplemental gaze data in which the supplemental occurrence was detected.

In a further aspect, systems may be implemented to actively search supplemental gaze data that is pre-recorded, such as image libraries that are accessible via a network. For example, a server or an associated system may be configured to analyze images from one or more online image albums to determine supplemental user-view data for the advertisement space. In such an embodiment, the supplemental user-view data is based on occurrences of the advertisement space in the plurality of images.

For example, the system may search image albums on a photo-sharing website, social-network website, or another network source, for occurrences of the ad space in the images. When an occurrence is found, the system may generate supplemental user-view data for the occurrence. For instance, many such websites require that users open a user-account in order to create photo albums and/or share photos. Accordingly, the system may store data linking the occurrence of the ad space to the user-account via which the image was shared.

In a further aspect, one image of an ad space may be indicative of a more valuable view of the ad space than another image. As such, each image that includes the ad

space may be evaluated for indications of how significant the occurrence is, so that the occurrence may be weighted accordingly when determining the ad value.

For example, an exemplary method may involve analyzing one or more images from one or more image albums to detect any occurrences of the advertisement space in the images. Then, for each image where an advertisement space is detected, the system may determine a gaze-value contribution and/or an advertising-value contribution for the given image (note that in some instances, the advertising-value contribution may be used as the gaze-value contribution). As a specific example, determining a prominence value corresponding to a prominence of the advertisement space in the given image (e.g., a size and/or location of the ad space in the image), and then use the prominence value as a basis for determining a gaze-value contribution and/or an advertising-value contribution for the given image. The system may then use any gaze-value contributions from these images, such as any prominence values that are determined for any of the images, when determining the gaze value for the associated user-account. Similarly, the system may use any advertising-value contributions from these images, such as any prominence values that are determined for any of the images, when determining the advertisement value for the advertisement space.

In a further aspect, data from GPS systems and other sensors, such as magnetometers, accelerometers, and/or gyroscopes may provide supplemental gaze data. In some embodiments, GPS on a wearable computing device or another device (e.g., a mobile phone or tablet) may provide an indication of location, and a magnetometer on the same device may provide an indication of orientation, such that it may be inferred that a user of the device is in a certain location and is facing a certain direction. Further, the location of an ad space may also be determined as described herein. Thus, if the user is inferred to be facing a location where and space is located, this may be considered a view of the ad space, and thus may be factored into the wearer-view data. Other examples of using GPS and/or sensor data to infer supplemental gaze data are also possible.

IX. Wearable-Computer-Based Functionality

The above-described methods and systems are generally described with reference to examples where wearable computing devices collect and send gaze data to an ad-valuation server system, such that the server system provides most of the functionality as far as detecting ad spaces in gaze data, determining individual gaze values, and/or determining ad values for ad spaces. This arrangement may be referred to as a “cloud-based” embodiment. However, it should be understood that wearable-computer-based embodiments and partially cloud-based embodiments are also possible. Thus, it should be understood that some or all of the functionality that is described herein as being carried out by a server may alternatively be carried out at one or more wearable computing devices.

For example, FIG. 6 is a flow chart illustrating a method that may be carried out at a wearable computing device, according to an exemplary embodiment. As shown by block 602, method 600 involves a wearable computing device receiving gaze data that is indicative of a wearer-view associated with the wearable computing device. At block 604, the wearable computing device analyzes the gaze data to detect occurrences of advertisement spaces in the gaze data. Based at least in part on the detected advertisement-space occurrences, the wearable computing device may

determine an individual gaze value for a user-account that is associated with the wearable computing device, as shown by block 606.

The wearable computing device may then send the individual gaze value for display, as shown by block 608. For example, the wearable computing device may simply send the gaze value for display in its own display (e.g., in an HMD), or may send the gaze value to another computing device of the wearer (e.g., a mobile phone, tablet, or laptop computer). Other examples are also possible.

The function of analyzing gaze data for occurrences of advertisement spaces may be accomplished in various ways. In some embodiments, the wearable computing device may monitor gaze data as it is generated, so that occurrences of ad spaces can be detected in real-time. In other embodiments, the wearable computing device may store gaze data and then later search for occurrences of advertisement spaces in the gaze data. In yet other embodiments, a wearable computing device may implement a combination of real-time analysis and after-the-fact analysis of stored gaze data. For instance, the wearable computing device may search for some ad spaces in real-time, as well as storing some or all gaze data so that, if needed, the wearable computing device can search for other ad spaces at a later time. Other techniques are also possible.

In a further aspect, the processing and/or storage capabilities of individual wearable computing devices may be limited as compared to a server system. As such, when a wearable computing device stores gaze data for analysis at a later time, the wearable computing device may take various actions to reduce the size of the gaze data before storing the gaze data. For instance, rather than store full-size point-of-view video that is captured at the wearable computing device, the wearable computing device may periodically generate and store screenshots from the point-of-view video. Alternatively, the wearable computing device may apply compression, and/or may reduce the resolution and/or frame-rate of the POV video, before storing the video. As another alternative, the wearable computing device may implement real-time analysis of the gaze data for occurrences of ad spaces, and also send the gaze data to the server to be stored in the event further analysis is needed at a later time.

In a variation on method 600, the wearable computing device may notify the server each time it detects an occurrence of ad space in its gaze data, so that the server can use the detected occurrence when determining a gaze value for the user-account associated with the wearable computing device. In such an embodiment, the server may determine a gaze-value contribution of each occurrence that is indicated by the wearable computing device, and use all or a subset of these gaze-value contributions to determine the individual gaze value for the user-account.

In a further aspect, when a wearable computing device detects an ad space in its gaze data, the wearable computing device may determine whether it is being worn when the ad space is detected. The wearable computing device may then adjust or change the manner in which such an occurrence of an ad space is used to determine the individual gaze value, in the event the wearable computing device is not being worn at the time of the detected occurrence. For example, if the wearable computing device is not being worn when an ad space is detected, this may be interpreted to mean that the gaze data that included the ad space is unlikely to represent what the wearer is actually viewing. Accordingly, the wearable computing device may reduce the significance of the occurrence when determining the individual gaze value for

the occurrence, or may ignore the occurrence entirely (e.g., by refraining from generating any wearer-view data based on the occurrence). Alternatively, in an embodiment where the wearable computing device notifies the server system of detected occurrences, the wearable computing device may notify the server system that the wearable computing device was not being worn when the particular occurrence was detected, or may simply refrain from notifying the server of ad-space occurrence while the wearable computing device is not being worn.

Further, in embodiments where the wearable computing device notifies the server of detected ad-space occurrences, the wearable computing device may notify the server when the wearable computing device is not being worn during an occurrence. Alternatively, in such an embodiment, the wearable computing device may simply refrain from notifying the server about the particular occurrence and/or may refrain from sending wearer-view data to the server that is based on the particular occurrence.

Yet further, the wearable computing device may use various techniques to determine whether or not the wearable computing device is being worn. For example, the wearable computing device may use various sensors, such as accelerometers, gyroscopes, and/or compasses, to determine whether the position and/or motions of the wearable computing device are characteristic of the wearable computing device being worn. Additionally or alternatively, the wearable computing device may use various sensors to determine whether the wearable computing device is in contact with a wearer's skin or positioned on the wearer's face. More generally, the wearable computing device may use any technique that is now known or later developed to determine whether the wearable computing device is being worn.

X. Exemplary Wearable Computing Systems

FIG. 7A illustrates a wearable computing system according to an exemplary embodiment. In FIG. 7A, the wearable computing system takes the form of a head-mounted device (HMD) 702 (which may also be referred to as a head-mounted display or a heads-up display (HUD)). It should be understood, however, that exemplary systems and devices may take the form of or be implemented within or in association with other types of devices, without departing from the scope of the invention. As illustrated in FIG. 7A, the head-mounted device 702 comprises frame elements including lens-frames 704, 706 and a center frame support 708, lens elements 710, 712, and extending side-arms 714, 716. The center frame support 708 and the extending side-arms 714, 716 are configured to secure the head-mounted device 702 to a user's face via a user's nose and ears, respectively.

Each of the frame elements 704, 706, and 708 and the extending side-arms 714, 716 may be formed of a solid structure of plastic and/or metal, or may be formed of a hollow structure of similar material so as to allow wiring and component interconnects to be internally routed through the head-mounted device 702. Other materials may be possible as well.

One or more of each of the lens elements 710, 712 may be formed of any material that can suitably display a projected image or graphic. Each of the lens elements 710, 712 may also be sufficiently transparent to allow a user to see through the lens element. Combining these two features of the lens elements may facilitate an augmented reality or heads-up display where the projected image or graphic is

superimposed over a real-world view as perceived by the user through the lens elements.

The extending side-arms 714, 716 may each be projections that extend away from the lens-frames 704, 706, respectively, and may be positioned behind a user's ears to secure the head-mounted device 702 to the user. The extending side-arms 714, 716 may further secure the head-mounted device 702 to the user by extending around a rear portion of the user's head. Additionally or alternatively, for example, the HMD 702 may connect to or be affixed within a head-mounted helmet structure. Other possibilities exist as well.

The HMD 702 may also include an on-board computing system 718, a video camera 720, a sensor 722, and a finger-operable touch pad 724. The on-board computing system 718 is shown to be positioned on the extending side-arm 714 of the head-mounted device 702; however, the on-board computing system 718 may be provided on other parts of the head-mounted device 702 or may be positioned remote from the head-mounted device 702 (e.g., the on-board computing system 718 could be wire- or wirelessly-connected to the head-mounted device 702). The on-board computing system 718 may include a processor and memory, for example. The on-board computing system 718 may be configured to receive and analyze data from the video camera 720 and the finger-operable touch pad 724 (and possibly from other sensory devices, user interfaces, or both) and generate images for output by the lens elements 710 and 712.

The video camera 720 is shown positioned on the extending side-arm 714 of the head-mounted device 702; however, the video camera 720 may be provided on other parts of the head-mounted device 702. The video camera 720 may be configured to capture images at various resolutions or at different frame rates. Many video cameras with a small form-factor, such as those used in cell phones or webcams, for example, may be incorporated into an example of the HMD 702.

Further, although FIG. 7A illustrates one video camera 720, more video cameras may be used, and each may be configured to capture the same view, or to capture different views. For example, the video camera 720 may be forward- or outward-facing to capture at least a portion of the real-world view perceived by the user. This forward facing image captured by the video camera 720 may then be used to generate an augmented reality where computer generated images appear to interact with the real-world view perceived by the user.

The sensor 722 is shown on the extending side-arm 716 of the head-mounted device 702; however, the sensor 722 may be positioned on other parts of the head-mounted device 702. The sensor 722 may include one or more of a gyroscope or an accelerometer, for example. Other sensing devices may be included within, or in addition to, the sensor 722 or other sensing functions may be performed by the sensor 722.

The finger-operable touch pad 724 is shown on the extending side-arm 714 of the head-mounted device 702. However, the finger-operable touch pad 724 may be positioned on other parts of the head-mounted device 702. Also, more than one finger-operable touch pad may be present on the head-mounted device 702. The finger-operable touch pad 724 may be used by a user to input commands. The finger-operable touch pad 724 may sense at least one of a position and a movement of a finger via capacitive sensing, resistance sensing, or a surface acoustic wave process, among other possibilities. The finger-operable touch pad 724

may be capable of sensing finger movement in a direction parallel or planar to the pad surface, in a direction normal to the pad surface, or both, and may also be capable of sensing a level of pressure applied to the pad surface. The finger-operable touch pad **724** may be formed of one or more translucent or transparent insulating layers and one or more translucent or transparent conducting layers. Edges of the finger-operable touch pad **724** may be formed to have a raised, indented, or roughened surface, so as to provide tactile feedback to a user when the user's finger reaches the edge, or other area, of the finger-operable touch pad **724**. If more than one finger-operable touch pad is present, each finger-operable touch pad may be operated independently, and may provide a different function.

FIG. **7B** illustrates an alternate view of the wearable computing device illustrated in FIG. **7A**. As shown in FIG. **7B**, the lens elements **710**, **712** may act as display elements. The head-mounted device **702** may include a first projector **728** coupled to an inside surface of the extending side-arm **716** and configured to project a display **730** onto an inside surface of the lens element **712**. Additionally or alternatively, a second projector **732** may be coupled to an inside surface of the extending side-arm **714** and configured to project a display **734** onto an inside surface of the lens element **710**.

The lens elements **710**, **712** may act as a combiner in a light projection system and may include a coating that reflects the light projected onto them from the projectors **728**, **732**. In some embodiments, a reflective coating may not be used (e.g., when the projectors **728**, **732** are scanning laser devices).

In alternative embodiments, other types of display elements may also be used. For example, the lens elements **710**, **712** themselves may include: a transparent or semi-transparent matrix display, such as an electroluminescent display or a liquid crystal display, one or more waveguides for delivering an image to the user's eyes, or other optical elements capable of delivering an in focus near-to-eye image to the user. A corresponding display driver may be disposed within the frame elements **704**, **706** for driving such a matrix display. Alternatively or additionally, a laser or LED source and scanning system could be used to draw a raster display directly onto the retina of one or more of the user's eyes. Other possibilities exist as well.

FIG. **8A** illustrates another wearable computing system according to an exemplary embodiment, which takes the form of an HMD **802**. The HMD **802** may include frame elements and side-arms such as those described with respect to FIGS. **7A** and **7B**. The HMD **802** may additionally include an on-board computing system **804** and a video camera **806**, such as those described with respect to FIGS. **7A** and **7B**. The video camera **806** is shown mounted on a frame of the HMD **802**. However, the video camera **806** may be mounted at other positions as well.

As shown in FIG. **8A**, the HMD **802** may include a single display **808** which may be coupled to the device. The display **808** may be formed on one of the lens elements of the HMD **802**, such as a lens element described with respect to FIGS. **7A** and **7B**, and may be configured to overlay computer-generated graphics in the user's view of the physical world. The display **808** is shown to be provided in a center of a lens of the HMD **802**, however, the display **808** may be provided in other positions. The display **808** is controllable via the computing system **804** that is coupled to the display **808** via an optical waveguide **810**.

FIG. **8B** illustrates another wearable computing system according to an exemplary embodiment, which takes the

form of an HMD **822**. The HMD **822** may include side-arms **823**, a center frame support **824**, and a bridge portion with nosepiece **825**. In the example shown in FIG. **8B**, the center frame support **824** connects the side-arms **823**. The HMD **822** does not include lens-frames containing lens elements. The HMD **822** may additionally include an on-board computing system **826** and a video camera **828**, such as those described with respect to FIGS. **7A** and **7B**.

The HMD **822** may include a single lens element **830** that may be coupled to one of the side-arms **823** or the center frame support **824**. The lens element **830** may include a display such as the display described with reference to FIGS. **7A** and **7B**, and may be configured to overlay computer-generated graphics upon the user's view of the physical world. In one example, the single lens element **830** may be coupled to the inner side (i.e., the side exposed to a portion of a user's head when worn by the user) of the extending side-arm **823**. The single lens element **830** may be positioned in front of or proximate to a user's eye when the HMD **822** is worn by a user. For example, the single lens element **830** may be positioned below the center frame support **824**, as shown in FIG. **8B**.

FIG. **9** illustrates a schematic drawing of a wearable computing device according to an exemplary embodiment. In system **900**, a device **910** communicates using a communication link **920** (e.g., a wired or wireless connection) to a remote device **930**. The device **910** may be any type of device that can receive data and display information corresponding to or associated with the data. For example, the device **910** may be a heads-up display system, such as the head-mounted devices **102**, **802**, or **822** described with reference to FIGS. **7A-9**.

Thus, the device **910** may include a display system **912** comprising a processor **914** and a display **916**. The display **910** may be, for example, an optical see-through display, an optical see-around display, or a video see-through display. The processor **914** may receive data from the remote device **930**, and configure the data for display on the display **916**. The processor **914** may be any type of processor, such as a micro-processor or a digital signal processor, for example.

The device **910** may further include on-board data storage, such as memory **918** coupled to the processor **914**. The memory **918** may store software that can be accessed and executed by the processor **914**, for example.

The remote device **930** may be any type of computing device or transmitter including a laptop computer, a mobile telephone, or tablet computing device, etc., that is configured to transmit data to the device **910**. The remote device **930** and the device **910** may contain hardware to enable the communication link **920**, such as processors, transmitters, receivers, antennas, etc.

In FIG. **9**, the communication link **920** is illustrated as a wireless connection; however, wired connections may also be used. For example, the communication link **920** may be a wired serial bus such as a universal serial bus or a parallel bus. A wired connection may be a proprietary connection as well. The communication link **920** may also be a wireless connection using, e.g., Bluetooth® radio technology, communication protocols described in IEEE 802.11 (including any IEEE 802.11 revisions), Cellular technology (such as GSM, CDMA, UMTS, EV-DO, WiMAX, or LTE), or Zigbee® technology, among other possibilities. The remote device **930** may be accessible via the Internet and may include a computing cluster associated with a particular web service (e.g., social-networking, photo sharing, address book, etc.).

It should be understood that for situations in which the embodiments discussed herein collect and/or use any personal information about users or information that might relate to personal information of users, the users may be provided with an opportunity to opt in/out of programs or features that involve such personal information (e.g., information about a user's preferences or a user's contributions to social content providers). In addition, certain data may be anonymized in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be anonymized so that no personally identifiable information can be determined for the user and so that any identified user preferences or user interactions are generalized (for example, generalized based on user demographics) rather than associated with a particular user.

XI. Listing Suggestions for Gaze-Based Advertisement Marketplace

FIG. 10 is a flow chart illustrating a method according to an exemplary embodiment. The method 1000 shown in FIG. 10 is described by way of example as being carried out by a server system in order to provide advertisement-marketplace functionality. However, it should be understood that exemplary methods, such as method 1000, may be carried out by other systems or combinations of systems, without departing from the scope of the invention.

Method 1000 involves the server system receiving a listing request that indicates an offer to sell advertisement rights to an advertisement space, where the listing request corresponds to a first user-account, as shown by block 1002. The server system also determines an advertisement value for the advertisement space, as shown by block 1004. Further, in response to the listing request, the server system updates an advertising-space database with a listing for the advertisement space, where the listing is at a listing price that is based on the determined advertisement value, as shown by block 1006. The server system may also make the listing for the advertisement space available via a network-based advertisement marketplace, as shown by block 1008.

In an exemplary embodiment, the advertisement value that is determined at block 1004 is based on detected occurrences of the advertisement space in gaze data that is received from a plurality of wearable computing devices. Further, the gaze data from each wearable computing device is indicative of the respective wearer-view associated with the wearable computing device. For example, the gaze data from each wearable computing device may take the form of point-of-view video that is captured at the wearable computing device. As such, the gaze data upon which an advertisement value is based may include a number of point-of-view videos (e.g., a respective point-of-view video from each of the wearable computing devices).

Gaze data may additionally or alternatively take forms other than point-of-view video. For example, the gaze data from a given wearable computing device may take the form of point-of-view images captured by a forward- or outward-facing camera on the wearable computing device. As a specific example, a given wearable computing device may periodically take a picture, and then send the picture to the server system for use in generating wearer view data. To do so, the wearable computing device may analyze point-of-view video for one or more advertisement spaces, and generate a screen capture of the video when and advertise-

ment space detected. The wearable computing device may then send the screen capture to the server system. Other examples are also possible.

Since the gaze data from a given wearable computing device is generally indicative of the wearer-view of the wearable computing device's wearer, the gaze data is generally indicative of what the wearer of the device is actually looking at. Further, since the wearer-view data is based on the gaze data, the wearer-view data is indicative of actual views of the advertisement space by wearers. For instance, the wearer-view data may provide an indication of how many people are looking at a particular advertisement space, which people are actually looking at a particular advertisement space, when people are looking at a particular advertisement space, and/or how long people are actually looking at a particular advertisement space, among other information. As such, the wearer-view data may help to more accurately determine what an advertising space is worth.

Since the gaze data from a given wearable computing device is generally indicative of the wearer-view of the wearable computing device's wearer, the gaze data is generally indicative of what the wearer of the device is actually looking at. Further, since the wearer-view data is based on the gaze data, the wearer-view data is indicative of actual views of the advertisement space by wearers. For instance, the wearer-view data may provide an indication of how many people are looking at a particular advertisement space, which people are actually looking at a particular advertisement space, when people are looking at a particular advertisement space, and/or how long people are actually looking at a particular advertisement space, among other information. As such, the wearer-view data may help to more accurately determine what an advertising space is worth. Methods and systems for determining an advertisement value using gaze data will be described in greater detail with reference to FIGS. 17A, 17B, and 18.

In some cases, the advertisement value for the advertisement space may have already been calculated when the listing request is received at block 1002, using methods such as those described in reference to FIGS. 17A, 17B, and 18. In this case, block 1004 may simply involve querying an ad-value database to retrieve the predetermined advertisement value.

In other cases, the advertisement value may not yet have been calculated when the listing request is received at block 1002. In such cases, block 1004 may involve the server system responsively analyzing gaze data in order to determine the advertisement value, using methods such as those described in reference to FIGS. 17A, 17B, and 18. Note that in some of these cases, analyzing the gaze data may take a considerable amount of time (e.g., more time than a typical user will want to wait, and possibly days, weeks, or even longer). As such, the server system may indicate that time is needed to determine the advertisement value so that the user can choose whether or not to wait for the advertisement value to be determined. If the user elects not to wait, then the server system may notify the user once the advertisement value is determined (e.g., via an indication sent to the user's user-account).

Further, in some cases, even when the advertisement value has been calculated before receipt of the listing request at block 1002, the server system may still analyze gaze data at block 1004 in order to update the predetermined advertisement value. This may be useful in the event it has a significant amount of time since the advertisement value was determined (e.g., long enough that the advertisement value is likely to have changed based on gaze data received in the

interim). This may be the default procedure, or may be triggered when a certain amount of time has passed since the advertisement value was determined (e.g., long enough that the advertisement value is likely to have changed based on gaze data received in the interim). In the latter case, the amount of time that triggers an update to the advertisement value may be set depending upon the goals of the particular implementation.

XII. Exemplary Terms and Conditions of Sale in an Advertisement Marketplace

In an exemplary method, such as method 1000, a listing request may take various forms and/or include various different types of information, depending upon the embodiment. For example, a listing request may include data that: (a) identifies and/or defines the advertisement space to be listed, (b) identifies the user-account to be associated with the listing, (c) data demonstrating that the user-account is authorized to list the advertisement space, (d) data defining the terms and conditions under which the advertisement rights are being offered, and/or (c) other types of data related to the requested listing and/or the user account.

In a further aspect, an exemplary system may allow for certain conditions to be placed on a listing. Accordingly, an exemplary method may further involve receiving, in conjunction with the listing request, one or more condition indications which each indicate a condition of the advertisement rights offered in the listing for the advertisement space. Such condition indications may be included in the listing request, or received separately from the listing request. Examples of conditions that may be specified may include: (a) a time period for which the advertisement rights will be provided, (b) a portion of the advertisement space to be provided, (c) one or more timeshare conditions, and (d) an advertisement-type restriction for the advertisement space. Others types of conditions are also possible.

Furthermore, an exemplary system may allow a listing to be created for a certain type of contract or agreement. Functionally, the contract may specify a set of conditions that define the advertisement rights that will be provided when an advertiser purchases the advertisement rights. For example, a listing may specify a contract for:

- (a) A sale transferring ownership of the advertisement space from the first user-account to the second user-account.
- (b) An exclusive lease of the advertisement space (e.g., an agreement conveying all advertisement rights for the term of the lease, such as a day, month, year, and so on).
- (c) A partial exclusive lease of the advertisement space (e.g., an agreement conveying all advertisement rights for a portion or portions of the term of the lease, such as between 4:00 pm and 6:00 pm every day during a month-long lease).
- (d) A timeshare lease of the advertisement space (e.g., an agreement to conveying the right to have an advertisement displayed in a rotation with other advertisements, without any specific time period being set aside for a given one of the advertisements in the rotation).

Other types of contracts or agreements are also possible. Further, in some embodiments, an advertisement marketplace may provide pre-defined contracts, which can be selected by a user for a listing. Additionally or alternatively, an advertisement marketplace may allow a user to create a custom contract by specifying the terms and conditions desired by the user. Further, in some embodiments, an advertisement marketplace may allow a user to create a

custom contract by modifying a pre-defined contract. In a further aspect, an exemplary system may be configured to facilitate a transaction for the advertisement rights to an advertisement space that was listed using a method such as method 100. For example, an exemplary system may be configured to receive a purchase request from a second user-account, which indicates a desire to purchase a listing from a first user-account. In response to such a request, an exemplary system may facilitate a transaction for the advertisement rights to the advertisement space by, e.g., creating a contract for the advertisement rights as specified in listing, between the first user-account and the second user-account.

XIII. Exemplary Pricing in an Advertisement Marketplace

In an exemplary advertisement marketplace, various types of pricing may be implemented. In some embodiments, all advertisement spaces may be listed with the same pricing structure. In other embodiments, different advertisement spaces may be listed with different pricing structures. Further, in some embodiments, a single listing for an advertisement space may offer multiple types of pricing structures, so that a purchaser of the advertisement space can select the pricing structure that fits their needs.

Examples of possible pricing structures include: (a) various types of fixed-rate pricing, (b) various types of variable-rate pricing, and/or (c) various types of auction-based pricing. It should be understood that other pricing structures may additionally or alternatively be incorporated, without departing from the scope of the invention.

In a further aspect, in order to help provide more accurate pricing for advertisement spaces, an exemplary system may extrapolate from the occurrences of an ad-space that are detected in gaze data to estimate how many views occurred in the viewing population as a whole. The advertisement value and/or the price based on the advertisement value may therefore take into account the estimated number of views by the viewing population. This may be useful as there may be many cases where those that view an advertisement space are not wearing a wearable computer that is configured to provide gaze data, and/or where a wearable computer does not capture an advertisement that is viewed by its wearer. As one specific example, consider the case where it is assumed that one out of every thousand views will be captured in gaze data. In this case, if the value of ten occurrences that are detected in gaze data during a one-month period is determined to be \$100, the server system may estimate are 10,000 views that are worth \$10 each. As such, the advertisement value based on all views may be determined to be \$100,000 per month. Other examples are also possible.

Note that in some instances, the extrapolation from the occurrences in gaze data may assume that the users who provide gaze data are generally representative of the viewing population as a whole. However, it is also possible that the users who provide gaze data (e.g., those who own wearable computers and opt in to a program to share gaze data) may not be representative of the viewing population as a whole. In this case, an exemplary system may be configured to account for such differences when extrapolating from the gaze data. The particular techniques used to determine a total number of ad-space views based on the occurrences detected in gaze data may vary from application to application, depending on the characteristics of the viewing population, the demographics of those who provide gaze data, and/or other factors.

In a further aspect, demographic biases created by differences between wearable-computing-device wearers and the population as a whole (or a certain subset of the population) may also be accounted for when calculating individual ad-value contributions upon which the advertisement value is ultimately based. For example, consider a town or another geographic region where an “over-55” age group is known to make up 60% of the viewing population as a whole, but only makes up 12% of the wearable-computing-device users in that area (as indicated by registered wearers of the given age group and/or observed occurrences of an ad space in gaze data from wearers in the age group). Accordingly, when determining the value for an advertisement space in this town, the data from the 12% of wearers who are in the over-55 age group may be given a 60% weight, while the data from the other 88% of wearers from which gaze data is received may be given a 40% weight. Many other examples are also possible.

In yet another aspect, users who are selling advertisement rights to their ad spaces may be provided options to control and/or affect what types of advertisements are displayed in their ad space and/or the terms under which certain advertisers can advertise in their advertisement space. For example, a certain user might not want a certain type of advertisement in their advertisement space for moral or ethical reasons, or for any other reason, and might specify that there advertisement space cannot be used for this type of advertisement, or might indicate that the listing price and amount charge should be higher for this type of advertisement. For example, a user who does not like the idea of an advertisement for an alcoholic beverage in their advertisement space, may specify that such an advertisement should cost three times the standard rate (e.g., the relative advertisement value) for their ad space. As another example, a user who is a fan of a particular electronics company may want an advertisement from the company in their ad space, and therefore may specify that this company should be charged half of the standard rate for their advertisement space. Many other examples are also possible.

A. Fixed-Rate Pricing

As noted, fixed-rate pricing may be used for advertisement spaces in an exemplary advertisement marketplace. For example, in some instances, the server system may simply set the listing price for advertisement rights at the gaze-data based advertisement value. According, when a method such as that described in reference to FIGS. 17A, 17B, and 18 is used to determine the advertisement value for an advertisement space, performing the method may effectively determine the listing price at which the advertisement space can be listed.

In some cases, the price at which an advertisement space can be listed may differ from the determined advertisement value. However, the listing price may still be set to a fixed value or fixed rate that is based on the advertisement value for the advertisement space. In particular, the rate for a fixed-price listing may be adjusted according to the terms and conditions specified by the listing. For example, if an advertisement value is determined in terms of dollars per month, and a listing request specifies that the advertisement space can be purchased on a day-to-day basis, the system may calculate a daily rate that is based at least on part on the monthly advertisement value.

As another example, an advertisement value may be determined with the assumption of exclusive use for a certain term. In this case, if the listing specifies that the advertisement space can be purchased on a timeshare basis (e.g., placing an advertisement in a rotation with other

advertisements), the rate at which the timeshare rights to the advertisement space can be listed may be derived from the advertisement value for exclusive use. For example, the advertisement value may be divided by the number of shares to determine the rate for one share of the timeshare. For instance, consider an advertisement space for which exclusive use is valued at \$30.00 per month. If the owner wishes to offer this advertisement space as a timeshare to be split between ten advertisers, each share may be listed at \$3.00 per month. Other techniques for determining a timeshare price based on an advertisement value that assumes exclusive use are also possible.

More generally, many different techniques may be implemented to account for variations from the advertisement value due to terms and conditions of a listing. Furthermore, the manner in which the advertisement value is used to determine the price for a listing may vary based on other factors and/or for other reasons, without departing from the scope of the invention.

B. Variable-Rate Pricing

As noted, variable-rate pricing may be used for advertisement spaces in an exemplary advertisement marketplace. In particular, the listing price may be a variable rate that is based on the advertisement value for the advertisement space.

For example, at an initial time of the listing, the variable rate may be set equal to or based on the advertisement value. The server system may subsequently update the variable rate based at least in part on gaze data that is received after the initial time of the listing.

FIG. 11A is a flow chart illustrating a method for updating a variable rate for advertisement rights, according to an exemplary embodiment. The method 1100 shown in FIG. 11A is described by way of example as being carried out by a server system to provide the described functionality. However, it should be understood that method 1100 may be carried out by other systems or combinations of systems, without departing from the scope of the invention.

Method 1100 may be carried out by a server system after the initial time when a listing for advertisement rights to an advertisement space was made available. More specifically, at some point after the initial listing, the server system may receive additional gaze data from a plurality of wearable computing devices, as shown by block 1102. The server system may analyze the additional gaze data to detect occurrences of the advertisement space in the additional gaze data, as shown by block 1104. The server may then update the variable rate based at least in part on occurrences of the advertisement space in the additional gaze data, as shown by block 1106.

The server may use various techniques to update the variable rate at block 1106. In some embodiments, the server may use the occurrences of the advertisement space in the additional gaze data to update the advertisement value for the advertisement space. The server can in turn use the updated advertisement value as a basis for updating the variable rate. For instance, taking into account the additional gaze data, the server may implement a method such as that described in FIGS. 17A, 17B, and 18 to update the advertisement value for the advertisement space.

FIG. 11B is a flow chart illustrating another method for updating a variable rate for advertisement rights, according to an exemplary embodiment. The method 1150 shown in FIG. 11B is described by way of example as being carried out by a server system to provide the described functionality. However, it should be understood that method 1150 may be

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carried out by other systems or combinations of systems, without departing from the scope of the invention.

Method **1150** may be implemented in a scenario where the advertisement value upon which the listing price is initially based, is based on wearer-view data. Further, the wearer-view data upon which the advertisement value is based, is in turn based on the detected occurrences of the of the advertisement space in gaze data. In addition, like method **1100** of FIG. **11A**, method **1150** of FIG. **11B** may be carried out by a server system after a listing for advertisement rights to an advertisement space is made available.

More specifically, after a listing has been made available in the advertisement marketplace, the server system may receive additional gaze data, as shown by block **1102**. The server may then analyze the additional gaze data to detect additional occurrences of the advertisement space in the gaze data, as shown by block **1104**. As such, the server may update the wearer-view data for the advertisement space based on the additional occurrences that are detected in the additional gaze data, as shown by block **1106**. The server may then use the updated wearer-view data as a basis for updating the variable rate, as shown by block **1108**.

In an exemplary embodiment, at block **1108**, the server may use the updated wearer-view data to update the advertisement value, and in turn use the updated advertisement value to update the variable rate. For instance, taking into account the additional gaze data, the server may implement a method such as that described in FIGS. **17A**, **17B**, and **18** to update the wearer-view data and in turn the advertisement value for the advertisement space.

In a further aspect of an exemplary method, such as method **1100** or **1150**, the updated advertisement value may be based solely on the additional gaze data, or may take into account some or all of previously-received gaze data as well as the additional gaze data. As such, updating the advertisement value may involve re-determining the advertisement value using only data received between the current and previous instance in which the advertisement value was determined. By using this approach, the server is effectively “starting from scratch” each time it updates the advertisement value, as none of the gaze data that was previously used to determine the advertisement value, will be used to when re-determining the advertisement value. Alternatively, the server may combine the additional gaze data with some or all of the previously received data, and use this combination of gaze data to update the advertisement value.

In some implementations, an advertiser may be charged for an advertisement space on a per-occurrence basis. In such an embodiment, the server may search gaze data to detect any occurrences of an advertisement space, and determine an ad-value contribution of each detected occurrence. The server may then use the ad-value contributions for detected occurrences of the advertisement space to determine the amount to be charged for the user-account.

For example, consider an implementation where an advertiser pays for advertisement rights on a per-occurrence basis, and is billed on a predefined billing period or cycle (e.g., weekly or monthly). In this scenario, the server may determine the amount owed for a given billing period by summing the ad-value contributions of the ad-space occurrences that are detected during the billing period. Other billing techniques may also be used to charge advertisers on a per-occurrence basis.

When an advertiser is billed for advertisement rights on a per-occurrence basis, the advertiser may be charged a fixed amount for each occurrence, or may be charged an amount that can vary from occurrence to occurrence.

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More specifically, in some cases, it may be assumed that all views of an advertisement are of equal value (or that views of differing values may revert to a fixed average or median value over time). In this and other cases, the same amount may be charged per occurrence.

However, an exemplary method may also account for the fact that some views of an advertisement may be more valuable to an advertiser than other views. For example, the amount due according of a given occurrence may vary depending on factors such as the duration of occurrence, how focused the viewer was on the advertisement space (e.g., as indicated by a focus value), and/or characteristics of the particular user who viewed the ad, among others. Accordingly, an exemplary method may involve determining ad-value contributions for individual occurrences of an advertisement space, and determining an amount to be paid in a billing cycle based on the individual ad-value contributions during the billing cycle. Methods for determining ad-value contributions of individual occurrences of an advertisement space are described in greater detail with reference to FIGS. **9B** and **10**.

C. Auctions

As noted, pricing for an advertisement space may also be set using an auction process. For example, FIG. **12** is a flow chart illustrating an auction process for advertisement rights, according to an exemplary embodiment.

More specifically, method **1200** may involve the server system receiving an auction-listing request to offer advertisement rights to an advertisement space via an auction, where the auction-listing request corresponds to a first user-account, as shown by block **1202**. The server may also determine an advertisement value for the advertisement space, where the advertisement value is based on detected occurrences of the advertisement space in gaze data, as shown by block **1204**. Further, in response to the auction-listing request, the server system may add an auction listing for the advertisement space to an advertising-space database, where the auction listing indicates that the advertisement rights are available for purchase via an auction process that is based at least in part on the determined advertisement value, as shown by block **1206**. The server may then make the auction listing available via a network-based advertisement marketplace, as shown by block **1208**.

The auction process may be based on the determined advertisement value in various ways. For example, a minimum bid may be set to a certain percentage of the advertisement value. For instance, the auction process may require the first bid to be at least 60% of the advertisement value. Similarly, a reserve price for the auction may also be set to a percentage of the advertisement value. For instance, in the above example where the minimum bid is 60% of the advertisement value, the reserve may be set at 80% percent of the advertisement value. Therefore, while bidding may start at 60% of the advertisement value, the seller is not obligated to sell until bidding reaches at least 80% of the advertisement value. Many other examples are also possible.

XIV. Valuation Requests

As noted an exemplary advertisement marketplace may support valuation-request functionality, via which a user can request the advertisement value and/or the potential listing price for an advertisement space. For example, FIG. **13** is a flow chart illustrating a method for providing an advertisement space valuation in an advertisement marketplace, according to an exemplary embodiment.

More specifically, method **1300** involves the server system receiving a valuation request that indicates an advertisement space for valuation, where the request is associated with a certain user-account (e.g., received from a computing device that is associated with the user-account), as shown by block **1302**. The system then determines the advertisement value for the advertisement space, which is based on detected occurrences of the advertisement space in gaze data, as shown by block **1304**. Then, based at least in part on the determined advertisement value, the system may determine a listing price for the advertisement space, as shown by block **1306**. The system can then respond to the valuation request by sending an indication of the determined listing price, as shown by block **1308**.

The valuation request may indicate the advertisement space by way of data that may itself be used to search the gaze data for the advertisement space and/or by way of data from which search criteria may be derived. The server may accordingly use the data that identifies the advertisement space to search for occurrences of the advertisement space in subsequently-received gaze data and/or in past gaze data that has been stored. In particular, the server may use a method such as that described in FIGS. **17A**, **17B**, and **18** to determine the advertisement value based on such gaze data. The server may then send an indication of the advertisement value to the user that requested the valuation. Additionally or alternatively, the server may send an indication of a potential listing price if the user were to list the advertisement space.

In some cases, when a valuation request is received at block **1302**, the system may already have the data needed to determine the advertisement value, or in some instances, may even have the advertisement value available. In particular, the system may have already have generated wearer-view data based on previously-detected occurrences of the identified advertisement space in gaze data, and further, may have determined an advertisement value based on this wearer-view data. This may occur in a number of scenarios. For example, a user may have made a general request to search gaze data for advertisement spaces that can be listed via their user account, provided general authorization to search for such advertisement spaces, or previously requested that the system create wearer-view data specifically for the identified advertisement space. Other examples are also possible.

Therefore, in order to determine the advertisement value for the indicated advertisement space at block **1304**, the system may first query the advertisement-space database to determine if an entry for the advertisement space exists and if so, whether the entry indicates existing wearer-view data and/or an advertisement value. If an advertisement value is indicated, the system may simply retrieve the advertisement value. If wearer-view data exists, but no advertisement value is indicated or the advertisement value is determined to be out-of-date (e.g., not based on the latest available wearer-view data), then the system may use the existing wearer-view data to determine the advertisement value.

On the other hand, if no ad-value is available and no wearer-view data exists (or if existing wearer-view data is determined to be out-of-date), then the system may need to search gaze data for the advertisement space in order to generate the wearer-view data upon which the advertisement value can be based. To do so, the server may search stored gaze data for occurrences of the advertisement space and/or may search future gaze data for occurrences of the advertisement space. More specifically, in an exemplary embodiment, the server may carry out a method such as those

described in FIGS. **17A**, **17B**, and **18** in order to determine to the advertisement value for the indicated advertisement space.

In a further aspect, note that if a listing request is received for advertisement space that is not yet include in the ad-space database, such as in method **100** of FIG. **1**, then the system may automatically treat this listing as a valuation request and/or prompt the user as to whether they would like valuation to be performed. Once the advertisement value has been determined, the system may then allow the user to proceed with listing the advertisement space.

XV. Identifying Potential Advertisement Spaces

A. Automated Suggestion of Potential Advertisement Spaces

In some embodiments, an exemplary system may be configured to search gaze data for physical spaces that are not valued and/or that are not listed for sale in the advertisement marketplace. When such potential advertisement spaces are detected, an exemplary system may identify a user that is authorized to list the physical space for sale as an advertisement space, and/or may determine a potential listing price for the advertisement space. Further, the system may then notify the authorized user that the physical space can be listed as advertisement space in the advertisement marketplace.

For example, FIG. **14** is flow chart illustrating a method for locating potential advertisement spaces, according to an exemplary embodiment. In particular, method **1400** involves the server system searching gaze data for any occurrence of an advertisement space, as shown by block **1402**. Upon detecting one or more occurrences of a given advertisement space in the gaze data, the server may determine whether or not the given advertisement space is an unlisted advertisement space, as shown by block **1404**. If it is determined that the advertisement space is unlisted, then the system may determine a user-account that is associated with the given advertisement space, as shown by block **1406**. In addition, the system may determine an advertisement value for the advertisement space that is based on the detected occurrences of the advertisement space, as shown by block **1408**. The system may then send an indication of an ad-space suggestion message to the user-account, which indicates a listing price for at least the advertisement space that is based on the determined advertisement value, as shown by block **1410**.

B. Request for Identification of Advertisement Spaces that can be Listed

In an exemplary advertisement marketplace, method **1400** may be implemented to automatically identify potential advertisement spaces that are currently unlisted, and notify users that are authorized to list these potential advertisement spaces of this possibility. This feature may be automatic, and thus may be performed without targeting potential advertisement spaces for any particular user. However, in some embodiments, features may be provided to allow a user to specifically request a search for unlisted advertisement spaces that can be listed by the user. This feature may help a user to provide advertisement spaces that they may have otherwise been unaware of, and may be helpful in other scenarios as well.

For example, FIG. **15** is flow chart illustrating a method for providing a search-request feature, according to an exemplary embodiment. More specifically, method **1500** involves the server system receiving a search request associated with a first user-account, as shown in block **1502**. In response to the search request, the server system searches

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gaze data to detect any occurrences of any advertisement space associated with the first user-account, as shown by block 1504. In response to detecting one or more associated advertisement spaces in the gaze data, the server may send the first user-account a search-result message that indicates the associated advertisement spaces that were located in the search, as shown by block 1506.

At block 1506, the server may wait for a certain time period and/or wait until a certain number of associated advertisement spaces are detected, before sending the search-result message. Accordingly, the search-result message may indicate a number of potential advertisement spaces for the user-account. Alternatively, the server may send a separate search-result message upon detection of each associated advertisement space.

In a further aspect of an exemplary embodiment, the search and/or the search results may be limited to unlisted advertisement spaces. Accordingly, if unlisted advertisement spaces are detected at block 1504, the server may provide the requesting user-account with an indication of the unlisted advertisement spaces at block 1506.

In another aspect, the search-result message may also include a listing price for some or all of the associated advertisement spaces. As such, method 1500 may further involve the system determining an advertisement value for at least one associated advertisement space, using gaze-data-based methods such as those described herein. The listing price may then be determined based on the advertisement value, and included in the search-result message. Provided with the listing price for an associated advertisement space, the user may send a listing request for the advertisement space. When the server receives such a listing request, the server may create a listing using an exemplary method, such as method 100.

C. Advertisement-Interest Indication Functionality

In another aspect, an exemplary advertisement marketplace may include features that allow a user (i.e., an “advertiser”) that is interested in an unlisted advertisement space to indicate their interest. For example, the advertisement marketplace may build a database of unlisted advertisement space using techniques similar to those described in blocks 1402 to 1406 of Figure, or using other techniques. The advertisement marketplace may then provide an interface via which advertisers can browse and/or search the unlisted advertisement spaces. This interface may allow an advertiser to indicate their interest in purchasing an unlisted advertisement space and/or to request that the user who is authorized to list the advertisement space be notified of this interest, in hopes that the user will then chose to list the advertisement space in the advertisement marketplace.

In other cases, an exemplary advertisement marketplace may allow an advertiser to indicate a interest in purchasing advertisement rights to a physical space that the system has not yet identified as an advertisement space. When the system receives such a request, the system may first need to determine which user is authorized to list the advertisement space (if any), so that an indication of the advertiser's interest can be sent to the user.

FIG. 15 is flow chart illustrating a method that provides advertiser-request functionality, according to an exemplary embodiment. In particular, method 1500 involves the server system receiving an advertisement-interest indication associated with a first user-account, where the advertisement-interest indication indicates an advertisement space, as shown by block 1502. The system then determines an advertisement value for the advertisement space, which is based on detected occurrences of the advertisement space in

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gaze data, as shown by block 1504. The system then identifies a second user-account that is associated with the advertisement space, as shown by block 1506. The system then sends an advertisement-interest notification, which indicates a listing price for the advertisement space that is based on the determined advertisement value, to the second user-account, as shown by block 1508.

In an exemplary embodiment, the associated user-account identified at block 1506 is a user-account of the user or one of multiple users that are authorized to list the advertisement space (e.g., the owner of the advertisement space or a representative of the owner). Further, in some implementations, multiple user-accounts may be identified at block 1506 (e.g., both the user-account of the owner and a user-accounts for a representative of the owner).

XVI. Detecting Advertisement Spaces in Gaze Data

As noted above, various embodiments involve analysis of gaze data to detect and/or identify when advertisement spaces occur in gaze data. Referring back to FIG. 2, in order to detect ad-space occurrences, an exemplary server system 200 may employ various types of video and/or image-processing techniques. For instance, advertisement server system 204 may implement various well known and yet-to-be-developed techniques for object recognition in video and/or still images in the process of recognizing advertising spaces.

In some cases, an advertisement space may be identified in gaze data by way of the advertisement that is displayed in the advertisement space. For example, ad-space database 210 may include data related to which specific advertisements are being displayed in which advertisement spaces (and may further indicate when there is no advertisement being displayed in a given advertisement space). As such, ad-valuation server 212 may search for advertisements that are currently being displayed in gaze data. To do so, the ad-valuation server may use various visual search techniques that are now known or yet to be developed in order to identify an advertisement in gaze data.

In other cases, an advertisement space may itself be identified, without necessarily relying on the particular advertisement that is being displayed in the advertisement space. (Note that this functionality may be particularly useful in cases where an advertisement space is empty.) In such an embodiment, detecting that an advertisement space occurs in gaze data may involve recognizing when the gaze data includes an object or a certain combination of objects that are associated with a particular advertisement space. For example, to recognize advertisement space on the bumper of a particular car, gaze data may be analyzed for an object shaped and/or having coloration that is characteristic of a car bumper. Further, the gaze data may be analyzed for an object having such a shape and/or coloration in conjunction with a license plate having a certain license plate number. In such an embodiment, the server system may consider an occurrence of a bumper in combination with the license plate number for the specific car to be an occurrence of the advertisement space on the car's bumper. Many other examples are also possible.

In some cases, searching gaze data from a large number of wearable computing devices for a large number advertisement spaces may be data intensive. Accordingly, an exemplary server and/or wearable computing devices may implement pre-processing techniques to tag and ID certain types of objects or certain types of information in gaze data, which may help to speed up the process of detecting

advertisement spaces. In some instances, wearable computing devices and/or the server may also store gaze data for processing when, e.g., a wearable computing device is offline, or when the amount of real-time data being collected is generally less. For example, a server may use certain processing resources to receive incoming gaze data during the day, when more gaze data may be received, and then re-assign these processing resources to analyze stored gaze data for advertisement spaces at night, when less new gaze data may be received.

In some embodiments, a server system may utilize location data to detect an occurrence of an advertisement space in gaze data. For example, a server system may determine or be provided with the geographic location of a particular advertisement space (e.g., the GPS coordinates of the advertisement space). Then, when the advertisement space is detected in gaze data from a particular wearable computing device, the server may determine the location of the wearable computing device. If this wearable computing device is located such that the advertisement space could be visible to the wearer of the wearable computing device (e.g., within a predetermined distance from the location of the advertisement space), then the server system may consider this an occurrence of the advertisement space. However, if the wearable computing device that provided the gaze data is located such that the advertisement space could not be viewed by the wearer (e.g., not within a predetermined distance from the location of the advertisement space), then the server system may not consider this an occurrence of the advertisement space.

As another example, a server system may use the geographic location of a particular advertisement space to limit the gaze data that is monitored for the advertisement space. For instance, the server may determine the locations of wearable computing devices from which gaze data is received. As such, the server may only monitor gaze data that is received from wearable computing devices that are located within a predetermined distance from the advertisement space. Other methods that utilize the location of an advertisement space when detecting occurrences of the advertisement space in gaze data are also possible.

In some embodiments, radio frequency identification (RFID) may be used to help detect occurrences of an advertisement space in gaze data. In particular, an advertisement space may be associated with a certain RFID tag, and wearable computing devices may be configured with RFID readers. As such, when a wearable computing device detects an RFID tag from an advertisement space, the wearable computing device may relay this to the server system. For instance, when the wearable computing device detects an RFID that is associated with an advertisement space, it may insert metadata into the gaze data which indicates the RFID tag and the time at which the RFID tag was detected. Alternatively, the wearable computing device may send a separate message indicating that the RFID tag was detected at a particular time. In either case, the server system can then search for the associated advertisement space in gaze data that is received from the wearable computing device at or near the time when the RFID tag is detected. This may help the server system to more efficiently detect occurrences of advertisement spaces, as the timing with which the RFID tags are detected may indicate, for example, times in corresponding point-of-view video where the advertisement space is likely to occur. Further, various types of RFID may be utilized, such as near-field communications (NFC) and/or other types of RFID, depending upon the implementation.

In some embodiments, barcodes may be used to help detect occurrences of an advertisement space in gaze data. For instance, a barcode that identifies an advertisement space may be displayed within or near to an advertisement space. The server system may then search for barcodes within gaze data. When a barcode associated with a particular advertisement space is detected, the server may consider this to be an occurrence of the advertisement space, or may treat this as a factor that, along with other factors, can indicate that there is an occurrence of the advertisement space in the gaze data. Various types of barcodes, such as high capacity color barcodes (HCCBs) and/or quick response (QR) codes may be utilized in such an embodiment. Other types of barcodes are possible as well.

In a further aspect, machine-readable codes may also be associated with a particular ad that is displayed in an ad space. For example, a certain advertisement may be associated with a certain QR code. As such, the advertisement may be detected when the QR code is detected in gaze data.

Further, in some embodiments, machine-readable codes may be associated with a certain combination of a particular advertisement and a particular user-account. As such, each copy of the same advertisement may have a different QR code, which uniquely identifies the user-account that is authorized to sell the advertisement space. As a result, detecting a QR code in gaze data may enable an exemplary server system to positively identify not only the advertisement that is displayed in the advertisement space, but also which copy of the advertisement has been detected. This may be useful in a scenario where the advertisement space itself is difficult to detect, and in other scenarios as well.

It should be understood that the above techniques for detecting occurrences of advertisement spaces are not intended to be limiting. Other techniques are also possible.

XVII. Determining the Value of an Advertisement Space

As noted, in some embodiments, an exemplary system may be configured to use gaze data to determine advertisement values for advertisement spaces. Further, an exemplary system may be configured to determine an advertisement value for many types of physical spaces; many of which may not have been valued using traditional advertisement valuation techniques.

FIG. 17A is a flow chart illustrating a method according to an exemplary embodiment. This method may be implemented by a computing device, and in particular, by a server system, in order to value an advertisement space based on point-of-view gaze data received from a number of wearable computing devices (which may be referred to interchangeably as wearable computing devices). Note that wearable computing devices may also be referred to as wearable computers herein. Further, a server system that implements an exemplary method may be referred to as an ad-valuation system, as an ad-valuation server, or simply as a server.

As shown by block 1702, method 1700 involves a server system receiving gaze data from a number of wearable computing devices. The server system analyzes the gaze data from the wearable computing devices to detect occurrences of an advertisement space in the gaze data, as shown by block 1704. The server system then generates wearer-view data for the advertisement space, which is based on detected occurrences of the advertisement space in the gaze data, as shown by block 1706. The wearer-view data can then be used as a basis for determining an advertisement value for the advertisement space, as shown by block 1708.

Once the advertisement value is determined, the server system may cause a computing system to make the advertisement space available for purchase at the determined advertisement value, as shown by block 1710.

In an exemplary method 1700, the gaze data is received from a number of wearable computing devices. Further, the gaze data from each wearable computing device is generally indicative of a respective wearer-view associated with the given wearable computing device. For example, the gaze data from each wearable computing device may take the form of point-of-view video that is captured at the wearable computing device. As such, the gaze data that is analyzed by the server system may include a number of point-of-view videos (e.g., a respective point-of-view video from each of the wearable computing devices).

The gaze data from some or all of the wearable computing devices that provide gaze data may additionally or alternatively take forms other than point-of-view video. For example, the gaze data from some or all of the wearable computing devices may take the form of respective point-of-view images captured by a forward- or outward-facing camera on the respective wearable computing device. As a specific example, a given wearable computing device may periodically take a picture, and then send the picture to the server system for use in generating wearer view data. To do so, the wearable computing device may analyze point-of-view video for one or more advertisement spaces, and generate a screen capture of the video when an advertisement space is detected. The wearable computing device may then send the screen capture to the server system. Other examples are also possible.

Since the gaze data from a given wearable computing device is generally indicative of the wearer-view of the wearable computing device's wearer, the gaze data is generally indicative of what the wearer of the device is actually looking at. Further, since the wearer-view data is based on the gaze data, the wearer-view data is indicative of actual views of the advertisement space by wearers. For instance, the wearer-view data may provide an indication of how many people are looking at a particular advertisement space, which people are actually looking at a particular advertisement space, when people are looking at a particular advertisement space, and/or how long people are actually looking at a particular advertisement space, among other information. As such, the wearer-view data may help to more accurately determine what an advertising space is worth.

As noted above, when occurrences of an advertisement space are detected in gaze data, an exemplary method 1700 may involve generating wearer-view data that is based on the detected occurrences. As such, an exemplary server system 204 may be configured to carry out an exemplary method 1700 or portions thereof for many different advertisement spaces. Generally, the accuracy of the ad-space valuation will typically increase as the number of wearable computing devices providing gaze data increases. However, the gaze data may be collected from any number of wearable computing devices without departing from the scope of the invention.

To facilitate determining an advertisement value for a given advertisement space, the wearer-view data may provide various types of information. For example, the wearer-view data for a given advertisement space may include, for each detected occurrence of the given advertisement space: (a) data indicating the particular wearable computing device that provided the gaze data in which the advertisement space occurred, (b) data indicating a user-profile associated with the particular wearable computing device, (c) data indicating

a time of the detected occurrence, (d) a duration of the detected occurrence, and/or (e) other information.

Generally, the function of generating wearer-view data for the advertisement space, as shown in block 1706 of method 1700, may vary depending upon the information to be included in the wearer-view data. In an exemplary embodiment, detecting an occurrence of an advertising space in the gaze data may serve as a trigger for the server system to generate wearer-view data recording the fact that the occurrence was detected. Further, to generate the wearer-view data for a given occurrence, the server system may extract information from the gaze data in which the occurrence was detected. The extracted information (or information derived from the extracted information) may be included in the wearer-view data generated for the detected occurrence.

A. Per-Occurrence Data for an Advertisement Space

In some embodiments, the server system 204 may update the wearer-view database 1108 upon each detected occurrence of an advertisement space. For example, the server system may generate a record in the wearer-view database for each detected occurrence of an advertisement space. In such an embodiment, the record for a given occurrence of an advertisement space may include: (a) an indication of the particular wearable computing device that provided the gaze data in which the advertisement space occurred, (b) an indication of a user-profile associated with the particular wearable computing device, (c) a time of the occurrence, and/or (d) a duration of the occurrence.

The wearer-view data for a given occurrence of an advertisement space may indicate the corresponding wearable computing device that provided the gaze data in which the advertisement space occurred. In such an embodiment, the server system may determine the corresponding wearable computing device in various ways. For instance, consider an embodiment where the server system receives point-of-view (POV) video stream from a number of wearable computing devices. In such an embodiment, the server system may establish a communication session to receive the video stream from a given one of the wearable computing devices, and as part of establishing and/or participating in the session, may receive an identifier of the wearable computing device. (Note that various protocols, which are well known in the art, may be used to receive a POV video stream and/or to receive other forms of gaze data.) Additionally or alternatively, metadata in the gaze data itself may include an identifier of the wearable computing device that is providing the gaze data. Other techniques for determining which wearable computing device corresponds to a particular occurrence of an advertisement space are also possible.

As further noted above, the wearer-view data for a given occurrence of an advertisement space may indicate an associated user-profile, which is associated with the wearable computing device that provided the gaze data having the particular occurrence. The server system may determine the associated user-profile in various ways. For example, the server may determine the identifier for the corresponding wearable computing device in a manner such as described above or otherwise. The server may then look up a user-profile of a user that is registered to use or is otherwise associated with the corresponding wearable computing device (e.g., by querying a user database that indicates which users are associated with which wearable computing devices). Alternatively, a user-identifier may be provided in the course of receiving the gaze data (e.g., in a communication session or in metadata). In such an embodiment, the server system may use the user-identifier to access a user-profile for the user. As another alternative, the user-profile

itself may be received directly from the device (e.g., during the communication session in which the gaze data is received, as metadata included in the gaze data, or in a separate message that is associated with the gaze data). Other techniques for determining a corresponding user-profile for a particular occurrence of an advertisement space are also possible.

In a further aspect, when the wearer-view data for a given occurrence indicates the associated user-profile, the wearer-view data may simply include an identifier of the associated user-profile. In such an embodiment, the data from such user-profiles may be stored in one or more separate user-profile databases. In this case, the server may use the identifiers of the associated user-profiles to retrieve the data from the actual user-profiles. Alternatively, some or all of the data from the associated user-profile may be included in the wearer-view data for the advertisement space (e.g., in wearer-view database 1108).

In a further aspect, the server system may include a time stamp in the wearer-view data that is generated for a given occurrence. The timestamp may indicate the time at which the occurrence of the advertisement space was detected. Additionally or alternatively, the timestamp may indicate a time that is derived from time data included in the gaze data. For example, point-of-view video from a given wearable computing device may include time data indicating when the video was recorded by the wearable computing device. As such, the server system may use this time data to generate a timestamp for an occurrence that is detected in such point-of-view video. For instance, the server system may determine a frame or frames of the video that include the advertisement space, and use a time stamp or time stamps of the frame or frames to generate the timestamp for the detected occurrence. Other techniques for generating a timestamp for a particular occurrence of an advertisement space are also possible.

In another aspect, the wearer-view data for a given occurrence of an advertisement space may indicate the duration of the given occurrence. Accordingly, the server system may be configured to determine the duration of a given occurrence of an advertisement space. For instance, in the above example where POV video includes time data, the server system may use timestamps on frames of the video to determine the duration of time the first frame of the video that includes the advertisement space and the last subsequent and consecutive frame that includes the advertisement space. Alternatively, the server system may implement its own timer to determine the duration of a given occurrence of an advertisement space. Other techniques for determining the duration of a particular occurrence of an advertisement space are also possible.

In a further aspect, when generating wearer-view data for a given occurrence, the server may consider whether the wearable computing device that corresponds to a given occurrence was being worn during the occurrence. In particular, if the corresponding wearable computing device is not being worn at the time of the detected occurrence, the server may adjust or change the wearer-view data that is generated in response to detecting the occurrence. For example, when the wearable computing device is not being worn, the server may interpret this to mean that the gaze data from the wearable computing device is unlikely to represent what the wearer is actually viewing. Accordingly, the server may include an indication that the wearable computing device was not being worn in the wearer-view data that is created for such an occurrence. Further, server may adjust the wearer-view data so as to decrease the weight of such an

occurrence when determining the advertisement value for the advertisement space, or may ignore the occurrence entirely (e.g., by refraining from generating any wearer-view data for the occurrence).

5 B. Summary Data for an Advertisement Space

In some embodiments, the wearer-view data for a given advertisement space may include summary data for the advertisement space such as: (a) a list of which wearable computing devices viewed the advertisement space (e.g., which wearable computing devices provided gaze data in which one or more occurrences were detected), (b) a list of the user-accounts or the user-profiles that are associated with the wearable computing devices that have viewed the advertisement space, (c) a total view count indicating the total number of detected occurrences of the advertisement space, (d) a total view duration of the advertisement space, (e) an average view duration for occurrences of the advertisement space, and/or (f) a view rate that indicates how frequently the advertisement space occurs in the gaze data (e.g., occurrences/hour, occurrences/month, etc.). The wearer-view data for a given advertisement space may additionally or alternatively include other types of summary data for the advertisement space.

In order to keep the above and other such summary data substantially current, the server system may update the wearer-view data for an advertisement space each time the advertisement space is detected in gaze data. For example, when the server system detects an advertisement space in gaze data from a given wearable computing device, the server system may update the wearer-view data by: (a) adding the given wearable computing device to a list of wearable computing devices that have viewed the advertisement space (if the wearable computing device is not on the list already), (b) adding the user-account or the user-profile that is associated with the given wearable computing device to a list of user-accounts or user-profiles that have viewed the advertisement space, (c) incrementing the total view count for the advertisement space, (d) determining the duration of the occurrence and adding the determined duration to a total view duration for the advertisement space, and/or (e) determining the duration of the occurrence and adding the determined duration and recalculating the average view duration to account for the determined duration. Other examples are possible as well.

In some embodiments, the wearer-view data for each advertisement space may include only summary data such as that described above, and thus may not include per-occurrence data for each detected occurrence of an advertisement space. However, it is also possible that the wearer-view data for a given advertisement space may include only per-occurrence data, or may include both per-occurrence data and summary data for the advertisement space.

55 C. Focus Data for an Occurrence

In some embodiments, the wearer-view data for a given advertisement space may include focus data, which is generally indicative of the amount of attention paid to an advertisement space by viewers of the advertisement space. The focus data may help to provide a more accurate valuation for the advertisement space by helping take into account the fact that not all views are necessarily equal, since the amount of attention paid to the advertisement space may vary between views. In such an embodiment, the server system may determine a focus value for a detected occurrence (as described above) when it generates wearer-view data for the occurrence, or may determine a focus value at a later time.

D. Use of Summary Data for Advertisement Valuation

As noted above, an exemplary method 1700 may involve using the wearer-view data for an advertisement space to determine an advertisement value for the advertisement space. Various types of wearer-view data may be utilized when determining an advertisement value. For instance, various types of the summary data described above and/or various types of the per-occurrence data described above may be used to determine the advertisement value for a given advertisement space. An exemplary valuation method may also incorporate other types of data in addition to wearer-view data. Further, the manner in which a given type of wearer-view data is used to determine an advertisement value may vary depending upon the implementation.

In some embodiments, the advertisement value for a given advertisement space may be based on summary data for the advertisement space. For example, the advertisement value may be based at least in part on the total view count for an advertisement space (e.g., the total number of occurrences that are detected in the gaze data). In such an embodiment, the total number of occurrences may be tracked over all time. Alternatively, the total number of occurrences may be tracked over a predetermined period of time (e.g., a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates total view count, the determined advertisement value will typically increase as the total number of occurrences increases. Further, the manner in which the total view count is used to determine advertisement value may vary, depending upon the implementation.

As another example, the advertisement value for a given advertisement space may be based at least in part on a view rate for the advertisement space (e.g., the rate at which occurrences of the advertisement space are detected in the gaze data). For instance, the wearer-view data may indicate a number of views per month, per week, per day, per hour, etc. In such an embodiment, the rate may be based on detected occurrences over all time. Alternatively, the rate may be based on occurrences during a predetermined period of time (e.g., during a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates view rate, the determined advertisement value will typically increase as the view rate increases. Further, the manner in which the view rate is used to determine advertisement value may vary, depending upon the implementation.

In the above examples, the advertisement value is determined based on summary data that generally does not differentiate one detected occurrence from another. However, some embodiments may apply further intelligence to account for the fact that some views of an advertisement space may be more valuable to an advertiser than others.

For example, the advertisement value for a given advertisement space may be based at least in part on a total view duration and/or an average view duration for the advertisement space. In such an embodiment, the total view duration and/or the average view duration may be calculated from all detected occurrences of the advertisement space or from a representative sample of occurrences. In either case, the total view duration and/or the average view duration may be calculated over all time, or may be calculated over a predetermined period of time (e.g., a year, a month, a week, or a custom-defined time period). In an exemplary embodiment that incorporates total view duration and/or the average view duration, the determined advertisement value will typically increase as the total view duration and/or the average view duration increases. Accordingly, views that

last longer will generally contribute more to the advertisement value and/or be weighted more heavily when determining the advertisement value. It should be understood that the manner in which the total view duration and/or the average view duration is used to determine advertisement value may vary, depending upon the implementation.

As another example, the server may determine focus values for all or a representative sample of the detected occurrences of an advertisement space. The server may then average the focus values for the detected occurrences to determine an average focus value for the advertisement space. The server can then use the average focus value to determine the advertisement value for the advertisement space.

In a further aspect, an exemplary embodiment may help account for the fact that views of an advertisement space by certain people may be considered more valuable than views of the same advertisement space by other people. More specifically, in an exemplary embodiment, wearers may opt-in to a program or otherwise give permission for information from their user-profile to be used to value advertisement spaces. Various types of information from an associated user-profile may then be used to determine how valuable a given occurrence of an advertisement space is. For instance, a user-profile for a wearer may include: (a) consumer information such as spending habits, locations of purchases, amounts of purchases, types or categories of purchases, timing of purchases, etc., (b) demographic information such as age or age group, ethnicity, nationality, sex, location of residence, and/or location of workplace, (c) contact and/or social networking information such as a wearer's contacts, and possibly data indicating a purchasing influence of the wearer with regard to their contacts (e.g., data indicating any correlation of the wearer's purchasing history to the wearers' friends' purchasing histories), and/or (d) other information such as income, job or job type, other job details, hobbies, interests, and so on.

Therefore, since the occurrence of an advertisement space in gaze data from a given wearable computing device may be interpreted to mean that the wearer of the given wearable computing device has viewed or is viewing the advertisement space, information from user-profiles that wearer-view data associates with a given advertisement space may provide information about the type or types of people that an advertisement space reaches and/or of the characteristics of people that the advertisement space reaches. As a result, this information may be used to more accurately determine what types of people are viewing the advertisement space, and value the advertisement space accordingly. In particular, an exemplary server may place greater weight on occurrence of an advertisement space associated with certain people and/or certain types of people when determining the advertisement value for a given advertisement space.

For example, the server may determine a respective income level for the user-profile associated with each occurrence. The server may then average the determined income levels to calculate an average income level for viewers of the advertisement space, and use the average income level as input data to determine the advertisement value for the advertisement space. Alternatively, the server may determine an income range of the determined income levels, and use the income range as an input to the ad-value calculation for the advertisement space. Other examples are also possible.

It should be understood that the advertisement value for a given advertisement space may be based upon one type of summary data or a combination of various types of summary

data. For example, in one implementation, the total number of views, the view rate, the average view duration, and one or more characteristics of the associated user-profiles, could all be used as inputs when calculating advertisement value. Many other examples are also possible.

E. Use of Per-Occurrence Ad-Value Contributions for Advertisement Valuation

In some embodiments, a server system may determine an advertisement value for an advertisement space by first determining an individual advertisement-value contribution for each detected occurrence of the advertisement space. The advertisement-value contribution for a given occurrence may be based on information from the user-profile associated with the occurrence and/or on other information related to the occurrence. The collective knowledge provided by all the individual advertisement-value contributions may then be used to determine the advertisement value for the advertisement space and/or be used to determine summary data for the advertisement space, which may in turn be used to determine the advertisement value.

FIG. 17B is a flow chart illustrating a method for determining advertisement value, according to an exemplary embodiment. In particular, FIG. 17B illustrates a method 1750 in which the advertisement value for an advertisement space is based on individual ad-value contributions of occurrences of the advertisement space in gaze data.

More specifically, method 1750 involves monitoring gaze data from a number of wearable computing devices for occurrences of a particular advertisement space, as shown by block 1752. Each time an occurrence of the advertisement space is detected, as shown by block 1754, the server system may further determine an advertising-value contribution of the occurrence, as shown by block 1756. The server system may then determine advertising-value contributions for a number of occurrences by repeating blocks 1754 and 1756 for a number of detected occurrences of the advertisement space. The server system may then use the advertising-value contributions for the detected occurrences of the advertisement space as a basis for determining the advertisement value for the advertisement space, as shown by block 1758.

In some embodiments, such as method 1750 of FIG. 17B, the ad-value contribution for each occurrence of an advertisement space may be determined upon detecting the occurrence in the gaze data. However, it should be understood that the ad-value contribution for some or all occurrences of an advertisement space may be calculated at a later time, based on wearer-view data that is stored as the occurrences are detected in the gaze data.

A server may use various techniques to determine the individual ad-value contribution for a given occurrence of an advertisement space in gaze data. Such techniques may utilize various different factors to determine the ad-value contribution for a given occurrence. In some embodiments, the server may use a weighting value for an occurrence to determine the ad-value contribution of the occurrence. In particular, the server may determine a weighting value that generally indicates the particular occurrence's value relative to a "standard" occurrence of the advertisement space. The weighting value for the particular occurrence may then be applied to a base advertising-value contribution to determine the advertising-value contribution for the particular occurrence. In such an embodiment, the weighting value may be based on various factors or combinations of factors, such as the particular wearable computing device from which the gaze data including the particular occurrence was received, the duration of the occurrence, characteristics of the person

who viewed (e.g., as indicated by the user-profile associated with the occurrence), the focus value of the occurrence, and/or other factors.

As a specific example, the ad-value contribution for each occurrence of the advertisement space may be a dollar amount that is attributed to the occurrence. As such, the server may determine a dollar amount for the advertisement value by summing the ad-value contributions. As another example, the ad-value contribution for each occurrence of the advertisement space may be a price rate (e.g., dollars per month, dollars per view, etc.) that is attributed to a respective occurrence of the advertisement space. As such, the server may determine the advertisement value by summing the ad-value contributions to get an overall price rate. Other examples are also possible.

Once a server system has determined the individual ad-value contributions for a number of occurrences of the particular advertisement space, the server may use various techniques to determine the advertisement value for the advertisement space. For example, in some embodiments, the server may determine an average advertising-value contribution by averaging the advertising-value contributions of some or all of the detected occurrences. The server may then use the average advertising-value contribution as a basis for determining the advertisement value for the advertisement space. As a specific example, the server may determine an ad-value contribution for each occurrence in the same manner as the overall price or rate for the advertisement space, but using the assumption that all occurrences of the advertisement space are identical to the occurrence. The server may then determine a dollar amount or price rate for the advertisement space by averaging the ad-value contributions determined in this manner. Other examples are also possible.

It should be understood that techniques described herein for determining an advertisement value based on the ad-value contributions are not intended to be limiting. Other techniques for determining an advertisement value based on the ad-value contributions of individual occurrences are also possible, without departing from the scope of the invention.

F. Valuation of an Advertisement Space on a Per-Advertisement Basis

Some embodiments may involve determining a value an advertisement space that is specific to a particular type of advertisement. For example, an advertisement server may determine a value for an advertisement space when the advertisement space is used to display an advertisement for a particular type of product (e.g., for clothing or for a movie).

Further, in some embodiments, an exemplary method may be implemented for ad-specific valuation of an advertisement space based on the extent to which the advertisement space reaches the target market of the advertisement. For instance, wearer-view data may be used to determine who is viewing an advertisement space. The advertisement space valuation may then be based on the correlation between those who view the advertisement space and the target market of the specific advertisement.

In such an embodiment, an exemplary method may utilize wearer-view data indicating user-profiles associated with occurrences of an advertisement space in the gaze data. As such, the server may analyze the associated user-profiles to determine one or more characteristics of those who have viewed the advertisement space. More specifically, an exemplary method may involve determining a group of user-profiles associated with the advertisement space (e.g., user-profiles that are associated with wearable computing devices that captured the advertisement space in their respective

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gaze data). Then, based on characteristics of the associated user-profiles, the server may determine one or more viewer characteristics of the group. The viewer characteristics of the group may be interpreted as indicative of a “standard” viewer of the advertisement space. As such, the viewer characteristics of the group may incorporate when determining the advertisement value for a specific type advertisement.

For example, some embodiments may involve determining both: (a) the viewer characteristics of the group of associated user-profiles and (b) one or more target-viewer characteristics for the particular advertisement. The server may then compare the viewer characteristics of the group to the target-viewer characteristics and, based on the comparison, determine the advertisement value for the particular advertisement in the advertisement space.

In some embodiments, the advertisement value for the particular advertisement may be further based on the location of the advertisement space. In particular, there may be a relationship between the characteristics of a particular advertisement and the location of advertisement space, and an exemplary method may help to account for such a relationship. In such cases, a weighting factor may be applied to increase or decrease the advertisement value depending upon the relationship between the location of the advertisement space and the characteristics of the advertisement.

For example, consider an advertisement for a clothing product and an advertisement space that is located near to a shopping area and/or near to a store where the clothing product can be purchased. This advertisement space may generally be considered more valuable when used to display the advertisement for the clothing product than when used to display an advertisement for a type of product that cannot be purchased nearby. Accordingly, a weighting factor may be applied to increase the advertisement value for the clothing product in the advertisement space. Similarly, the weighting factor may function to decrease the advertisement value for a product that cannot be purchased nearby.

As another example, consider an advertisement for a movie and an advertisement space that is located near to a movie theater that is showing the movie. This advertisement space may generally be considered more valuable when used to display the advertisement for the movie than when used to display an advertisement for a movie that is not in any nearby theaters. Accordingly, a weighting factor may be applied to increase the advertisement value of the advertisement space for a movie that is showing in the nearby theater. Similarly, the weighting factor may decrease the advertisement value for the movie that is not in any nearby theaters. Other examples are also possible.

In a further aspect, some implementations of methods **1700** and **1750** may utilize the type of advertisement as the characteristic of the advertisement upon which the advertisement value is based. In such an embodiment, all advertisements of the same type may be evaluated in the same way. As such, the advertisement value in such an embodiment may in effect be determined for the type of advertisement in the advertisement space (rather than specifically for an individual advertisement). Alternatively, the type of advertisement may be one of a number of factors, such that an advertisement space may be valued differently for different advertisements that are classified as being the same type of advertisement.

In yet another aspect, the advertisement value may vary for a particular advertisement, if it is determined that the advertisement is more or less likely to interest wearers who

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view it. In particular, some embodiments may increase the advertisement value for an advertisement in a given advertisement space (and/or the amount that an advertiser is ultimately charged, if this differs from the advertisement value), when the advertisement that is more likely to interest the people who typically view the particular advertisement space. However, in other embodiments, the opposite approach is also possible. In particular, an exemplary system may actually decrease the advertisement value for an advertisement in a given advertisement space (and/or the amount that an advertiser is ultimately charged, if this differs from the advertisement value) when the advertisement that is more likely to interest the people who typically view the particular advertisement space.

While this may seem counter-intuitive in one sense, this strategy of charging less for more valuable advertisement selection may provide long-term gains. In particular, by displaying advertisements that are likely to interest users, the advertising format as a whole gains credibility with the viewing public, which may result in more people viewing the advertisement space in the long-term. Put another way, if people are disinterested in what they see in an advertisement space, they may choose not to view and/or ignore what is displayed in the advertisement space in the future. Therefore, by reducing the amount charged for advertising that interests users, advertisers are incentivized to provide advertising that is interesting. Providing such incentivizes may in turn increase the chances for long-term success in the advertising space and similar types of advertising spaces, thus creating more long term value for the seller and/or a third-party that is brokering advertisement sales.

FIG. **18** is a flow chart illustrating a method for using multiple factors to determine the ad-value contribution of a given ad-space occurrence in gaze data, according to an exemplary embodiment. In method **1800** of FIG. **18**, a pre-determined base contribution for an occurrence of the advertisement space is weighted according to multiple factors in order to determine an ad-value contribution for each occurrence of an advertisement space in gaze data for a given user-account. The advertisement value for a given advertisement space may then be determined from the collective knowledge provided by the ad-value contributions of the all or a representative subset of its occurrences in gaze data. While method **1800** is described by way of example as being implemented by a server system, this by another device or system, or by a combination of the server system and/or other devices and systems.

Method **1800** involves the server system determining a weighting value (weighting_value) to be applied for an occurrence of an advertisement space (O_i) in gaze data. The weighting value is based on m different factors (f_1 to f_m). In particular, the server system may determine a value for a first factor (f_1) with respect to a given occurrence of an advertisement space in, as shown by block **1802**. The server system may then determine a benchmark value for the first factor (bm_f_1), as shown by block **1804**. This benchmark may be the average or a mean value for the given factor over a sample set of different occurrences, or may be a predefined standard value for the given factor. The benchmark value may be determined in other ways as well.

In any case, the server system may determine a relationship between the determined value of the first factor for the given occurrence and the benchmark value with respect to the first factor (e.g., f_1/bm_f_1), as shown by block **1806**. If there are additional factors to consider (e.g., if m is greater than one), as indicated by block **1808**, then the server may repeat blocks **1802** to **1806** for the additional factors, until

the relationship between the given user-account and relationship between the value for the given occurrence and the average value has been determined for each factor. Once all the factors have been evaluated, the server system may use the determined relationships as a basis for determining a weighting value for the occurrence, as shown by block 1810.

Thus, for a given occurrence O_i , and values for a set of n factors f_1 to f_n , method 1800 may be implemented to determine a weighting_value; as a function of the respective relationships between the values of factors for the respective occurrence f_1 to f_m , and the respective benchmark value bm_f_1 to bm_f_m . For example, weighting_value; may be calculated as:

weighting_value; =

$$F[(f_1/bm_f_1), (f_2/bm_f_2), \dots (f_m/bm_f_m)]$$

Note that the particular function of the relationships used may vary from implementation to implementation, depending upon the design goals.

Once the server has determined the weighting_value; for a given occurrence O_i of an advertisement space, the server may use a base contribution for an occurrence of the advertisement space detected in occurrence O_i to calculate the ad-value contribution for the occurrence as:

$$ad_value_contribution_i = base_contribution_i * weighting_value_i$$

Further, the server system may repeat the above process to determine an ad-value contribution for each of n occurrences of the advertisement space in the gaze data. As such, the advertisement value for the advertisement space may be determined as a function of $ad_value_contribution_i$ for i equal 1 to n . For example, the advertisement value for the advertisement space may be calculated by summing $ad_value_contribution$ to $ad_value_contribution_n$. Other examples are also possible.

It should be understood that many other types of information provided by and/or related to a given user-account may be considered, alone or in combination, when determining the ad-value contribution of an advertisement space in gaze data for the given user-account. Further, information provided by and/or related to a given user-account may be considered in combination with other factors, such as duration of an occurrence and/or a focus value associated with the occurrence, when determining the ad-value contribution for an occurrence.

G. Adjusting the advertisement Value Based on Other Factors

In some embodiments, advertisement valuation may be based on other types of data, in addition to wearer-view data. In such an embodiment, the advertisement server may determine a base value for the advertisement, or a weighting to be applied to the advertisement value based on an intrinsic value of the advertisement space, which accounts for the characteristics of the advertisement space itself, and then adjust the intrinsic value according to the wearer-view data.

For example, in some embodiments, an exemplary method may use the geographic location of the advertisement space as a further basis for determining the advertisement value for the advertisement space. For example, an advertisement that is located in a shopping area may have a

greater intrinsic value than one that is located in an alley. Accordingly, an advertisement value that is determined based on wearer-view data may be adjusted based on the location of the advertisement space. Other examples are also possible. Generally, the type and/or the amount of adjustment that is applied due to the location of an advertisement space may vary depending upon the particular implementation.

Further, in some embodiments, the server may consider the type of advertisement space and/or the format in which advertisements can be displayed in the advertisement space when determining the advertisement value for the advertisement space. For example, an LCD billboard might generally be considered more valuable than an equivalent print billboard. As such, when an advertisement value is determined for a billboard based on wearer-view data, the determined advertisement value may be adjusted based on whether the billboard is an LCD or a print billboard. Other examples are also possible. Generally, the type and/or the amount of adjustment that is applied based on the type of advertisement space may vary depending upon the particular implementation. Further, other adjustments, based on other characteristics of an advertisement space, are also possible.

In another aspect, an advertisement space may be blank (e.g., not displaying an advertisement) during some or all of the period in which gaze data is being collected for purposes of determining the advertisement value. The fact that an advertisement space is blank, as opposed to displaying an advertisement, may affect the gaze data for the advertisement space because a blank space might attract less attention from nearby people. Further, different advertisements may attract different amounts of attention. Therefore, when an advertisement is displayed while gaze data is being collected, the particular advertisement may itself affect the gaze data. As such, it is possible that wearer-view data for an advertisement space may be effected by whether or not an advertisement space is blank, and if something is displayed in the advertisement space, what specifically is displayed.

Accordingly, an exemplary method may further involve determining a pre-sale weighting factor for an advertisement space, which is based on: (a) whether the advertisement space is blank while gaze data is being collected and/or (b) the characteristics of what is displayed in the advertisement space while gaze data is being collected. A server may then use the pre-sale weighting factor for the advertisement space as a further basis for determining the advertisement value for the advertisement space.

As a specific example, an exemplary method may further the server determining whether or not the advertisement space had an advertisement in place while receiving and analyzing the gaze data. Then, if the advertisement space had an advertisement in place, the server may apply a first adjustment to the wearer-view data before using the wearer-view data to determine the advertisement value (e.g., an adjustment or weighting factor that corresponds to the particular advertisement that is displayed). On the other hand, if an advertisement was not displayed in the advertisement space, then the server may apply a second adjustment to the wearer-view data (e.g., an adjustment that accounts for the fact that no advertisement was displayed).

In such an embodiment, the server may determine whether or not the advertisement space had an advertisement in place in various ways. For example, the server may query an advertisement space database to determine whether the advertisement space is in use and if so, what advertisement is being displayed. Additionally or alternatively, the server may analyze the gaze data itself (e.g., by analyzing point-

of-view video in which the advertisement space is detected). Other examples are also possible.

In yet another aspect, some embodiments may implement gaze-data requirements that require a certain amount of gaze data be analyzed before an advertisement space can be offered for sale in an ad-marketplace system. For instance, an ad-marketplace system may require that gaze data from a threshold number of devices have been analyzed before the determined advertisement value is deemed accurate enough to offer the advertisement space for sale via the marketplace. Additionally or alternatively, ad-marketplace system may require that gaze data be monitored for a certain period of time (e.g., at least a week) before the determined advertisement value is deemed accurate enough to offer the advertisement space for sale.

Further, in some embodiments, wearer-view data requirements may require that a certain amount of wearer-view data be generated before an advertisement space can be offered for sale in an ad-marketplace system. For example, an ad-marketplace system may require that a certain number of occurrences of an advertisement space be detected before the determined advertisement value is deemed accurate enough to offer the advertisement space for sale via the marketplace (or in other words, require that the wearer-view data takes into account at least a threshold number of occurrences). Other examples are also possible.

XVIII. Use of Supplemental Gaze Data from Non-Wearable Computing Devices

In some embodiments, an exemplary method may incorporate supplemental gaze data from one or more non-wearable computing devices. In such an embodiment, the supplemental gaze data may include media captured by the non-wearable computing devices. For example, supplemental gaze data may be received from mobile phones, tablet computers, network-enabled video and/or still cameras, and/or other non-wearable computing devices.

Similar to the gaze data from wearable computing devices, the supplemental gaze data is generally indicative of a respective user-view associated with a device that provides supplemental gaze data. Accordingly, an exemplary method may further involve receiving supplemental gaze data from one or more non-wearable computing devices that are registered to a given user-account. A server may then detect additional occurrences of advertisement spaces in the supplemental gaze data, and factor the additional occurrences in when determining the advertisement value for a given advertisement space.

However, because supplemental gaze data is captured at non-wearable devices, supplemental gaze data may less reliably represent what the user actually sees, as compared to gaze data captured by a wearable device that is physically worn on the user's person. Accordingly, in an exemplary method, the server may weight supplemental occurrences that are detected in the supplemental gaze data in order to account for the increased probability that the supplemental gaze data does not represent what the user actually sees. For example, the server may weight a supplemental occurrence by a significance factor corresponding to the likelihood that the corresponding supplemental gaze data is indicative of a respective user-view associated with the non-wearable computing device that provided the supplemental gaze data in which the supplemental occurrence was detected.

In a further aspect, systems may be implemented to actively search supplemental gaze data that is pre-recorded, such as image libraries that are accessible via a network. For

example, a server or an associated system may be configured to analyze images from one or more online image albums to determine supplemental user-view data for the advertisement space. In such an embodiment, the supplemental user-view data is based on occurrences of the advertisement space in the plurality of images.

For example, the system may search image albums on a photo-sharing website, social-network website, or another network source, for occurrences of the advertisement space in the images. When an occurrence is found, the system may generate supplemental user-view data for the occurrence. For instance, many such websites require that users open a user-account in order to create photo albums and/or share photos. Accordingly, the system may store data linking the occurrence of the advertisement space to the user-account via which the image was shared.

In a further aspect, one image of an advertisement space may be indicative of a more valuable view of the advertisement space than another image. As such, each image that includes the advertisement space may be evaluated for indications of how significant the occurrence is, so that the occurrence may be weighted accordingly when determining the advertisement value.

For example, an exemplary method may involve analyzing one or more images from one or more image albums to detect any occurrences of the advertisement space in the images. Then, for each image where an advertisement space is detected, the system may determine an ad-value contribution and/or an advertising-value contribution for the given image (note that in some instances, the advertising-value contribution may be used as the ad-value contribution). As a specific example, determining a prominence value corresponding to a prominence of the advertisement space in the given image (e.g., a size and/or location of the advertisement space in the image), and then use the prominence value as a basis for determining an ad-value contribution and/or an advertising-value contribution for the given image. The system may then use any ad-value contributions from these images, such as any prominence values that are determined for any of the images, when determining the advertisement value for a given advertisement space. Similarly, the system may use any advertising-value contributions from these images, such as any prominence values that are determined for any of the images, when determining the advertisement value for the advertisement space.

In a further aspect, data from GPS systems and other sensors, such as magnetometers, accelerometers, and/or gyroscopes may provide supplemental gaze data. In some embodiments, GPS on a wearable computing device or another device (e.g., a mobile phone or tablet) may provide an indication of location, and a magnetometer on the same device may provide an indication of orientation, such that it may be inferred that a user of the device is in a certain location and is facing a certain direction. Further, the location of an advertisement space may also be determined as described herein. Thus, if the user is inferred to be facing a location where and space is located, this may be considered a view of the advertisement space, and thus may be factored into the wearer-view data. Other examples of using GPS and/or sensor data to infer supplemental gaze data are also possible.

While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

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The invention claimed is:

1. A method comprising:
 - determining a first location of a real-world surface in image data received from a wearable device, the image data including point-of-view image data received from the wearable device;
 - determining that a second location of the wearable device is within a distance of the real-world surface; and
 - determining a focus value for the real-world surface based on the first location, the focus value associated with an amount of attention provided by a user to the real-world surface within a field of view.
2. The method of claim 1, further comprising:
 - identifying an amount of movement for the real-world surface between a first frame of the image data and a second frame of the image data,
 - the focus value for the real-world surface is based on the amount of movement.
3. The method of claim 1, further comprising:
 - determining a gaze-location of the user associated with the wearable device,
 - the focus value for the real-world surface is based on the gaze-location.
4. The method of claim 1, wherein determining the first location of the real-world surface in the image data comprises determining coordinates of the real-world surface in at least one frame of the image data.
5. The method of claim 1, wherein the point-of-view image data comprises images captured from a forward or outward-facing camera on the wearable device.
6. The method of claim 1 further comprising:
 - identifying a size of the real-world surface in at least one frame of the image data,
 - the focus value for the real-world surface is based on the size of the real-world surface in the at least one frame.
7. The method of claim 1, wherein the second location is a geographic location associated with the wearable device.
8. The method of claim 1, wherein the real-world surface is an advertisement space.
9. A non-transitory computer-readable medium storing instructions executable by one or more processors for executing a method, comprising:
 - determining a first location of a real-world surface in image data received from a wearable device, the image data including point-of-view image data received from the wearable device;
 - determining that a second location of the wearable device is within a distance of the real-world surface; and
 - determining a focus value for the real-world surface based on the first location, the focus value associated with an amount of attention provided by a user to the real-world surface within a field of view.
10. The non-transitory computer-readable medium of claim 9, wherein the method further comprises:
 - identifying an amount of movement for the real-world surface between a first frame of the image data and a second frame of the image data,
 - the focus value for the real-world surface is based on the amount of movement.
11. The non-transitory computer-readable medium of claim 9, wherein the method further comprises:
 - determining a gaze-location of the user associated with the wearable device,
 - the focus value for the real-world surface is based on the gaze-location.
12. The non-transitory computer-readable medium of claim 9, wherein determining the first location of the real-

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world surface in the image data comprises determining coordinates of the real-world surface in at least one frame of the image data.

13. The non-transitory computer-readable medium of claim 9, wherein the point-of-view image data comprises images captured from a forward or outward-facing camera on the wearable device.

14. The non-transitory computer-readable medium of claim 9, wherein the method further comprises:

- identifying a size of the real-world surface in at least one frame of the image data,
- the focus value for the real-world surface is based on the size of the real-world surface in the at least one frame.

15. The non-transitory computer-readable medium of claim 9, wherein the second location is a geographic location associated with the wearable device.

16. The non-transitory computer-readable medium of claim 9, wherein the real-world surface is an advertisement space.

17. A computing apparatus comprising:

- computer-readable storage media;
- at least one processor operatively coupled to the computer-readable storage media; and

- program instructions stored on the computer-readable storage media that, when executed by the at least one processor, perform a method, the method comprising:
 - detecting a real-world surface in image data received from a wearable device, the image data including point-of-view image data received from the wearable device;

- identifying an amount of movement for the real-world surface between a first frame of the image data and a second frame of the image data; and
- determining a focus value for the real-world surface based on the amount of movement, the focus value associated with an amount of attention provided by a user to the real-world surface.

18. The computing apparatus of claim 17, wherein the method further comprises:

- determining a geographical location associated with the wearable device; and
- determining that the geographical location is within a threshold distance of the real-world surface, wherein detecting the real-world surface in the image data occurs in response to determining that the geographical location is within the threshold distance of the real-world surface.

19. The computing apparatus of claim 17, wherein the method further comprises:

- determining a location of the real-world surface relative to a center of a frame of the image data,
- the focus value for the real-world surface is based on the location.

20. The computing apparatus of claim 17, wherein the method further comprises:

- determining a gaze-location of a user associated with the wearable device,
- the focus value for the real-world surface is based on the gaze-location.

21. The computing apparatus of claim 17, wherein the point-of-view image data comprises images captured from a forward or outward-facing camera on the wearable device.

22. The computing apparatus of claim 17, wherein the method further comprises:

- identifying a size of the real-world surface in at least one frame of the image data,

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the focus value for the real-world surface is based on the size of the real-world surface in the at least one frame of the image data.

23. The computing apparatus of claim **17**, wherein the real-world surface is an advertisement space.

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